



Earth Observation for Sustainable Development



Urban Development Project

Service Operations Report - Dhaka

ESA Ref:
Doc. No.:
Issue/Rev.:
Date:

AO/1-8346/15/I-NB
EO4SD_Urban_DHAKA_Operations_Report
2.6
23.03.2018

Consortium Partners

No.	Name	Short Name	Country
1	GAF AG	GAF	Germany
2	Système d'Information à Référence Spatiale SAS	SIRS	France
3	GISAT S.R.O.	GISAT	Czech Republic
4	Egis SA	EGIS	France
5	Deutsche Luft- und Raumfahrt e. V	DLR	Germany
6	Netherlands Geomatics & Earth Observation B.V.	NEO	The Netherlands
7	JOANNEUM Research Forschungsgesellschaft mbH	JR	Austria
8	GISBOX SRL	GISBOX	Romania

Disclaimer:

The contents of this document are the copyright of GAF AG and Partners. It is released by GAF AG on the condition that it will not be copied in whole, in section or otherwise reproduced (whether by photographic, reprographic or any other method) and that the contents thereof shall not be divulged to any other person other than of the addressed (save to the other authorised officers of their organisation having a need to know such contents, for the purpose of which disclosure is made by GAF AG) without prior consent of GAF AG.

Summary

This document contains information about EO4SD-Urban service operations performed for providing to stakeholders the full geospatial dataset required over the Area of Interest of Dhaka located in Bangladesh. The context as well as the product requirements and specifications are firstly detailed, before providing the relevant information about all steps of the service operations including the input data, the processing methods, the accuracy assessment and Quality Control procedures. Finally, the relevant analysis of mapping results is presented with explicit visual inputs.

	Affiliation/Function	Name	Date
Prepared	GISAT S.R.O.	J.Kolomaznik, V.Stonacek, M.Jindrova	08/09/2017
Reviewed	GISAT S.R.O.	T.Soukup	15/09/2017
Approved	GAF AG, Project Coordinator	T. Haeusler	22/09/2017

The document is accepted under the assumption that all verification activities were carried out correctly and any discrepancies are documented properly.

Distribution

Affiliation	Name	Copies
ESA	Z. Bartalis	electronic copy
WBG	Jon Ker Kaw & WBG working team	electronic copy
WBG	Luisa M. Mimmi & WBG working team	electronic copy
Partners	Consortium Members	electronic copy

Document Status Sheet

Issue	Date	Details
1.0	15/09/2017	Document Initial Issue
2.0	11/10/2017	Final version
2.3	06/02/2018	Updated and extended statistics for Open and green product
2.4	23/03/2018	Updated according to comments from WBG
2.4	31/12/2018	Final version including full Open and green assessment
2.6	04/10/2019	Updated chapters Executive Summary and Stakeholders and Requirements

Document Change Record

#	Date	Request	Location	Details
	11/10/2017			Added chapter 3.5, 4.1, 4.6
	06/02/2018			Updated chapter 4.5
	31/12/2018			Updated chapter 4.5
	04/10/2019			Updated chapters Executive Summary and Stakeholders and Requirements

Executive Summary

The European Space Agency (ESA) has been working closely together with the International Finance Institutes (IFIs) and their client countries to demonstrate the benefits of Earth Observation (EO) in the IFI development programmes. Earth Observation for Sustainable Development (EO4SD) is an ESA initiative, which aims to achieve an increase in the uptake of satellite based information in the regional and global IFI programmes. The overall aim of the EO4SD Urban project is to integrate the application of satellite data for urban development programmes being implemented by the IFIs or Multi-Lateral Development Banks (MDBs) with the developing countries. The overall goal will be achieved via implementation of the following main objectives:

- To provide a service portfolio of Baseline and Derived urban-related geo-spatial products
- To provide the geo-spatial products and services on a geographical regional basis
- To ensure that the products and services are user-driven

The report describes the methodological approach to produce geospatial products and results of urban analytics derived from the products as implemented as a part of the ESA funded EO4SD Urban project for Dhaka in Bangladesh in collaboration with the World Bank. EO-based information support for Dhaka has been provided to two teams/activities:

- 1) World Bank Urban GP supported UrbanScapes Community-of-Practice (CoP) platform under an umbrella of ASA programme: **“Transforming Cities through Public Spaces”**
- 2) World Bank Water GP - Water Supply and Sanitation (WSS) Global Solutions Group’s **“Analytical and Planning Tool for Integrating Multi-dimensional Data in Urban Slums”**.

The services delivered included both standard products from the EO4SD Urban product portfolio and specific products tailored to requirements of the teams in support of Open & Green Spaces and Slum deprivation analysis. The Report provides a Service Description by referring to the user driven service requirements and the associated product list with the detailed product specifications. The following products were requested and delivered:

- Global Urban Products
- Urban Extent as of 2017
- Urban & Peri-Urban Land Use / Land Cover status and change over the observed period (2006 – 2017)
- Informal Settlements/Slum Areas as of 2006 and 2017
- Urban Green Areas and Open Spaces as of 2017
- Transportation network as of 2017
- Flood History, Hazard and Risk based on last 10 years

A description of the user requirements process and overall approach adopted are provided in the initial sections of the document followed by a presentation of the results obtained for the each service and the main findings based on analytics included, whilst a detailed description of the methodology used for each product is provided in the annexes (Annex 1) as well as detailed quality control sheets (Annex 2).

Table of Contents

1	INTRODUCTION GENERAL BACKGROUND OF EO4SD-URBAN	9
2	SERVICE DESCRIPTION	9
2.1	STAKEHOLDERS AND REQUIREMENTS.....	9
2.2	SERVICE AREA SPECIFICATION	12
2.3	PRODUCT LIST AND PRODUCT SPECIFICATIONS.....	13
2.4	LAND USE/LAND COVER NOMENCLATURE.....	15
2.5	LAND USE/LAND COVER CHANGE NOMENCLATURE.....	16
2.6	NOMENCLATURES OF OTHER PRODUCTS	17
2.7	TERMS OF ACCESS:.....	17
3	SERVICE OPERATIONS.....	18
3.1	SOURCE DATA	18
3.2	PROCESSING METHODS	19
3.3	ACCURACY ASSESSMENT OF MAP PRODUCTS	19
3.3.1	The Applied Sampling Design	19
3.3.2	The Applied Response Design	21
3.3.3	The Applied Analysis Design.....	22
3.4	QUALITY CONTROL/ASSURANCE	22
3.5	OVERVIEW OF ACCURACY ASSESSMENT PER PRODUCT.....	23
3.6	METADATA	24
4	ANALYSIS OF MAPPING RESULTS	25
4.1	GLOBAL URBAN PRODUCTS	25
4.2	URBAN EXTENT - STATUS AND TRENDS	26
4.3	LAND USE / LAND COVER - STATUS AND TRENDS.....	30
4.4	INFORMAL SETTLEMENTS - STATUS AND TRENDS	37
4.5	URBAN GREEN AREAS AND OPEN SPACES.....	48
4.6	TRANSPORTATION NETWORK – STATUS.....	54
4.7	FLOOD HISTORY, HAZARD AND RISK.....	57

Annexes

Annex 1: Processing Methods for EO4SD-Urban Products

Annex 2: QC Sheet

List of Figures

Figure 1: Dhaka, Bangladesh - Core City (green), Service Area (red).....	12
Figure 2: VHR data of the city of Dhaka of the year 2006 (left) and year 2017 (right) overlaid with randomly distributed sample points for accuracy assessment.	21
Figure 3: Quality Control process for EO4SD-Urban product generation. At each intermediate processing step output properties are compared against pre-defined requirements.	23
Figure 4: Global Settlement Extent Status (left) and Level of Imperviousness (right) - 2015.....	25
Figure 5: Spatial distribution of Urban Extent and Growth	26
Figure 6: Urban Extent Status and Growth - overall, in Core City and in Larger Urban Zone in % (left) and km2 (right).....	27
Figure 7: Urban Extent in 2006 and Urban Extent Growth 2006-2017 in % of district area.....	28
Figure 8: Urban Extent 2017 (% urbanized area of total area; left) and Urban Extent Growth Intensity 2006-2017 (% increase comparing to initial 2006 area; right) per districts – spatial distribution.	29
Figure 9: Detailed Land Cover Land Use 2017- Spatial distribution.....	30
Figure 10: Detailed Land Cover Land Use 2017 structure - overall, in Core City and in Larger Urban Zone in % (left) and km2 (right).	31
Figure 11: Detailed Land Cover Land Use 2017 structure in % per districts.....	32
Figure 12: Land Cover Land Use Change Types - Spatial Distribution.	33
Figure 13: Land Cover Land Use Change Types 2006-2017 - overall, in Core City and in Larger Urban Zone in % (left) and km2 (right).	34
Figure 14: Land Cover Land Use Changes Types 2006-2017 structure in % per districts.	35
Figure 15: Urban Densification change type intensity (left) and Share of urban extension (LCF1) per Urban Fabric 2006 per ward (right) 2006-2017 (in %) - Spatial Distribution.	36
Figure 16: Consumption (left) and Formation structure (right) for Land Cover Land Use changes LCF1 (Urban extension) 2006-2017 - overall, in Core City and in Larger Urban Zone in %.....	36
Figure 17: Informal Settlements/Slum Areas (left) and Slum Areas Types represented by points (right) - Spatial Distribution 2017.	37
Figure 18: Informal Settlements/Slum Areas distribution per type - overall, in Core City.....	38
Figure 19: Informal Settlements/Slum Areas distribution per type in % per districts area – status 2017.	40
Figure 20: Informal Settlements/Slum Areas distribution per type in % per total urban fabric area in each district – status 2017.....	42
Figure 21: Informal Settlements/Slum Areas proportion in % of district area (left) and Informal Settlements/Slum Areas proportion in % of total urban fabric area in each district (right) – status 2017.	43
Figure 22: Informal Settlements/Slum Areas development: change rate of Informal Settlements/Slum Areas (calculated as ratio of slum area as of 2017 on slum area as of 2006) (left) and changes in Informal Settlements/Slum Areas proportion in % of total urban fabric area in each district (right) – period 2006-2017.....	44
Figure 23: Proportion of Informal Settlements/Slum Areas in flood prone areas (high/low risk) – overall, in Core City in % of slum area (left) and in km2 (right) – status 2017.....	45
Figure 24: Proportion of Informal Settlements/Slum Areas in flood prone areas (high/low risk) in % of slum area per districts – status 2017.....	46
Figure 25: List of spatial characteristics automatically generated for slums and slum neighborhoods.	47
Figure 26: Examples of Informal Settlements/Slum Areas Types	47
Figure 27: Map of open and green spaces based on extended nomenclature – overview and detail maps	49
Figure 28: Development of green patterns: change of greenness index (2006-2017) per urban fabric blocks – left; Cumulative change of greenness index (2006-2017) per ward – right.....	49
Figure 29: Share of OGS (potential public spaces) Total area of Open and green spaces per ward (ha) – left; Share of Open and green spaces on Urban fabric per ward (%) – right.....	50
Figure 30: Open Green Spaces <i>Connectivity</i> : Mean distance between nearest Open Green Spaces in ward (m) – left; Euclidian distance to the nearest Open Green Spaces calculated for each Open Green Space - right	51

Figure 31: Open Green Spaces <i>Accessibility</i> . Street path distance to the nearest Open Green Space (m) – left; Average street path distance to the nearest Open Green Space (m) per ward – right	52
Figure 32: Directional distribution and density (as % of Urban Fabric) of selected OGS as a function of distance from the CBD	53
Figure 33: Structure of OGS as a function of distance from the CBD	53
Figure 35: Current status of road network (left) and its density - mean length per area unit per ward (km/km ²).....	54
Figure 36: Density of street intersections: Number of intersections per 1 km ² of urban fabric per ward (left) and intersections per 1 km ² grid (right).....	55
Figure 37: Urban Street Density – Street Length in km per 1km ² per ward (left) and Street Length per 1km ² grid (right).....	56
Figure 38: Proportion of (all, aggregated) flood risk classes per district (left) and Proportion of Flood High Risk per district (right) in % as of 2017	57
Figure 39: Flooded areas within AOI for the historical remarkable flooding event in 2004 (left) and 2014 (right).....	58
Figure 40: Flooded areas within AOI for the flooding event on August 10 th , 2017 (left) and August 24 th , 2017 (right).....	59
Figure 41: Flooded areas of AOI for peri-urban (top) and core-urban area (below) in km ² for the year 2004, 2007, 2014 and 2016. The total area of peri-urban is 710.73km ² and core urban area is 324.34km ²	60
Figure 42: Flood hazard within AOI (top) and core urban area (bottom) based on flooding events from 2007-2017.....	61
Figure 43: Flood hazard statistics for peri-urban (a) and core urban (b) area proportion in % and in km ² (c).	62
Figure 44: Flood risk within AOI of peri-urban (top) and core urban area (bottom) based on flooding events from 2007-2017.....	63
Figure 45: Flood risk for peri-urban (a) and core urban area (b) within AOI in area proportion % and km ² (c)	64
Figure 46: Flood risk for agriculture land for peri-urban (a) and core urban area (b) in area proportion % and area km ² (c).....	65
Figure 47: Flood risk level for high density continuous and discontinuous residential areas (left and middle) and low density residential area (right) in area proportion %	65
Figure 48: Flood risk levels for different land use classes within urban region are demonstrated in km ²	66

List of Tables

Table 1: Summary of requirements.	10
Table 2: Product list	13
Table 3: LULC nomenclature.....	15
Table 4: LCF nomenclature.....	16
Table 5: Number of sampling points for the EO4SD-Urban mapping classes after applied sampling design for year 2017.	20
Table 6: Overview of accuracy assessment per product.....	23
Table 7: Overall LU/LC Statistics.....	33
Table 8: Area and Share of different Informal Settlements/Slums Types.....	38
Table 9: <i>Inclusivity</i> : Percentage of population living within 400m catchment areas.....	51

List of Abbreviations

ADB	Asian Development Bank
AOI	Area of Interest
CS	Client States
DEM	Digital Elevation Model
DLR	German Space Agency - Geospatial Service Provider, Germany
EDF	European Development Fund
EEA	European Environmental Agency
EGIS	Consulting Company for Environmental Impact Assessment and Urban Planning, France
EIB	European Investment Bank
EO	Earth Observation
EO4SD	Earth Observation for Sustainable Development
ESA	European Space Agency
EU	European Union
FAO	Food and Agriculture Organization
GAF	GAF AG, Geospatial Service Provider, Germany
GDB	ESRI Geodatabase
GEO	Group on Earth Observations
GIS	Geographic Information System
GISAT	Geospatial Service Provider, Czech Republic
GISBOX	Romanian company with activities of Photogrammetry and GIS
GUF	Global Urban Footprint
HR	High Resolution
IFI	International Financing Institute
INSPIRE	Infrastructure for Spatial Information in the European Community
ISO/TC 211	Standardization of Digital Geographic Information
JR	JOANNEUM Research, Austria
LCF	Land use / Land cover Change Flow
LIDAR	Light Detection and Ranging
LUCC	Land use / Land cover Change
LULC	Land use / Land cover
MDB	Multi-lateral Development Bank
MDG	Millennium Development Goal
MGCP	Multinational Geospatial Co-production Program
MMU	Minimum Mapping Unit
NDVI	Normalized Difference Vegetation Index
NEO	Geospatial Service Provider, The Netherlands
NGO	Non-Governmental Organisation
ODA	Official Development Assistance
OGC	Open Geospatial Consortium
OSP	Operational Service Providers
OSM	Open Street Map
PUMA	Web Based Platform for Urban Management and Analysis
QA	Quality Assurance

QC	Quality Control
R&D	Research and Development
RMSE	Root Mean Square Error
SAR	Synthetic Aperture Radar
SC	Service Cluster
SDG	Sustainable Development Goal
S.L.	(Soil) Sealing Level ~ Imperviousness
SO	Service Operations
SP	Service Provider
SRTM	Shuttle Radar Topographic Mission
TEP	Thematic Exploitation Platform
TNA	Training Need Assessment
ToC	Table of Contents
UN	United Nations
UNDP	United Nations Development Programme
UUA	User Utility Assessment
VHR	Very High Resolution
WB	World Bank
WBG	World Bank Group

1 Introduction General Background of EO4SD-Urban

Since 2008 the European Space Agency (ESA) has worked closely together with the International Finance Institutes (IFIs) and their client countries to harness the benefits of Earth Observation (EO) in their operations and resources management. Earth Observation for Sustainable Development (EO4SD) is a new ESA initiative, which aims to achieve an increase in the uptake of satellite based information in the regional and global IFI programmes. The EO4SD-Urban project initiated in May 2016 (with a duration of 3 years) has the overall aim to integrate the application of satellite data for urban development programmes being implemented by the IFIs with the developing countries. The overall goal will be achieved via implementation of the following main objectives:

- To provide the services on a regional basis (i.e. large geographical areas); in the context of the current proposal with a focus on S. Asia, SE Asia and Africa, for at least 35-40 cities.
- To ensure that the products and services are user-driven; i.e. priority products and services to be agreed on with the MDBs in relation to their regional programs and furthermore to implement the project with a strong stakeholder engagement especially in context with the validation of the products/services on their utility.
- To provide a service portfolio of Baseline and Derived urban-related geo-spatial products that have clear technical specifications, and are produced on an operational manner that are stringently quality controlled and validated by the user community.
- To provide a technology transfer component in the project via capacity building exercises in the different regions in close co-operation with the MDB programmes.

This report supports the fulfilment of the third objective which requires the provision of geo-spatial Baseline and Derived geo-spatial products to various stakeholders in the IFIs and counterpart City authorities. The report provides a Service Description, and then in Chapter 3 systematically reviews the main production steps involved and importantly highlights whenever there are Quality control (QC) mechanisms involved with the related QC forms in the Annexe of this report. The description of the processes is kept intentionally at a top level and avoiding technical details as the report is considered mainly for non-technical IFI staff and experts and City authorities. Finally Chapter 4 presents the standard analytical work undertaken with the products which can be inputs into further urban development assessments, modelling and reports.

2 Service Description

The following Section summarises the service as it has been realised for the city of Dhaka (Bangladesh) within the EO4SD-Urban Project and as it had been delivered to the user in September 2017.

2.1 Stakeholders and Requirements

EO-based information support for Dhaka has been provided to two teams/activities - World Bank Urban GP's Advisory Services and Analytics (ASA) Programme on **“Urban Spaces for City Transformation”** and World Bank Water GP - Water Supply and Sanitation (WSS) Global Solutions Group's **“Analytical and Planning Tool for Integrating Multi-dimensional Data in Urban Slums”**.

Following table provides summary of products identified during user capacity and requirements assessment phase and reflected by user requirement specification. The services delivered included both standard products from the EO4SD Urban product portfolio and specific products tailored to requirements of the teams in support of Open & Green Spaces and Slum deprivation analysis.

Table 1: Summary of requirements.

Required Product Name (naming adopted from D3 specification)	Specific requirements (beyond standard product specification in D3)	Producer
Urban & Peri-Urban Land Use / Land Cover (LULC)	None	GISAT
Urban & Peri-Urban Land Use / Land Cover Change (LUCC)	None	GISAT
Open Green Areas	Land use changes and identification of green areas and public spaces Typology of public / green / open space - Characterize and describe: with appropriate indicators, how urban spaces are physically organized and structured within cities vis-à-vis land use, housing, connected, hierarchy, distribution. (e.g. polycentric; monocentric; sprawling etc.) - Where possible, provide some quantitative analysis (standard metrics) of the quality of urban spaces in selected cities (e.g. amount of green spaces; urban areas that are susceptible to floods etc.)	GISAT
Informal Settlements	Density of housing in order to support population density estimation Create spatially-defined typology classes (relevant to predict slum severity as response variable) using satellite imagery. Informal settlements (typology) location, topography and morphology and growth patterns, including time of settlement, spread and density of population: - location: in the hearth of city, old city slums, in path of development (along roads, airport), along railways, in residential centres, at fringe, in suburban industrial areas - pattern: shape, internal structure (rate of homogeneity, heterogeneity, building size, height & spacing), density (-> population) - size: area - growth: increase / decrease, dating Level of vulnerability to disasters and climate change and water scarcity storms, floods and earthquakes, etc., depending on: - Location - natural hazards (flood, landslide), technical hazards (industrial, road/railroad accident, plane crash) - Housing structural typology – material (limited, height, congestion / density) Accessibility - road density and distance to main roads	GISAT, GISBOX
Transport Network	None	GISAT
Urban Extent	None	DLR
Imperviousness	None	DLR
Flood History, Hazard & Risk	None	NEO

Requirements for Open & Green Spaces analytics were collected from UrbanScapes Community-of-Practice (CoP) platform under WB's ASA programme, whose objective is to develop a framework to examine how urban spaces can transform urban environments by promoting inclusive green growth and enhancing livability in cities. Specifically, Urban Scapes required development of enhanced diagnostics based on spatial analysis of localization and characteristics of public spaces to facilitate:

- Prioritization of interventions improving livability;
- Evidence-based discussion with local city partners;
- City-wide and inter-city assessment of public spaces patterns.

GP Water activities in Dhaka are addressing operational slum upgrading initiatives aiming at **evidence-based planning** of public services including WSS and systematic **selection of sites and justification for priority investments**. Therefore, EO4SD Urban products were required to facilitate:

- Identification of the main (spatially-driven) factors and patterns that can predict levels of WSS coverage in the slums of a city of Dhaka;
- Integration of multi-dimensional data feeding analytical and planning tool for both identifying slums and for targeting and prioritizing slums for interventions.

2.2 Service Area Specification

The Service Area or Area of Interest (AOI) has been divided in two parts:

- *Core Urban Area* which corresponds to the most populated area including the historic city centre represented by the Dhaka Metropolitan (DMP) area.
- *Peri-Urban Area* which covers the surroundings of the city represented by regular buffer (6 km) beyond the Dhaka Metropolitan (DMP) area boundary. In this document, it is also call Large Urban Zone (LUZ).

This AOI geographical division has been set up for offering products specifically relevant for artificial areas only and which require the use of Very High Resolution satellite imagery (< 4 m spatial resolution, see chapter 3.1 for details).

Figure 1 shows the Core City (324 km²) and the whole AOI (710 km²) of Dhaka, Bangladesh.

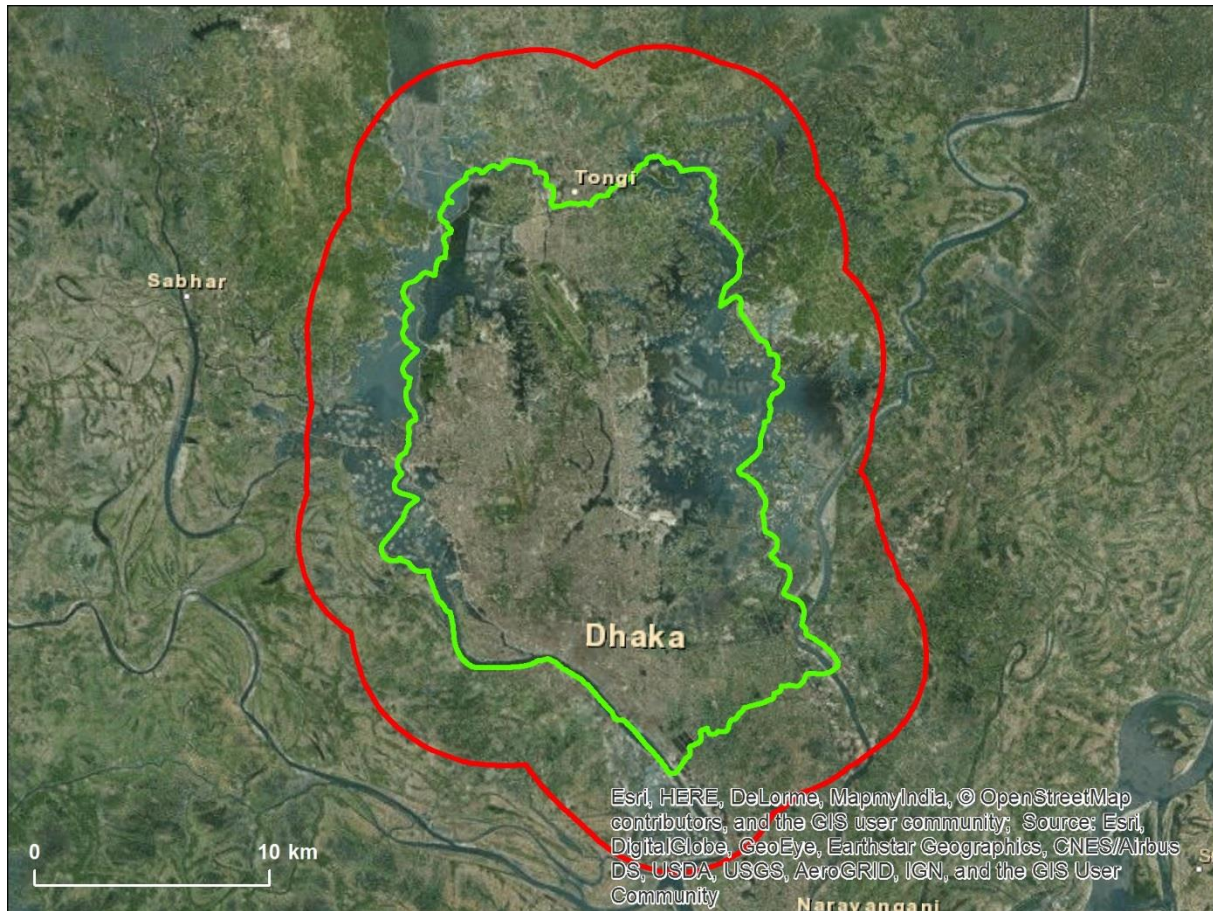


Figure 1: Dhaka, Bangladesh - Core City (green), Service Area (red).

2.3 Product List and Product Specifications

The following Table lists the products which have been produced within the EO4SD-Urban project for the production site in Dhaka.

Table 2: Product list

Product Name	Classes and Definitions	Resolution	
		Spatial (MMU, scale)	Temporal (revisit)
Urban & Peri-Urban Land Use / Land Cover (LULC)	<u>Land Cover Level 3:</u> 11100 - Formal high density continuous residential (Sealing level > 80%) 11200 - Formal high density discontinuous residential (Sealing level: 50% - 80%) 11300 - Formal low density discontinuous residential (Sealing level: 10% - 50%) 11500 - Village settlement (Sealing level < 10%) 12000 - Commercial and industrial units 12100 - Non-residential urban fabric 12200 - Roads and associated land 12300 - Airports 13000 - Construction sites 13100 - Sand-filled area 13200 - Land without current use 14100 - Urban greenery 14200 - Sport and leisure facilities 20000 - Agriculture 31000 - Trees 32000 - Other Natural and Semi-natural areas incl. wetlands 33000 - Bare land 50000 - Water bodies	Core Urban Area: 0.5 m (MMU – 0.25 ha) Peri-Urban Area: 10 m (MMU – 1 ha)	2006/2017 11-year interval
Urban & Peri-Urban Land Use / Land Cover Change (LUCC)	<u>Land Cover Flows Level 1:</u> LCF1 – Urban Extension LCF2 – Urban Internal changes LCF3 – Agricultural development LCF4 – Natural and semi-natural internal changes LCF5 – Water bodies development LCF6 – Natural and semi-natural development	Core Urban Area: 0.5 m (MMU – 0.25 ha) Peri-Urban Area: 10 m (MMU – 1 ha)	2006/2017 11-year interval
Open Green Areas	<u>Extended definition:</u> Green - Alley - Central / downtown park - Neighbourhood park - Mini pocket park - Sport & leisure facility - City woodland - Sub-urban forest - Airport green - Other / Unrecognized green area	Core Urban Area: 0.1 – 0.25 ha (MMU)	2017

	Market Square - Central - Neighbourhood Cemetery Living street Waterfront - Boardwalk / Berthing structure - Beach - Park Water		
Informal Settlements	<u>Single date (2017):</u> High density Low density Pocket Linear along railway Linear along rivers Under or mixed with trees <u>Change product:</u> 2006 / 2010 / 2017	Core Urban Area: 0.1 – 0.25 ha (MMU)	2006, 2010, 2017
Transport Network	Highway / Motorway Primary road Secondary road Tertiary / Local road	N/A	2017
Urban Extent	<u>Single date</u> Urban / artificial area	Core- and peri- Urban Area: 1 ha (MMU)	2017
Imperviousness	0 - Non-impervious area 1-100 – Impervious area (1 to 100% imperviousness degree) 254 - Unclassifiable (no satellite image available or presence of clouds, shadows) 255 - Outside area of interest	30 m	2017
Flood History, Hazard & Risk	<u>Product 1: Flood Extent</u> Water body Flooded area <u>Product 2: Flood Hazard</u> 0 - Water body 1-n - Number of flood occurrences in the past 10 years -999 - Remarkable historical flood (2000) <u>Product 3: Flood Risk</u> Flood risk level considering both flood hazard level (1-2-3) and land use damage cost level (A-B-C-D): Low risk (1A-1B-2A) Medium risk (1C-1D-2B-2C-3A-3B) High risk (2D-3C) Very high risk (3D)	30 m	2007 - 2017

2.4 Land Use/Land Cover Nomenclature

A pre-cursor to starting production was the establishment with the stakeholders on the relevant Land Use/Land Cover (LULC) nomenclature as well as class definitions. The approach taken was to use a standard remote sensing based LULC nomenclature and then adapt it to the user's LU requirements. Thus the remote-sensing based LULC classes in the urban context can be grouped into 5 Level 1 classes, which are Artificial areas, Natural/ Semi Natural, Agricultural, Wetland, Water bodies. These classes can then be sub-divided into several different more detailed classes such that the dis-aggregation can be down to Level 2-4. A depiction of the way the levels and classes are structured is presented as follows:

Table 3: LULC nomenclature

LULC Level I	LULC Level II	LULC Level III
Artificial surfaces	Urban Fabric	Continuous Urban Fabric (Sealing Layer-SL > 80%)
		Discontinuous Dense Urban Fabric (S.L. 50% - 80%)
		Discontinuous Low Density Urban Fabric (S.L. 10% - 50%)
		Village Settlements (S.L. < 10%)
	Industrial, commercial, public, military, private and transport unit	Industrial and commercial units
		Non-residential urban fabric
		Road and rail network and associated land (Open Street Map)
		Airport and port areas
	Urban greenery	Urban green
		Sport and leisure facilities
	Construction and other sites	Construction sites
		Sand-filled areas
		Land without current use
Non-artificial surface	Agriculture land	
	Forest	
	Natural and semi-natural land including wetlands	
	Bare land	
	Water bodies	

(Reference : Urban Atlas, 2012)

It should be noted that the Level II classes were used as the basis for classification of the Peri-urban areas using the High Resolution (HR) data such as Landsat or Sentinel-2. However, for the Core Urban areas using the Very High Resolution (VHR) data it was possible to go down to Level III. The different Levels, classes and sub-classes from the remote sensing based Urban classification, were adapted to the User requirements based on existing Master Plans for cities and/or direct advise from the User on critical classes required. The final LULC nomenclature had to be endorsed by the user before productions started.

2.5 Land Use/Land Cover Change Nomenclature

Beside the individual changes - individual Land Use/Land Cover classes combinations for both reference years, Land Use/Land Cover Changes are grouped into logical groups related to their causes and presented as LULC flows (LCF) for more straightforward user interpretation. For the flows groups specification see Table 4 below.

Table 4: LCF nomenclature

LCF level 1	LCF level 2	Description
LCF10: Urban extension	LCF11: Urban extension: extension of residential urban fabric	Formation of new residential urban fabric over non-artificial land over non-artificial land
	LCF12: Urban extension: extension of commercial & industrial units	Formation of new commercial & industrial fabric over non-artificial land
	LCF13: Urban extension: extension of transportation units	Formation of new transportation units over non-artificial land
	LCF14: Urban extension: extension of non-residential urban fabric	Formation of new non-residential urban fabric over non-artificial land over non-artificial land
	LCF15: Urban extension: extension of urban greenery	Formation of new urban greenery areas over non-artificial land
	LCF16: Urban extension: extension of construction sites	Formation of new construction sites over non-artificial land
LCF20: Urban internal changes	LCF21: Internal urban development: formation and densification of residential urban fabric	Internal conversion between artificial surfaces, formation and densification of residential urban fabric over other urban classes
	LCF22: Internal urban development: formation of commercial & industrial units	Internal conversion between artificial surfaces, formation of commercial and industrial units over other urban classes
	LCF23: Internal urban development: formation of transportation units	Internal conversion between artificial surfaces, formation of transportation units over other urban classes
	LCF24: Internal urban development: formation of non-residential urban fabric	Internal conversion between artificial surfaces, formation of non-residential urban fabric over other urban classes
	LCF25: Internal urban development: formation of urban greenery	Internal conversion between artificial surfaces, formation of urban greenery over other urban classes
	LCF26: Internal urban development: formation of construction sites	Internal conversion between artificial surfaces, formation of construction sites over other urban classes
LCF30: Agriculture development	LCF31: Agriculture uptake	Conversion of various types of natural and semi-natural land into agricultural land
	LCF32: Agriculture abandonment	Abandonment of agricultural land in favor of various types of natural and semi-natural land
	LCF33: Agriculture development: consumption of urban fabric	Conversion of various types of urban fabric into agricultural land

LCF40: Natural and semi-natural internal changes	LCF40: Natural and semi-natural areas internal changes	Internal conversion between various natural and semi-natural classes
LCF50: Riverbed and water bodies development	LCF51: Riverbed development: Extension of agriculture	Conversion of water body into agriculture (related mostly to riverbed development)
	LCF52: Riverbed development: consumption of agriculture	Conversion of agriculture into water body (related mostly to riverbed development)
	LCF53: Riverbed development: extension of natural and semi-natural areas	Conversion of water body into natural and semi-natural land (related mostly to riverbed development)
	LCF54: Riverbed development: consumption of natural and semi-natural areas	Conversion natural and semi-natural land into water body (related mostly to riverbed development)
	LCF55: Riverbed development: consumption of urban fabric	Conversion of various types of urban fabric into water body
LCF60: Natural and semi-natural development	LCF61: Trees development: consumption of urban fabric	Conversion of various types of urban fabric into trees
	LCF62: Natural areas development: consumption of urban fabric	Conversion of various types of urban fabric into natural areas
	LCF63: Vacant land development: consumption of urban fabric	Conversion of various types of urban fabric into vacant land
	LCF64: Bare land development: consumption of urban fabric	Conversion of various types of urban fabric into bare land

2.6 Nomenclatures of other products

Specification and nomenclatures for other specific products e.g. informal settlements, urban green and flood products are described in the Annex 1.

2.7 Terms of Access:

Standard product dissemination is done via online secured EO4SDU FTP solution. Other optional dissemination channels on request.

3 Service Operations

The following Sections present all steps of the service operations including the necessary input data, the processing methods, the accuracy assessment and Quality Control procedures.

3.1 Source Data

This Section presents the remote sensing and ancillary datasets that were used. Different types of data from several data providers have been acquired. A complete list of source data as well as a quality assessment is provided in Annex 2.

A summary of the main data used is provided in the following Sections.

High Resolution Optical EO Data

The data sources for the peri-urban current and historic mapping of urban LULC, urban extent and imperviousness were Landsat and Sentinel data which were accessible and downloadable free of charge.

More specifically, the following images have been downloaded, processed and used:

- Landsat-5: 2 scenes acquired on 21/12/2006 for the peri-urban historic mapping of LULC
 - MS (multispectral) bands 7 resolution 30, 120 (1x) m
- Sentinel-2: 1 scene acquired on 02/01/2017 for the peri-urban current mapping of LULC
 - MS (multispectral) bands 13 resolution 10 (4x), 20 (6x), 60 (3x) m
- Landsat-8: 280 scenes from 2017 for detection of urban extent and imperviousness within peri-urban area
 - MS (multispectral) bands 8 resolution 30 m
 - PAN (panchromatic) band 1 resolution 15 m

Very High Resolution Optical EO Data

The VHR data for the core urban area mapping had to be acquired and purchased through commercial EO Data Providers such as Airbus Defence and European Space Imaging.

It must be mentioned that under the current collaboration project the VHR EO data had to be purchased under **mono-license agreements** between GAF AG and the EO Data Providers. If EO data had to be distributed to other stakeholders, then further licences for multiple users would have to be purchased.

The following VHR sensor data have been acquired:

- Quickbird: 3 scenes acquired on 03/04/2017, 07/12/2006 and 20/12/2006
 - MS (multispectral) bands 4 resolution 2.4 m
 - PAN (panchromatic) band 1 resolution 0.6 m
- Pleiades: 1 scene acquired on 23/03/2017
 - MS (multispectral) bands 4 resolution 2.8 m
 - PAN (panchromatic) band 1 resolution 0.5 m

Ancillary Data

- **DEM:** Terrain heights for the Ortho-Image production were taken from the standard 1 arcsec SRTM DEM (SRTM = Shuttle Radar Topography Mission), providing a ground spatial resolution of 30m x 30m. The SRTM data meet the absolute horizontal and vertical accuracies of 20 meters (circular error at 90% confidence) and 16 meters (linear error at 90% confidence)
- **Data Provided by User:** Local administrative units (wards for core city)
- **Open Street Map (OSM):** Thematic vector data available through Open Street Map portal were mainly related to road network. This data has been downloaded and refined by visual interpretation for creating a full and consistent road network as solid basis before producing the baseline LULC mapping.

Detailed lists of the used ancillary data as well as their quality is documented in the attached Quality Control Sheets in Annex 2.

3.2 Processing Methods

Data processing starts at an initial stage with quality checks and verification of all incoming data. This assessment is performed in order to guarantee the correctness of data before geometric or radiometric pre-processing is continued. These checks follow defined procedures in order to detect anomalies, artefacts and inconsistencies. Furthermore all image and statistical data were visualised and interpreted by operators.

The main techniques and standards used for data analysis, processing and modelling for each product are described in **Annex 1**.

3.3 Accuracy Assessment of Map Products

Data and maps derived from remote sensing contain - like any other map - uncertainties which can be caused by many factors. The components, which might have an influence on the quality of the maps derived from EO include quality and suitability of satellite data, interoperability of different sensors, radiometric and geometric processing, cartographic and thematic standards, and image interpretation procedures, post-processing of the map products and finally the availability and quality of reference data. However, the accuracy of map products has a major impact on secondary products and its utility and therefore an accuracy assessment was considered as a critical component of the entire production and products delivery process. The main goal of the thematic accuracy assessment was to guarantee the quality of the mapping products with reference to the accuracy thresholds set by the user requirements.

The applied accuracy assessments were based on the use of reference data, and applying statistical sampling to deduce estimates of error in the classifications. In order to provide an efficient, reliable and robust method to implement an accuracy assessment, there are three major components that had to be defined: the **sampling design**, which determines the spatial location of the reference data, the **response design** that describes how the reference data is obtained and an **analyses design** that defines the accuracy estimates. These steps were undertaken in a harmonised manner for the validation of all the geo-spatial products.

3.3.1 The Applied Sampling Design

The sampling design specifies the sample size, sample allocation and the reference assessment units (i.e. pixels or image blocks). Generally, different sampling schemes can be used in collecting accuracy assessment data including: simple random sampling, systematic sampling, stratified random sampling, cluster sampling, and stratified systematic unaligned sampling. In the current project a single stage stratified random sampling based on the method described by Olofsson et al (2013¹) was applied which used the map product as the basis for stratification. This ensured that all classes even very minor ones were included in the sample.

However, in complex LULC products with many classes, this usually results in a large number of strata (one stratum per LULC classes), of which some classes cover only very small areas (e.g. sport fields, cemeteries) and not being adequately represented in the sampling. In order to achieve a

¹ Olofsson, P., Foody, G. M., Stehman, S. V., & Woodcock, C. E. (2013). Making better use of accuracy data in land change studies: Estimating accuracy and area and quantifying uncertainty using stratified estimation. *Remote Sensing of Environment*, 129, 122–131. doi:10.1016/j.rse.2012.10.031

representative sampling for the statistical analyses of the mapping accuracy it was decided to extend the single stage stratified random sampling. At the first stage the number of required samples was allocated within each of the Level I strata (see Table 5). In the second stage all Level III classes that were not covered by the first sampling, were grouped into one new stratum. Within that stratum the same number of samples was randomly allocated as the Level I strata received. To avoid a clustering of point samples within classes and to minimise the effect of spatial autocorrelation a minimum distance in between the sample points was set to be 150 m. The final sample size for each class can be considered to be as close as possible to the proportion of the area covered by each stratum considering that the target was to determine the overall accuracy of the entire map.

The total sample size per stratum was determined by the expected standard error and the estimated error rate based on the following formula which assumes a simple random sampling (i.e. the stratification is not considered):

$$n = \frac{P*q}{\left(\frac{E}{Z}\right)^2}$$

n = number of samples per strata / map class

p = expected accuracy

$q = 1 - p$

E = Level of acceptable (allowable) sample error

Z = z-value (the given level of significance)

Hence, with an expected accuracy of $p = 0.85$, a 95% confidence level and an acceptable sampling error of 5%, the minimum sample size is 196. A 10% oversampling was applied to compensate for stratification inefficiencies and potentially inadequate samples (e.g. in case of cloudy or shady reference data). For each Level I strata 215 samples have been randomly allocated. Afterwards, within all classes of Level III (see Figure 5) that did not received samples in the first run, additionally 215 samples were randomly drawn across all these classes.

Table 5: Number of sampling points for the EO4SD-Urban mapping classes after applied sampling design for year 2017.

Class Name	Class ID	No. of Sampling Points	Km ² coverage
Residential	1000	178	115,03
Village Settlements	1150	48	9,9
Industrial and commercial	1200	41	25,15
Non-residential	1210	39	25,64
Transportation	1220	19	7,04
Airport	1230	21	7,13
Construction	1300	37	21,24
Sand-filled areas	1310	23	8,07
Land without current use	1320	5	5,68
Urban greenery, sport and leisure facilities	1400	15	13,6
Agriculture	2000	52	47,03
Forest	3100	20	0,26
Natural and semi-natural areas	3200	21	12,75
Bare land	3300	3	0,04
Water	5000	13	25,8

3.3.2 The Applied Response Design

The response design determines the reference information for comparing the map labels to the reference labels. Collecting reference data on the ground by means of intensive fieldwork is both costly and time consuming and in most projects not feasible. The most cost effective reference data sources are VHR satellite data with 0.5 m to 1 m spatial resolution. Czaplewski (2003)² indicated that visual interpretation of EO data is acceptable if the spatial resolution of EO data is sufficiently better compared to the thematic classification system. However, if there are no EO data with better spatial resolution available, the assessment results need to be checked against the imagery used in the production process.

The calculated number of necessary sampling points for each mapping category was randomly distributed among the strata and overlaid to the VHR data of each epoch. The following Figure is showing the VHR data with the overlaid sample points.

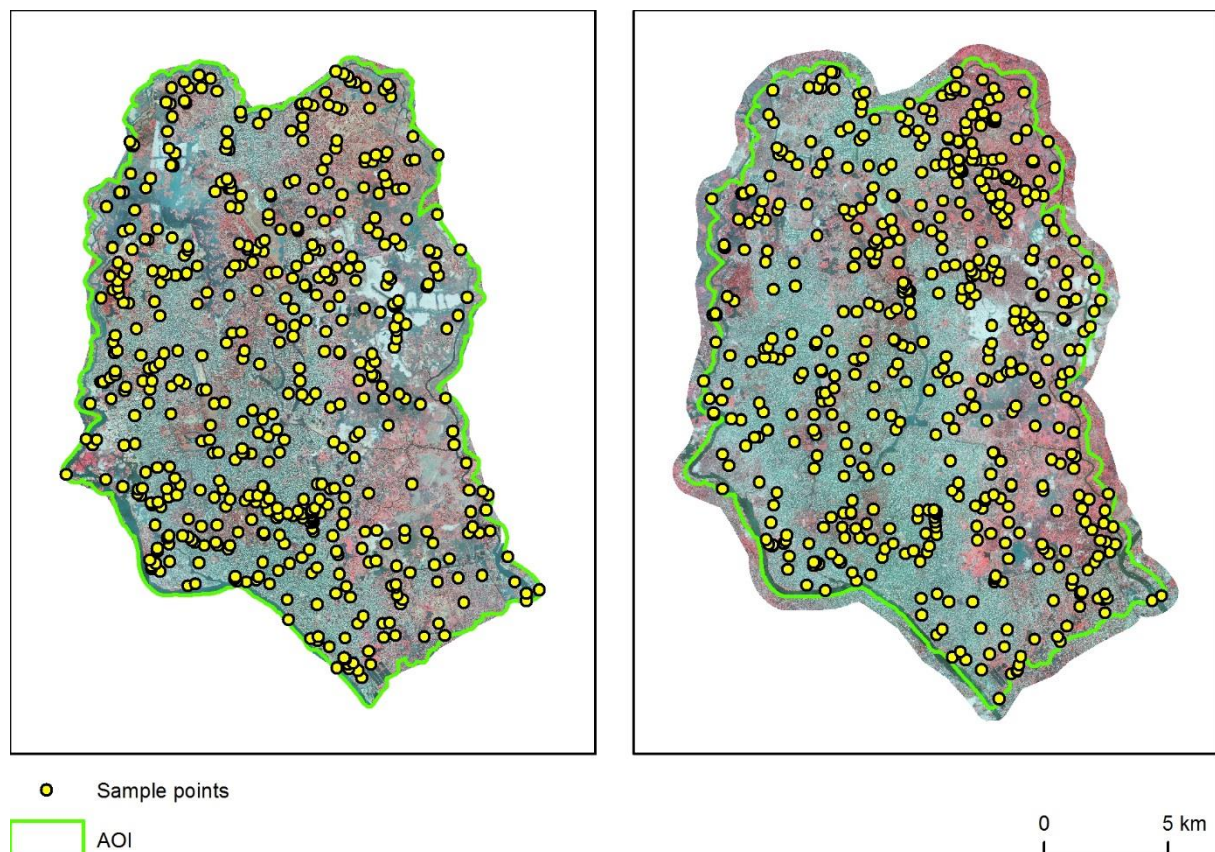


Figure 2: VHR data of the city of Dhaka of the year 2006 (left) and year 2017 (right) overlaid with randomly distributed sample points for accuracy assessment.

In this way a reference information could be extracted for each sample point by visual interpretation of the VHR data for all mapped classes. The size of the area to be observed had to be related to the MMU of the map product to be assessed. The reference information of each sampling point was compared with the mapping results and the numbers of correctly and not-correctly classified observations were recorded for each class. From this information the specific error matrices and statistics were computed (see next Section).

² Czaplewski, R. L. (2003). Chapter 5: accuracy assessment of maps of forest condition: statistical design and methodological considerations, pp. 115–140. In Michael A. Wulder, & Steven E. Franklin (Eds.), Remote sensing of forest environments: concepts and case studies. Boston: Kluwer Academic Publishers (515 pp.).

3.3.3 The Applied Analysis Design

Each class usually has errors of both omission and commission, and in most situations, these errors for a class are not equal. In order to calculate these errors as well as the uncertainties (confidence intervals) for the area of each class a statistically sound accuracy assessment was implemented.

The confusion matrix is a common and effective way to represent quantitative errors in a categorical map, especially for maps derived from remote sensing data. The matrices for each assessment epoch were generated by comparing the “reference” information of the samples with their corresponding classes on the map. The *Reference* represented the “truth”, while the *Map* provided the data obtained from the map result. Thematic accuracy for each class and overall accuracy is then presented in error matrices (see Tables below). Unequal sampling intensity resulting from the random sampling approach was accounted for by applying a weight factor (p) to each sample unit based on the ratio between the number of samples and the size of the stratum considered³:

$$\hat{p}_{ij} = \left(\frac{1}{M}\right) \sum_{x \in (i,j)} \frac{1}{\pi_{uh}^*}$$

Where i and j are the columns and rows in the matrix, M is the total number of possible units (population) and π is the sampling intensity for a given sample unit u in stratum h .

Overall accuracy and User and producer accuracy were computed for all thematic classes and 95% confidence intervals were calculated for each accuracy metric.

The standard error of the error rate was calculated as follows: $\sigma_h = \sqrt{\frac{p_h(1-p_h)}{n_h}}$ where n_h is the sample size for stratum h and p_h is the expected error rate. The standard error was calculated for each stratum and an overall standard error was calculated based on the following formula:

$$\sigma = \sqrt{\sum w_h^2 \cdot \sigma_h^2}$$

In which w_h is the proportion of the total area covered by each stratum. The 95% confidence interval is $\pm 1.96 \cdot \sigma$.

The confusion matrices are provided in the QAQC Sheets attached to the document in **Annex 2**, as well as the different accuracy indicators for both the overall product and each individual class. More precisely, the number of samples which are correctly and incorrectly classified is visible for each class, from which the user and producer accuracies as well as the 95% confidence levels are calculated based on the formula above.

The baseline Land Use/Land Cover product for Dhaka has an overall mapping accuracy of 92 %. The specific class accuracies are given in Annex 2.

3.4 Quality Control/Assurance

A detailed Quality Control and Quality Assurance (QC/QA) system has been developed which records and documents all quality relevant processes ranging from the agreed product requirements, the different types of input data and their quality as well as the subsequent processing and accuracy assessment steps. The main goal of the QC/QA procedures was the verification of the completeness, logical consistency, geometric and thematic accuracy and that metadata are following ISO standards on geographic data quality and INSPIRE data specifications. These assessments were recorded in Data Quality Sheets

³ Selkowitz, D. J., & Stehman, S. V. (2011). Thematic accuracy of the National Land Cover Database (NLCD) 2001 land cover for Alaska. *Remote Sensing of Environment*, 115(6), 1401–1407. doi:10.1016/j.rse.2011.01.020.

which are provided in Annex 2. The QC/QA procedures were based on an assessment of a series of relevant data elements and processing steps which are part of the categories listed below:

- Product requirements;
- Specifications of input data: EO data, in-situ data, ancillary data;
- Data quality checks: EO data quality, in-situ data quality, ancillary data quality;
- Geometric correction, geometric accuracy, data fusion (if applicable), data processing;
- Thematic processing: classification, plausibility checks;
- Accuracy: thematic accuracy, error matrices
- Delivery checks: completeness, compliancy with requirements

After each intermediate processing step a QC/QA was performed to evaluate products appropriateness for the subsequent processing (see Figure 3).

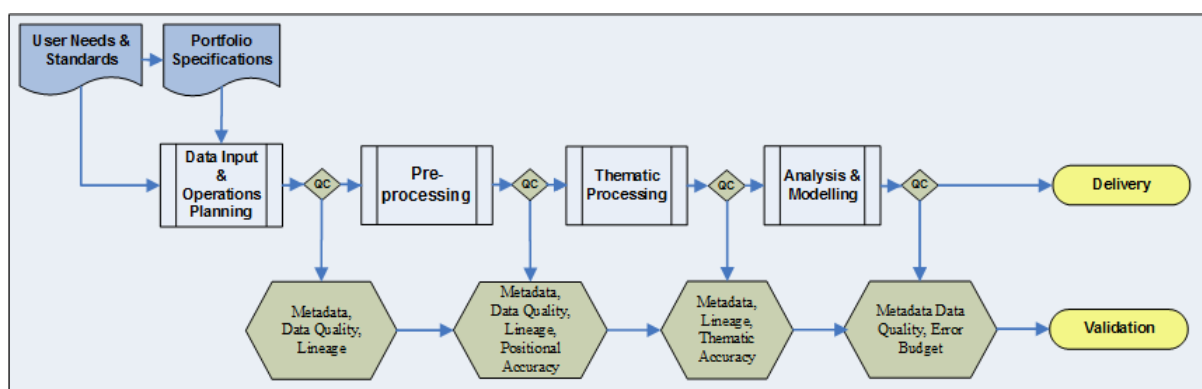


Figure 3: Quality Control process for EO4SD-Urban product generation. At each intermediate processing step output properties are compared against pre-defined requirements.

After the initial definition of the product specifications (output) necessary input data were defined and acquired. Input data include all satellite data and reference data e.g. in-situ data, reference maps, topographic data, relevant studies, existing standards and specifications, statistics. These input data were the baseline for the subsequent processing and therefore all input data had to be checked for **completeness**, **accuracy** and **consistency**. The evaluation of the quality of input data provides confidence of their suitability for further use (e.g. comparison with actual data) in the subsequent processing line. Data processing towards the end-product required multiple intermediate processing steps. To guarantee a traceable and quality assured map production the QC/QA assessment was performed and documented by personnel responsible for the Quality Control/Assurance. The results of all relevant steps provided information of the acceptance status of a dataset/product.

The documentation is furthermore important to provide a comprehensive and transparent summary of each production step and the changes made to the input data. With this information the user will be able to evaluate the provided services and products. Especially the accuracy assessment of map products and the related error matrices are highly important to rate the quality and compare map products from different service providers.

The finalised QC/QA forms are attached in Annex 2.

3.5 Overview of accuracy assessment per product

Table 6: Overview of accuracy assessment per product

Product	Overall Accuracy
Land Use / Land Cover	93 %
Land Use / Land Cover Change	92 %
Urban Green Areas	84 %
Informal Settlements	89 %

3.6 Metadata

Metadata provides additional information about the delivered products to enable it to be better understood. In the current project a harmonised approach to provide metadata in a standardised format applicable to all products and end-users was adopted. Metadata are provided as XML files, compliant to the ISO standard 19115 "Metadata" and ISO 19139 "XML Scheme Implementation". The metadata files have been created and validated by the GIS/IP-operator for each map product with the Infrastructure for Spatial Information in Europe (INSPIRE) Metadata Editor available at: <http://inspire-geoportal.ec.europa.eu/editor/>.

The European Community enacted a Directive in 2007 for the creation of a common geo-data infrastructure to provide a consistent metadata scheme for geospatial services and products that could be used not only in Europe but globally. The geospatial infrastructure called INSPIRE was built in a close relation to existing International Organization for Standardization (ISO) standards. These are ISO 19115, ISO 19119 and ISO 15836. The primary incentive of INSPIRE is to facilitate the use and sharing of spatial information by providing key elements and guidelines for the creation of metadata for geospatial products and services.

The INSPIRE Metadata provides a core set of metadata elements which are part of all the delivered geospatial products to the users. Furthermore, the metadata elements provide elements that are necessary to perform queries, store and relocate data in an efficient manner. The minimum required information is specified in the Commission Regulation (EC) No 1205/2008 of 3 December 2008 and contains 10 elements:

- Information on overall Product in terms of: Point of contact for product generation, date of creation
- Identification of Product: Resource title, Abstract (a short description of product) and Locator
- Classification of Spatial Data
- Keywords (that define the product)
- Geographic information: Area Coverage of the Product
- Temporal Reference: Temporal extent; date of publication; date of last revision; date of creation
- Quality and Validity: Lineage, spatial resolution
- Conformity: degree of conformance to specifications
- Data access constraints or Limitations
- Responsible party: contact details and role of contact group/person

These elements (not exhaustive) constitute the core information that has to be provided to meet the minimum requirements for Metadata compliancy. Each element and its sub-categories or elements have specific definitions; for example in the element "Quality" there is a component called "Lineage" which has a specific definition as follows: "a statement on process history and/or overall quality of the spatial data set. Where appropriate it may include a statement whether the data set has been validated or quality assured, whether it is the official version (if multiple versions exist), and whether it has legal validity. The value domain of this element is free text," (INSPIRE Metadata Technical Guidelines, 2013). The detailed information on the Metadata elements and their definitions can be found in the "INSPIRE Metadata Implementing Rules: Technical Guidelines," (2013). Each of the EO4SD-Urban products will be accompanied by such a descriptive metadata file. It should be noted that the internal use of metadata in these institutions might not be established at an operational level, but the file format (*.xml) and the web accessibility of data viewers enable for the full utility of the metadata.

4 Analysis of Mapping Results

This chapter will present and assess all results which have been produced within the framework of the current project, including the following sub-sections:

- Global Urban Products – Urban Extent and Imperviousness
- Urban Extent – Status and Trends
- Urban & Peri-Urban Land Use / Land Cover - Status and Trends
- Informal Settlements - Status and Trends
- Urban Green areas and Open Spaces - Status and Trends
- Transportation network - Status
- Flood History, Hazard & Risk

4.1 Global Urban Products

The 2014-2015 Global Urban Products - Settlement Extent and Imperviousness Level products are generated by exploiting multi-temporal Sentinel-1 (S1) radar and Landsat-8 optical data intersecting the investigated Area of Interest (AOI) in the period 2014-2015. These products based on land cover definition are supposed to provide means to cost-effective screening and global benchmarking of city extent between multiple cities thanks to availability of high resolution satellite data and their comparability. Hot-spots identified using the Global Urban Products can be then further explored and analysed in detail using tailored products from the EO4SD portfolio.

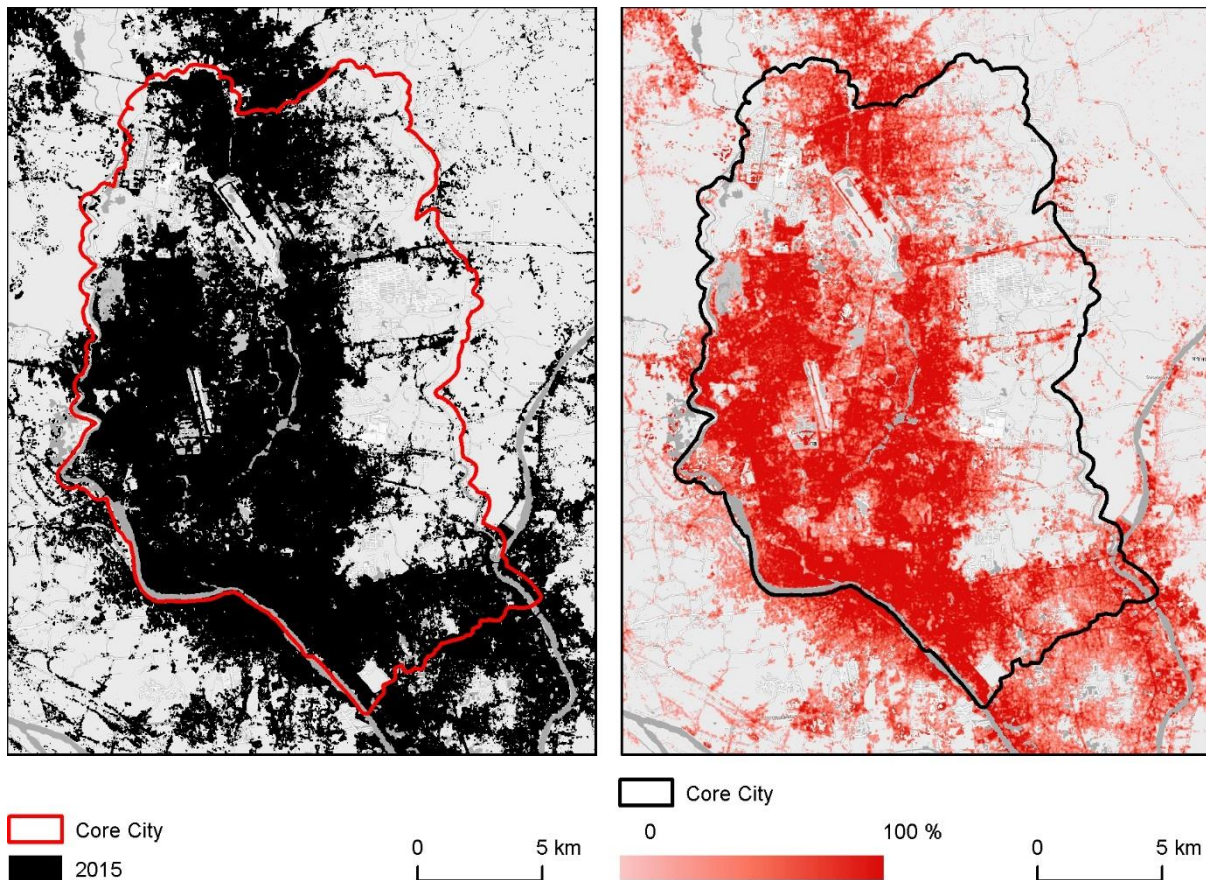


Figure 4: Global Settlement Extent Status (left) and Level of Imperviousness (right) - 2015

As documented in Figure 4, massive urbanisation and very high levels of imperviousness can be observed especially in central, southern and western part of the Dhaka Metropolitan (DMP) area.

Urbanized built-up areas stretch far beyond the DMP boundaries especially in direction to the north and south. In contrary, eastern part of the DMP reaches lower levels of urbanisation and imperviousness with development concentrated along tangential transport axes.

4.2 Urban Extent - Status and Trends

The assessment of the urban extent over time provides information on the changes in the urban development in terms of built-up area expansion or reduction. Urban Extent product delivered within EO4SD-Urban project has been directly extracted from the baseline LULC information product by aggregating classes to artificial and non-artificial areas. In contrast to information presented in Figure 4 which is more related to land cover showing predominantly built-up and sealed areas, urban extent status and growth presented in Figure 5 is based on land use nomenclature, i.e. the urban extent mask contains also artificial classes like construction sites and village settlements with high density of trees. No further analysis based on morphology or functional urban area concept has been applied, but the data products can provide a solid basis for such kind of analysis. Figure 5 shows the overall extent of artificial areas and distribution of urban expansion between 2006 and 2017.

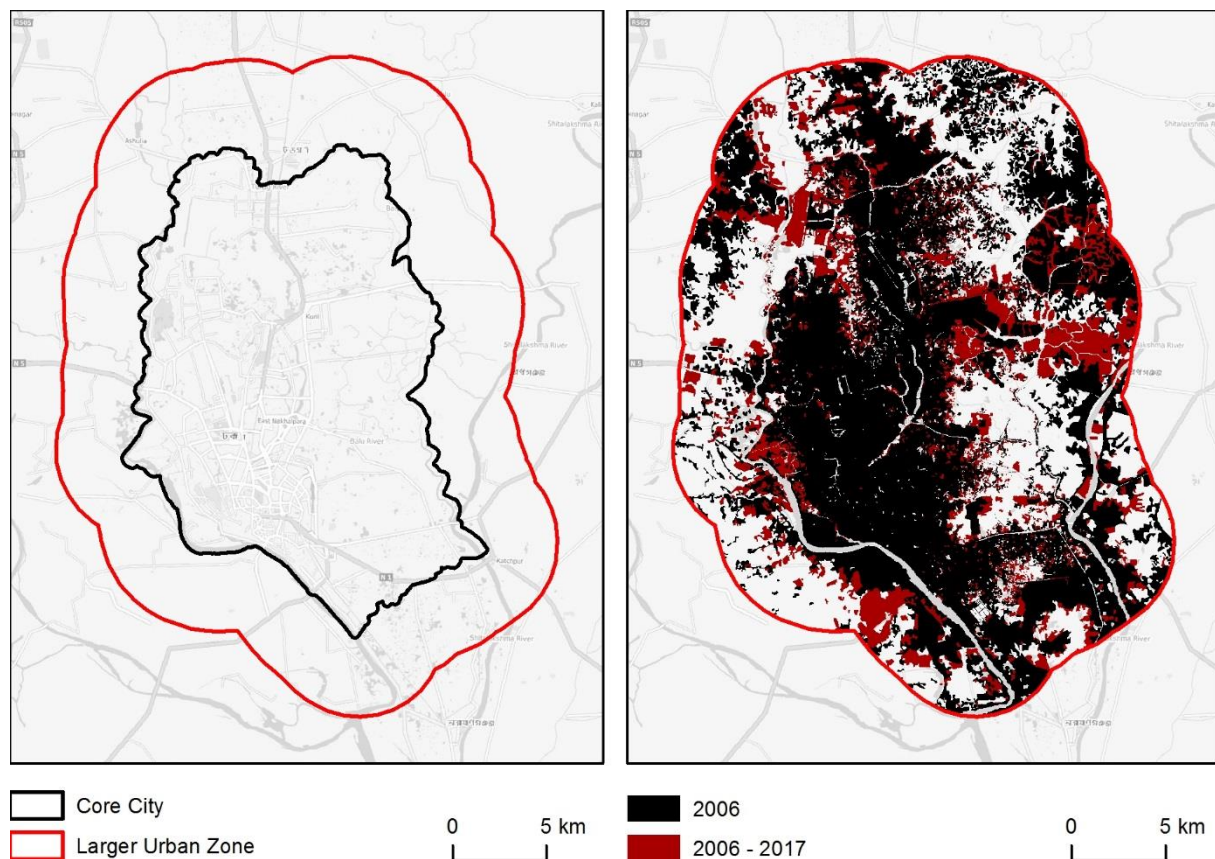


Figure 5: Spatial distribution of Urban Extent and Growth

Artificial urban areas have been mainly expanding in the east, south-west and north-west. Large areas of urban expansion are represented by sand-filled areas class which is considered as artificial surface (as well as construction sites). New built-up development is expected in these areas, which are located mostly to the east from the Dhaka central area.

Figure 6 shows the growth of urban area between years 2006 and 2017. The graph on the left displays the proportion of the total area divided into three classes: *Urban Extent 2006* which represent the urban area in 2006, *Urban Extent Growth 2006-2017* which shows the growth of urban area between these years, and *Non-urbanised* area which is the area in 2017 covered by non-urban classes. Bar plot on the right indicates absolute proportions in km². The graph is displayed for the whole area of Dhaka AOI (overall), and then separately for the area of the Core City and for the Larger Urban Zone.

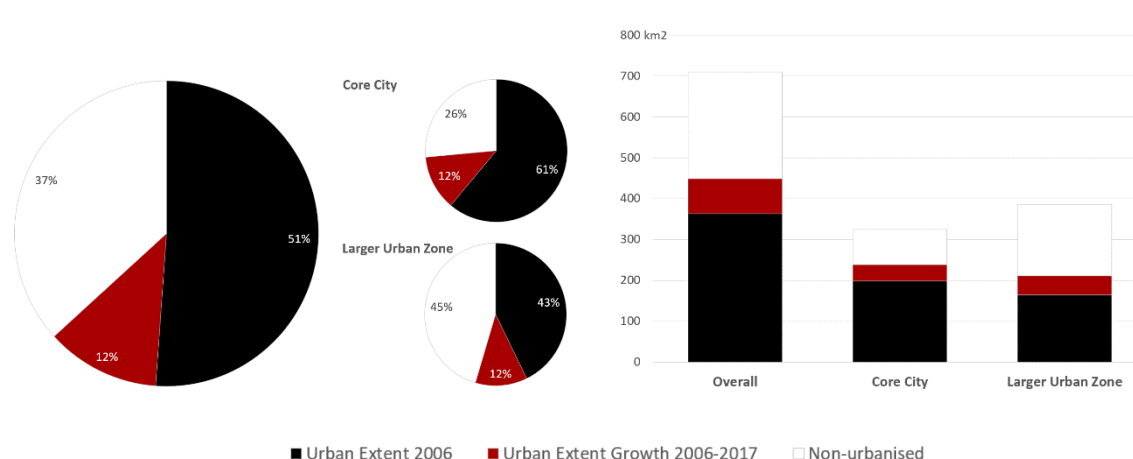


Figure 6: Urban Extent Status and Growth - overall, in Core City and in Larger Urban Zone in % (left) and km² (right).

As seen in the graphs above the Urban extent growth over the period of 11 years amounted to 12 % of the overall area of Dhaka, and this stands for both the Core City area (where urban extent growth is 40.3 km² in total) and the Larger Urban Zone (where urban extent growth is 45.6 km²). Whilst in 2006 urban area represented just over 50 % of the whole area, in 2017 urban area occupied 63 % which is nearly 2/3 of the area. In 2017 the urban area in the Core City represents 3/4 of the area, whilst in the Larger Urban Zone it represents just over a half of the area (55 %).

Figure 7 displays the urban extent and its growth between 2006 and 2017 for each of the Dhaka's district (wards). Because of the high number of districts the graph is divided into three parts. *Urban extent 2006* shows the % of overall district area covered by urban classes in 2006. *Urban extent growth 2006-2017* displays the % of area which in 2006 belonged to non-urban classes but which was urbanised and by 2017 it belongs to urban area. *The non-urbanised* area shows the proportion of the whole district area which is not covered by urban classes in 2017.

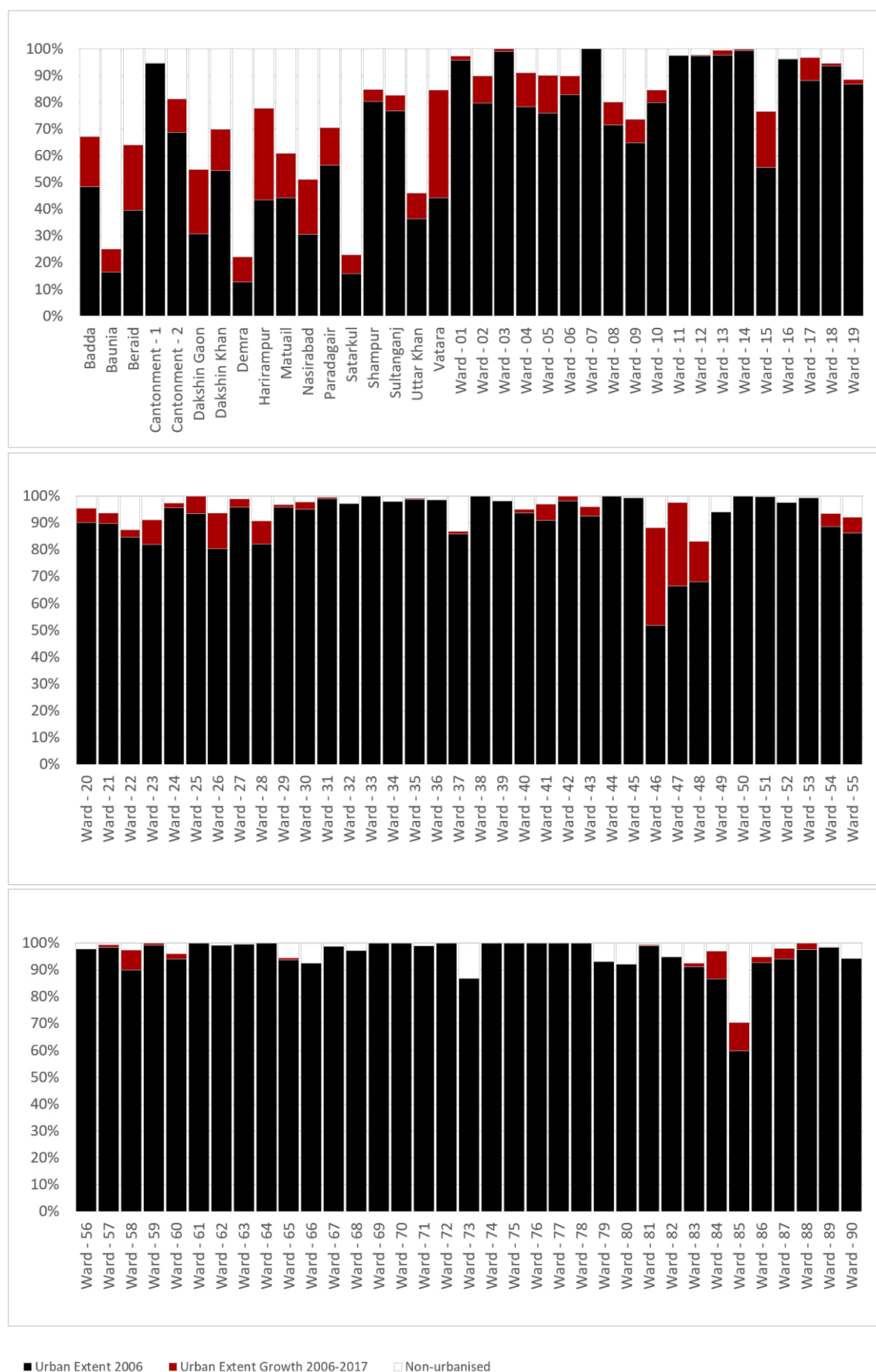


Figure 7: Urban Extent in 2006 and Urban Extent Growth 2006-2017 in % of district area

The red colour in Figure 7 displays the proportion (not total areas) of each of the district areas' which was urbanised between 2006 and 2017. The largest urban extent growth proportionate to the area of the district was recorded in the Vatara district where growth occurred on 40.6 % of its area. However, the largest total urban extent growth between 2006 and 2017 was recorded in Harirampur district where urban grew by 5.1 km². The largest non-urbanised area in 2017 is in the Uttar Khan district with 11.1 km². Considering the proportions the largest non-urbanised area can be found in the Demra district with 77.8 % of the area.

Figure 8 shows share of urbanized areas and intensity of urban area growth in wards.

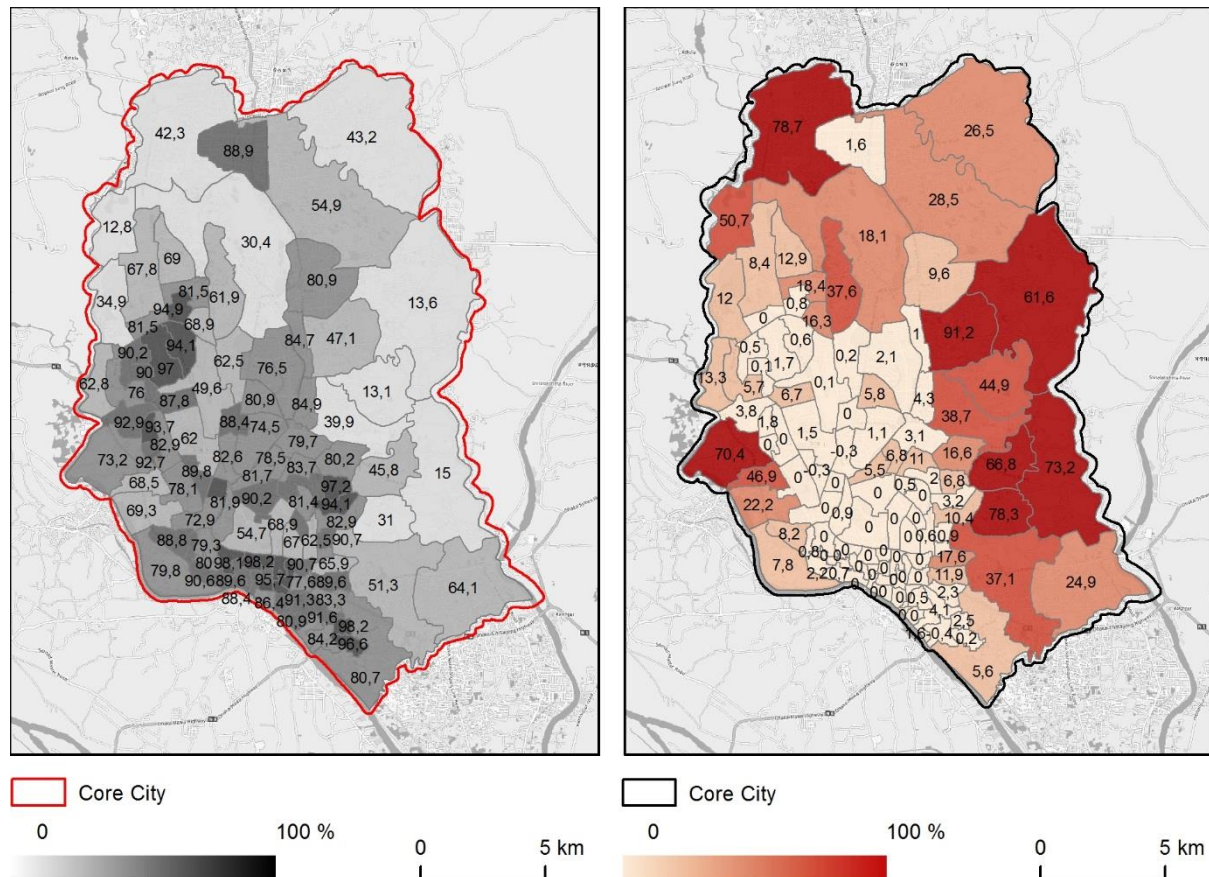


Figure 8: Urban Extent 2017 (% urbanized area of total area; left) and Urban Extent Growth Intensity 2006-2017 (% increase comparing to initial 2006 area; right) per districts – spatial distribution.

On the left map Figure 8 displays the distribution of urbanized area in districts. Results are in line with outcomes presented in figures 4 and 5.

4.3 Land Use / Land Cover - Status and Trends

Core city of Dhaka is a highly urbanized area which has been undergoing rapid development. Locally specific land use class in Dhaka is sand-filled areas, which serve as initial stage of preparation for construction site and, furthermore, flood protection measure. Peri-urban areas could be characterized as agricultural land with traditional village settlements. In parallel, these area is under pressure of growing city. As a result, non-urban classes are uptaken by urban ones.

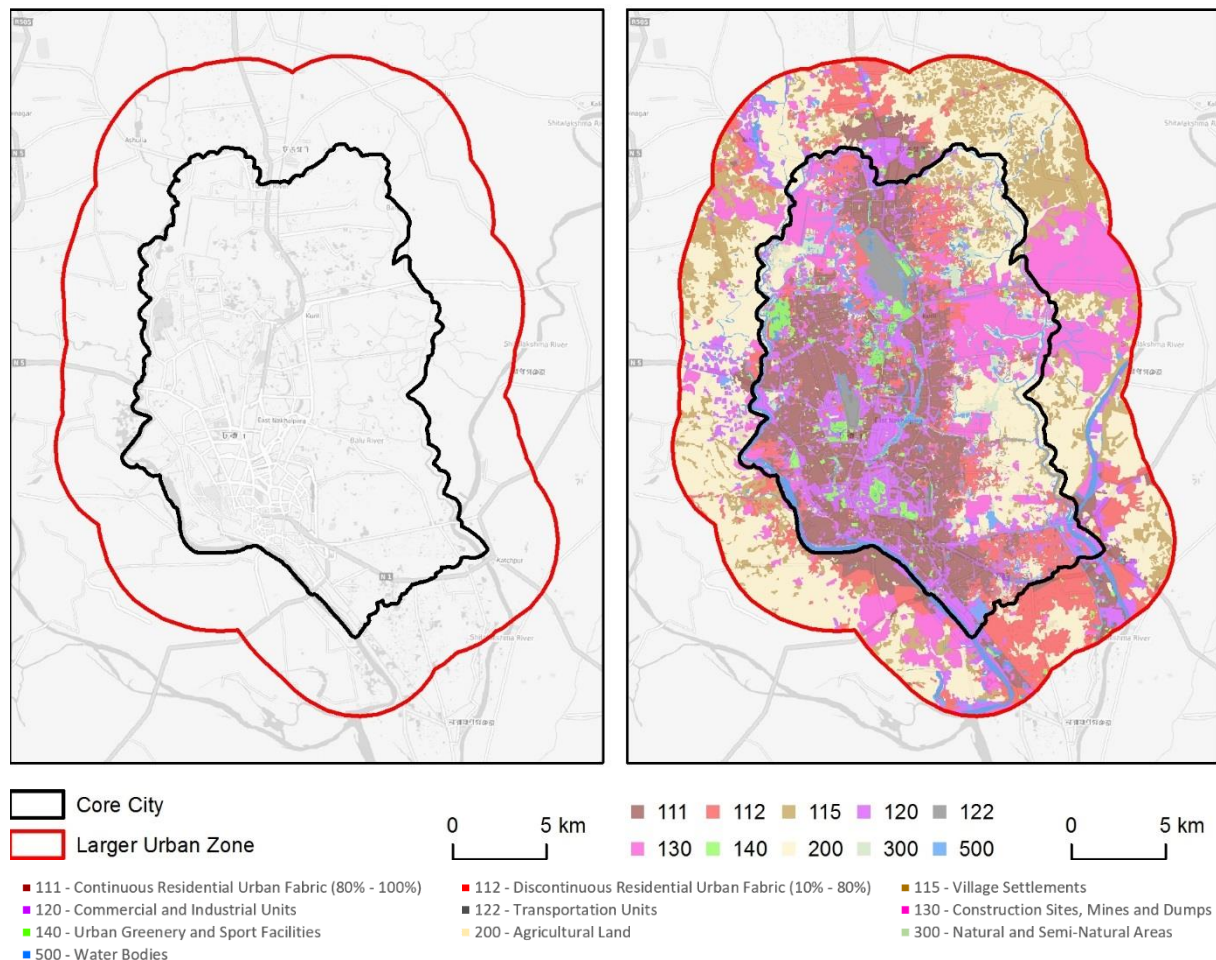


Figure 9: Detailed Land Cover Land Use 2017- Spatial distribution.

Figure 10 shows is distribution of the main Land Cover Land Use classes in Dhaka, both proportionally (on the left) and in actual area extent (on the right). In both cases, figures are presented for the whole area of Dhaka (overall) and then individually for the Core City area and for the Larger Urban Zone.

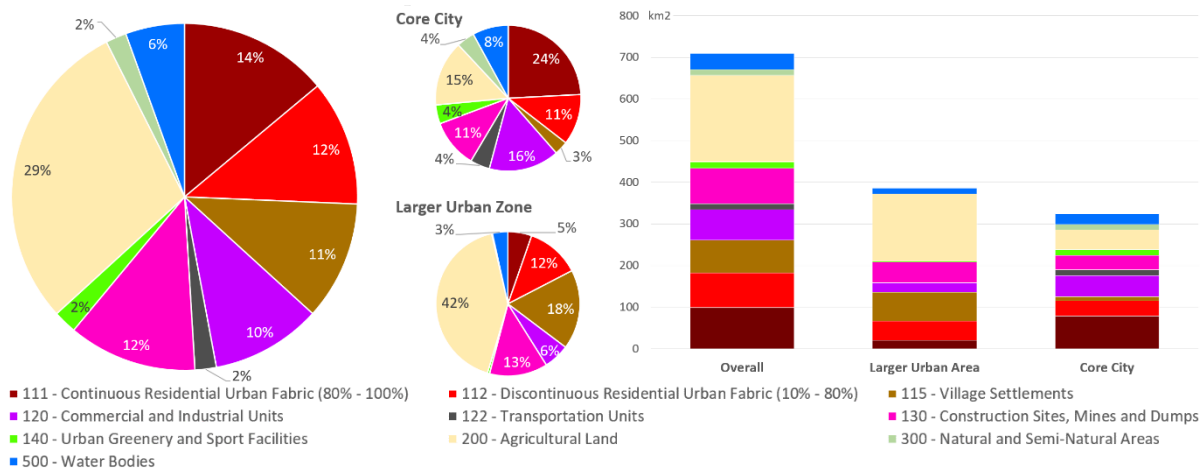
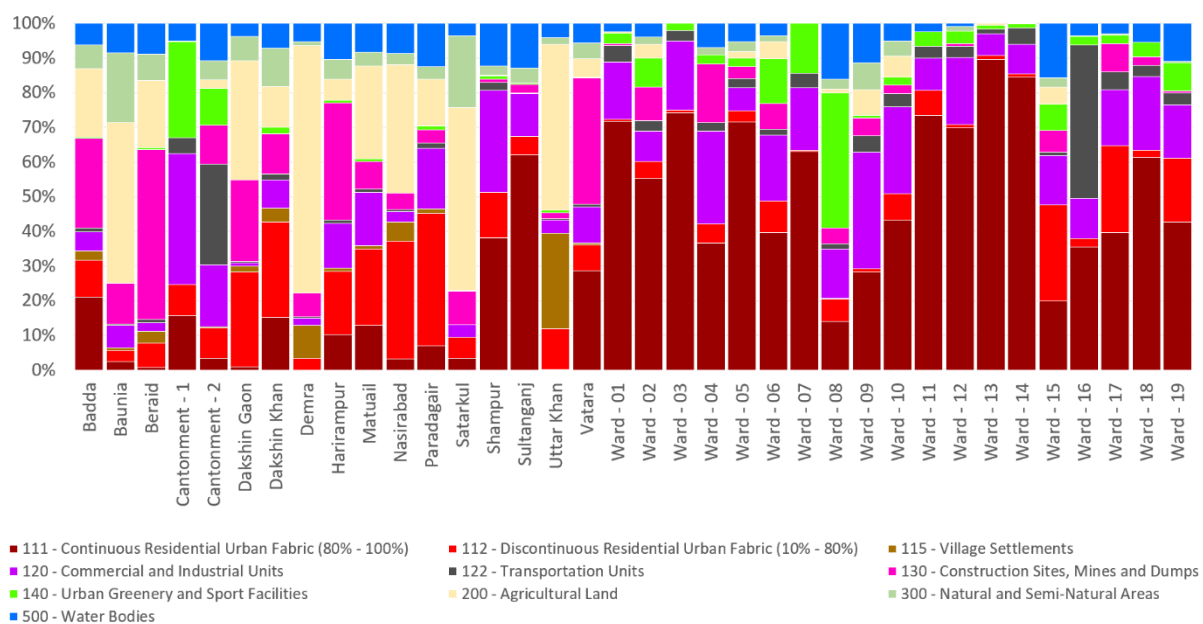


Figure 10: Detailed Land Cover Land Use 2017 structure - overall, in Core City and in Larger Urban Zone in % (left) and km2 (right).

The largest portion of the overall area of Dhaka is covered by agricultural land (29 %). The second largest share (14 %) is covered by the Continuous residential urban fabric (80% - 100% sealing level) area. As expected, in Core City area the largest part is taken by the Continuous residential urban fabric (80% - 100% sealing level) area which represents 24 % of the Core City area. In Larger Urban Zone the most common class is agricultural land (42 %), whilst Continuous residential urban fabric (80% - 100% sealing level) area only covers 5 %. The overall proportion of all urban classes is 63 % in the whole area of Dhaka, 73 % in Core City and 55 % in the Larger Urban Zone.

Figure 11 displays the proportion of each Land Cover Land Use class in each district. Given the large number of districts the graph has been divided into three parts.



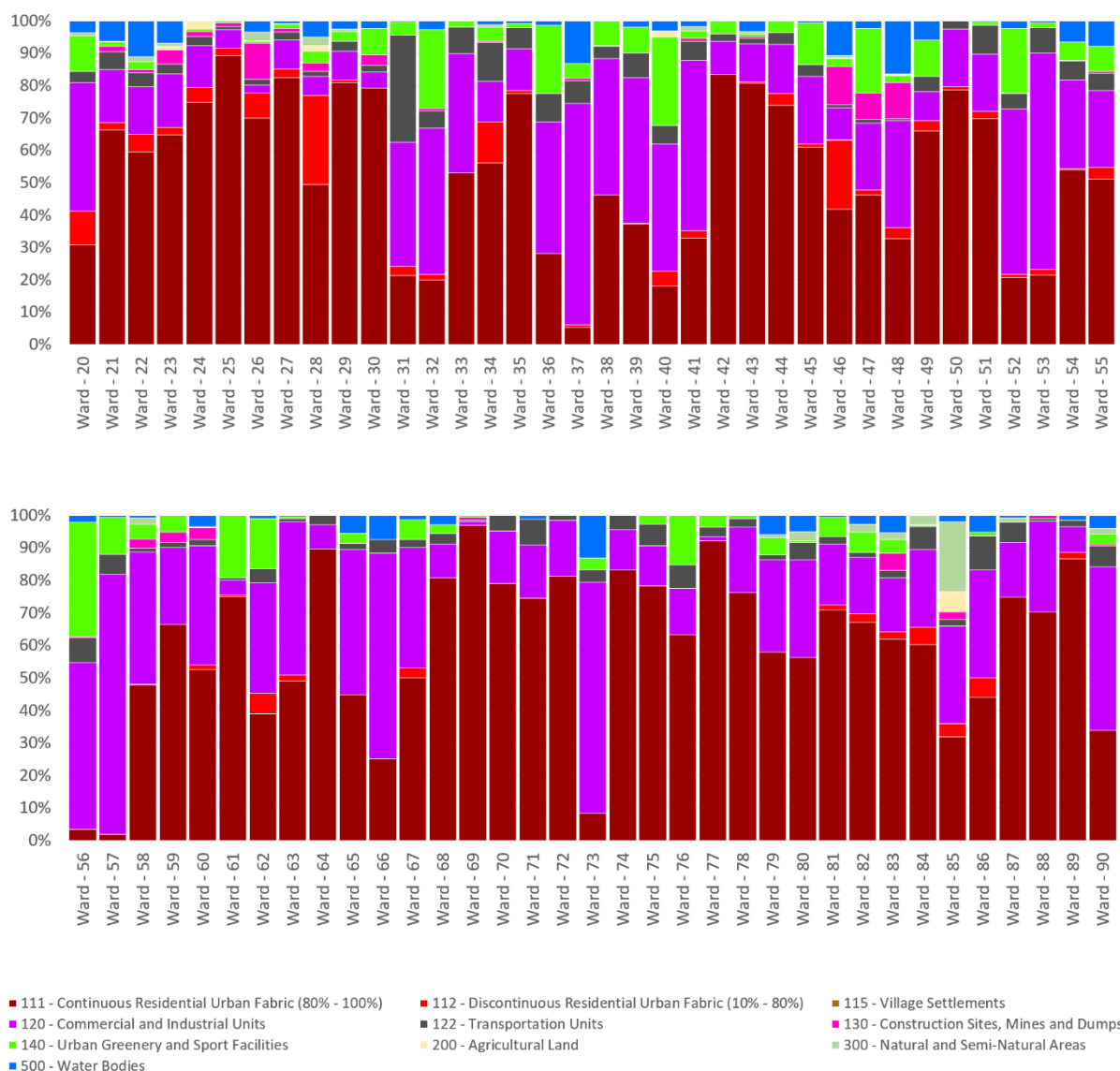


Figure 11: Detailed Land Cover Land Use 2017 structure in % per districts.

Based on the proportion of the district area the district where Continuous residential urban fabric (80% - 100% sealing level) area covers the largest proportion is Ward 69 where this area represents 97 % of the district area. Ward – 01 is the district where Continuous residential urban fabric (80% - 100% sealing level) class covers the largest area: 3.9 km².

When merging all the urban classes together the largest proportion of district area covered by urban classes can be found in Ward – 03, Ward – 42, Ward – 75, Ward – 78 and Ward – 88, in all cases urban area covers 100 % of the district area. When looking at the actual area covered by the urban classes Cantonment – 2 represents the district where the area covered by urban classes is the largest, i.e. 15.6 km².

Demra is the district where the highest proportion of the district area is covered by Agricultural land- 71.5 %. The largest area covered by Agricultural land can be found in Uttar Khan district where this area represents 9.9 km².

The following table displays the overall statistics of the Land Cover Land Use classes in Dhaka. It shows the area covered by each of the main classes in 2006 and in 2017 (both in km² and %). It also displays the total area which was changed from one class to another between years 2006 and 2017 in %. The last column shows the annual average change within each class in %.

Table 7: Overall LU/LC Statistics.

LU/LC Classes	2006 Area		2017 Area		Changed Area	Annual Change
	km ²	%	km ²	%	%	%
Urban	363.11	51.14	449.05	63.24	23.67	2.15
Agricultural	298.81	42.08	208.08	29.31	-30.36	-2.76
Natural-Semi Natural	6.61	0.93	13.74	1.93	108.00	9.82
Water	41.54	5.85	39.17	5.52	-5.71	-0.52

As foreseen the actual largest changed area was recorded within the Urban classes where the area in 2006 was 363.11 km² and in 2017 it is 449.05 km². The change represents a changed area of 23.67 %, i.e. 2.15 % per year on average.

The largest proportion of changed area was recorded in Natural and Semi-natural class where the change represents 108 % or the original area but this is mainly due to the fact that the actual area covered by this class is very small in comparison with other classes.

A negative change was recorded within Agricultural and Water classes meaning a reduction in the area in 2017 compared to the original state in 2006.

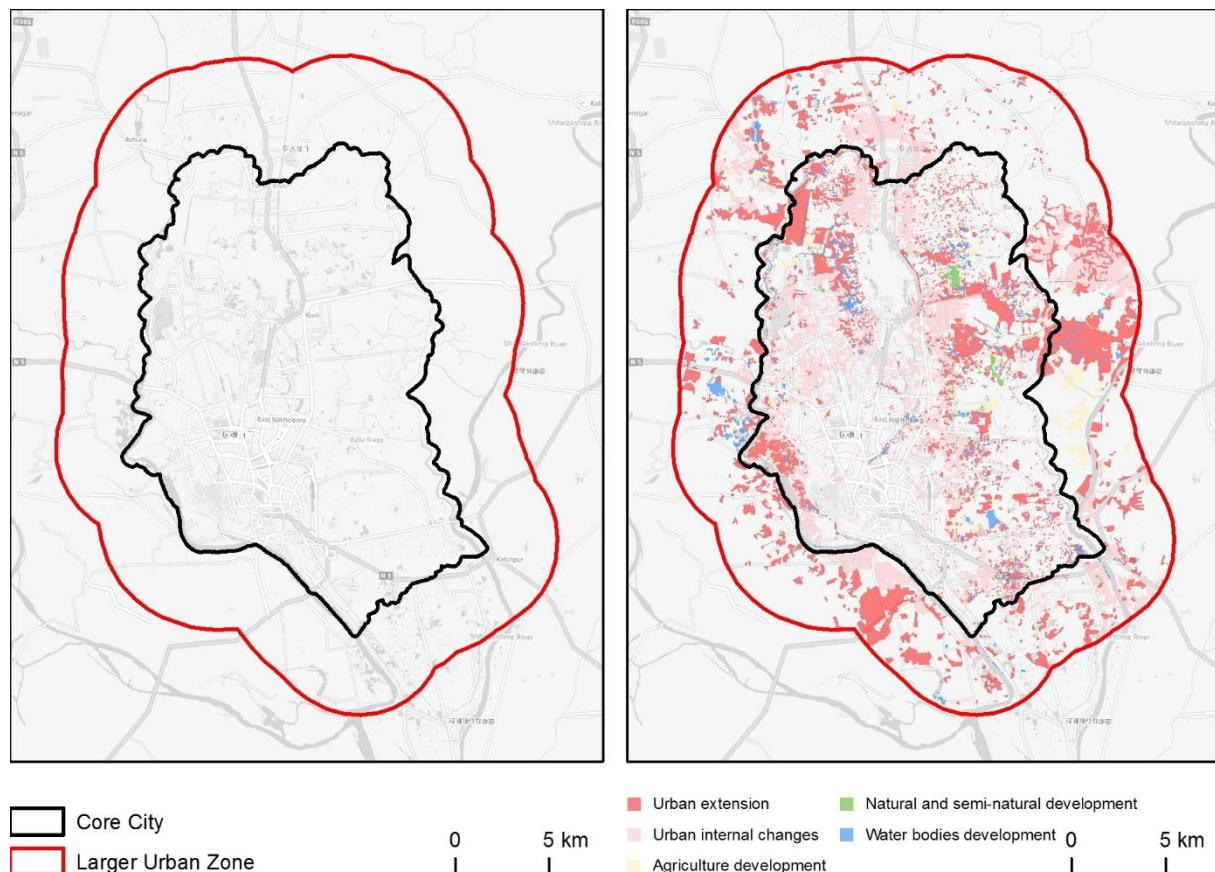


Figure 12: Land Cover Land Use Change Types - Spatial Distribution.

Figure 12 shows main developing areas, which are on east, north-west and south-west parts of AOI. Urban internal changes are located mainly in Core city and means the densification of these areas.

Figure 13 displays the change types between years 2006 and 2017. On the left the change types are shown proportionally in %, whereas on the right the actual area of the changes is displayed. In both cases the statistics is shown for the overall area of Dhaka, and then separately for the Core City area and for the Larger Urban Zone.

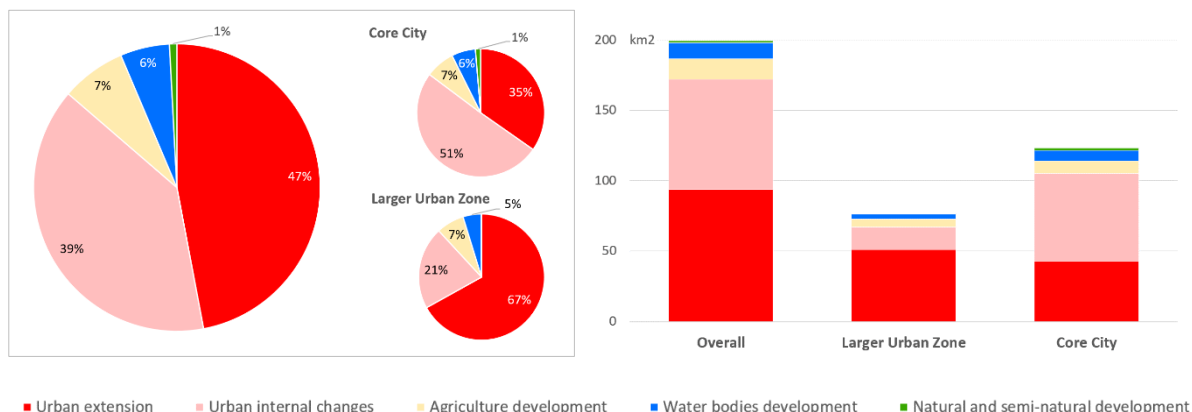


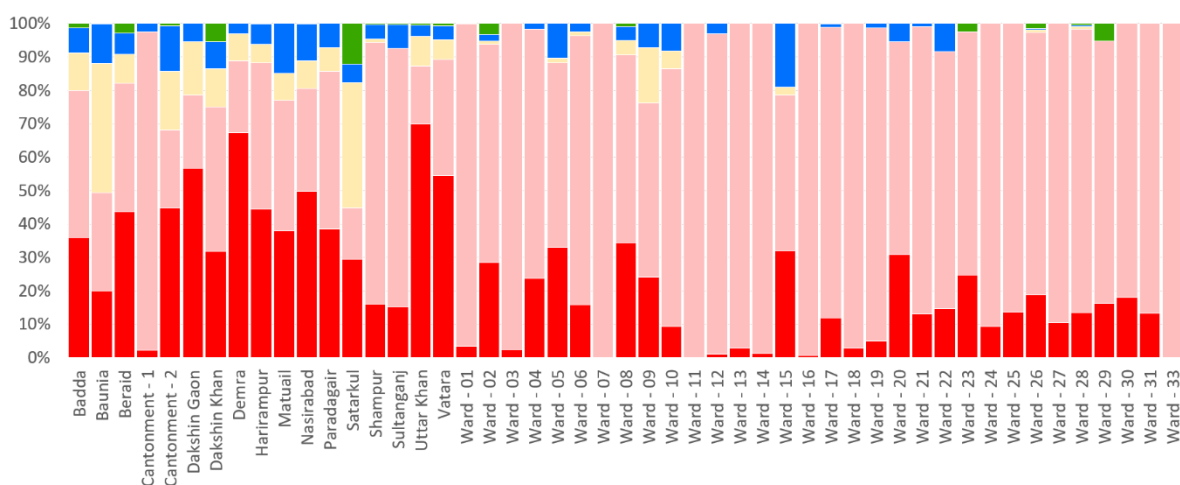
Figure 13: Land Cover Land Use Change Types 2006-2017 - overall, in Core City and in Larger Urban Zone in % (left) and km² (right).

Proportionally the largest change type considering the whole area of Dhaka is the *Urban extension* which represents 47 % (93.8 km²), closely followed by the *Urban internal changes* representing 39 % (78.4 km²) of all changes. These two change types together represent 86 % (172.2 km²) of all changes in Dhaka between 2006 and 2017.

Within the Core City the most common change type is the *Urban internal changes* which represents 51 % of all changes in the Core City (62.3 km²). Together with the *Urban extension* these two change types represent 86 % of all changes in the Core City (105.0 km²).

The situation is different in the Larger Urban Zone where the most common change type is the *Urban extension* which represents 67 % of all changes (51.0 km²). *Urban extension* and *Urban internal changes* change types represent 88 % of all the changes in the Larger Urban Zone area (67.1 km²).

The following figure shows the change types proportionately within each district. Due to the large number of districts the graph was divided into two parts. Within some of the districts there were no changes between 2006 and 2017 recorded and these districts are therefore not listed in the graph.



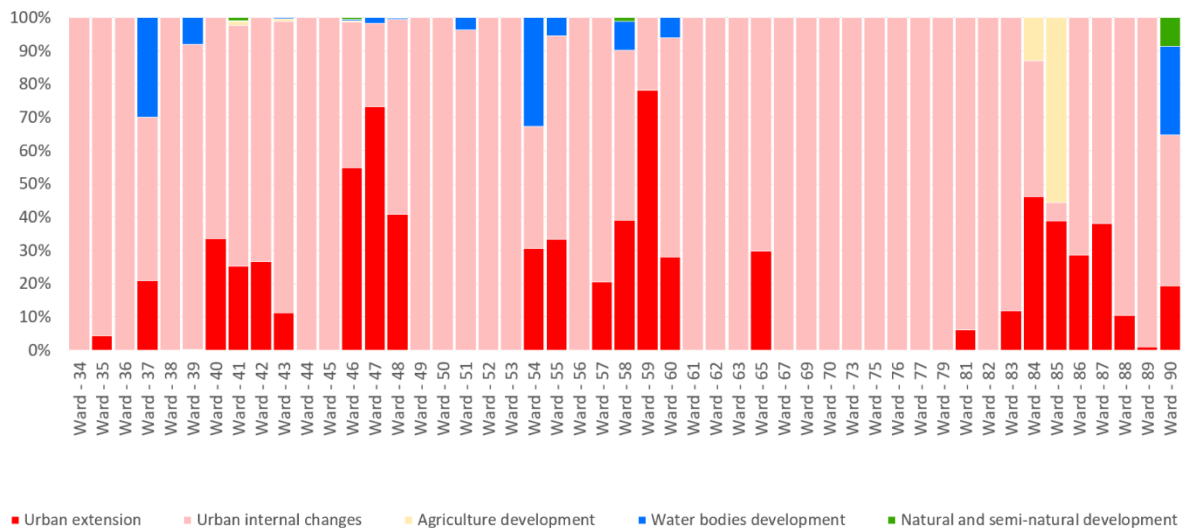


Figure 14: Land Cover Land Use Changes Types 2006-2017 structure in % per districts.

The actual largest area changed falling within *Urban extension* change type occurred in Beraid district where this type of change represents 5.5 km². Proportionally the largest *Urban extension* happened in Ward – 59 district where *Urban extension* represents 78 % of all the changes.

Urban internal changes occurred frequently and in many districts this change type represents 100 % out of all changes. Harirampur is the district with highest amount of *Urban internal changes* amounting to an area of 5.1 km².

As for *Agriculture development* the proportionally largest changes occurred in Ward – 85 representing 56 %. However, Dakshin Khan is the district where the largest *Agricultural development* occurred representing 1.3 km². In this district the largest *Water bodies development* was also recorded, together with the largest *Natural and Semi-natural development*, representing 0.9 km² and 0.6 km² respectively.

Next figure shows another point of view on changes in Core city of Dhaka. Figure on left represents difference of sealed areas between 2006 and 2017. Figure on the right on other hand shows share of LCF1 (Urban Extension) per Urban Fabric (2006) per ward.

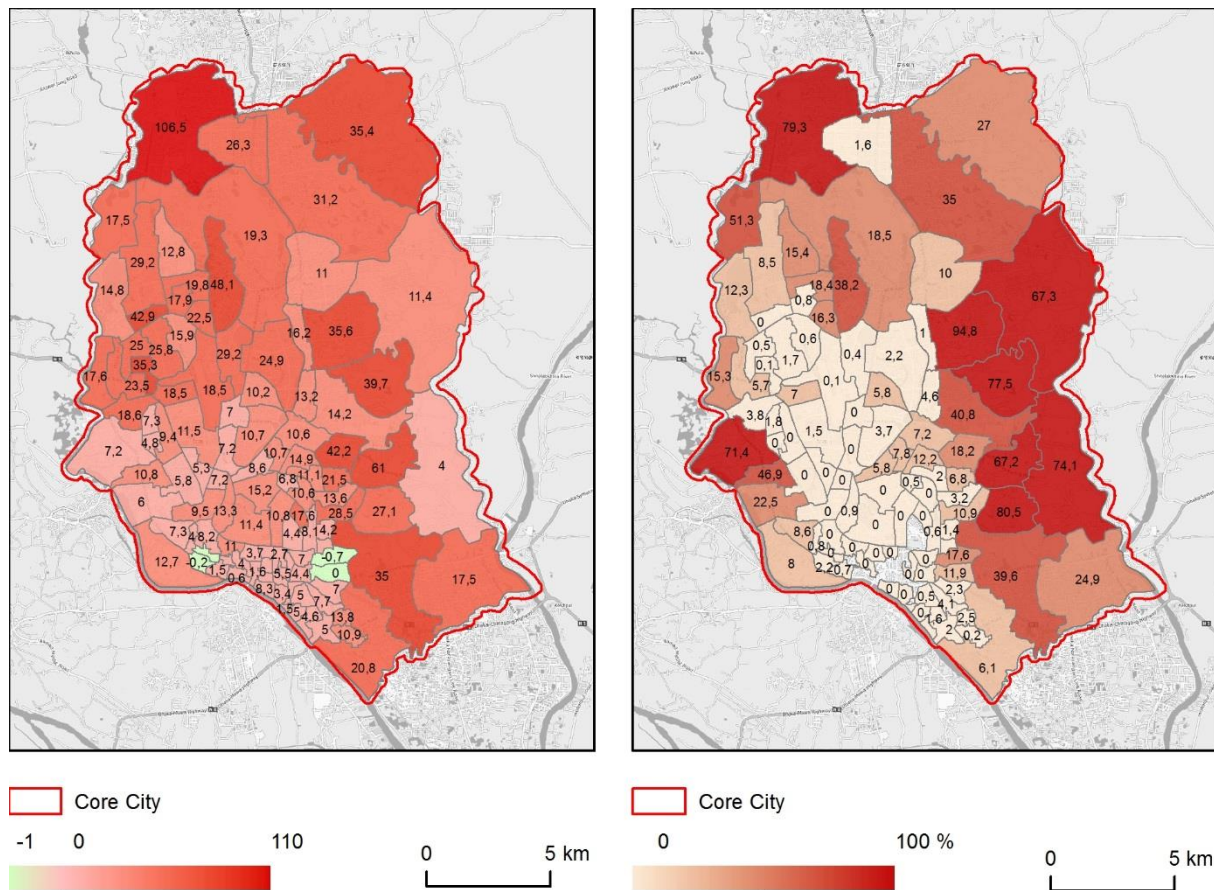


Figure 15: Urban Densification change type intensity (left) and Share of urban extension (LCF1) per Urban Fabric 2006 per ward (right) 2006-2017 (in %) - Spatial Distribution.

According to left figure the very center of the city is highly urbanized, so there is not a significant development. Districts affected by higher densification are located on the edge of core city. Right figure shows that the highest share of Urban extension is mainly in the east areas of AOI and that the City Center is already highly urbanized and no other extension is possible.

Figure 16 displays the proportionate consumption (on the left) and formation (on the right) within classes between 2006 and 2017. Both consumption and formation is shown for the overall Dhaka area, and individually for the Core City area and for the Larger Urban Zone.

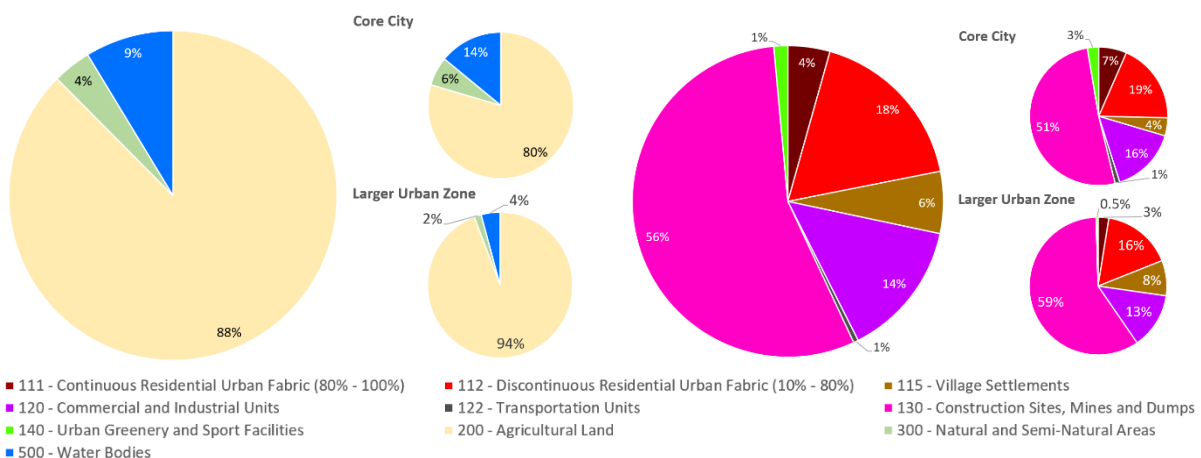


Figure 16: Consumption (left) and Formation structure (right) for Land Cover Land Use changes LCF1 (Urban extension) 2006-2017 - overall, in Core City and in Larger Urban Zone in %.

The largest consumption occurred within the *Agricultural Land*, representing 88 % for overall Dhaka area, 80 % within the Core City changes and 94 % in the Larger Urban Zone. Other consumption processes occurred in the *Natural and Semi-natural areas* and within the *Water bodies*, these representing 4 % and 9 % of consumption within the overall area respectively, 6 % and 14 % respectively in the Core City area, and 2 % and 4 % respectively within the Larger Urban Zone area.

As expected the proportionately largest formation process in all areas is presented by the *Construction sites, Mines and Dumps class*, representing 56 % in the overall Dhaka area, 51 % in the Core City and 59 % in the Larger Urban Zone area.

4.4 Informal Settlements - Status and Trends

Figure 17 shows spatial distribution of slums in Core city of Dhaka and their classification into several types regarding location and density.

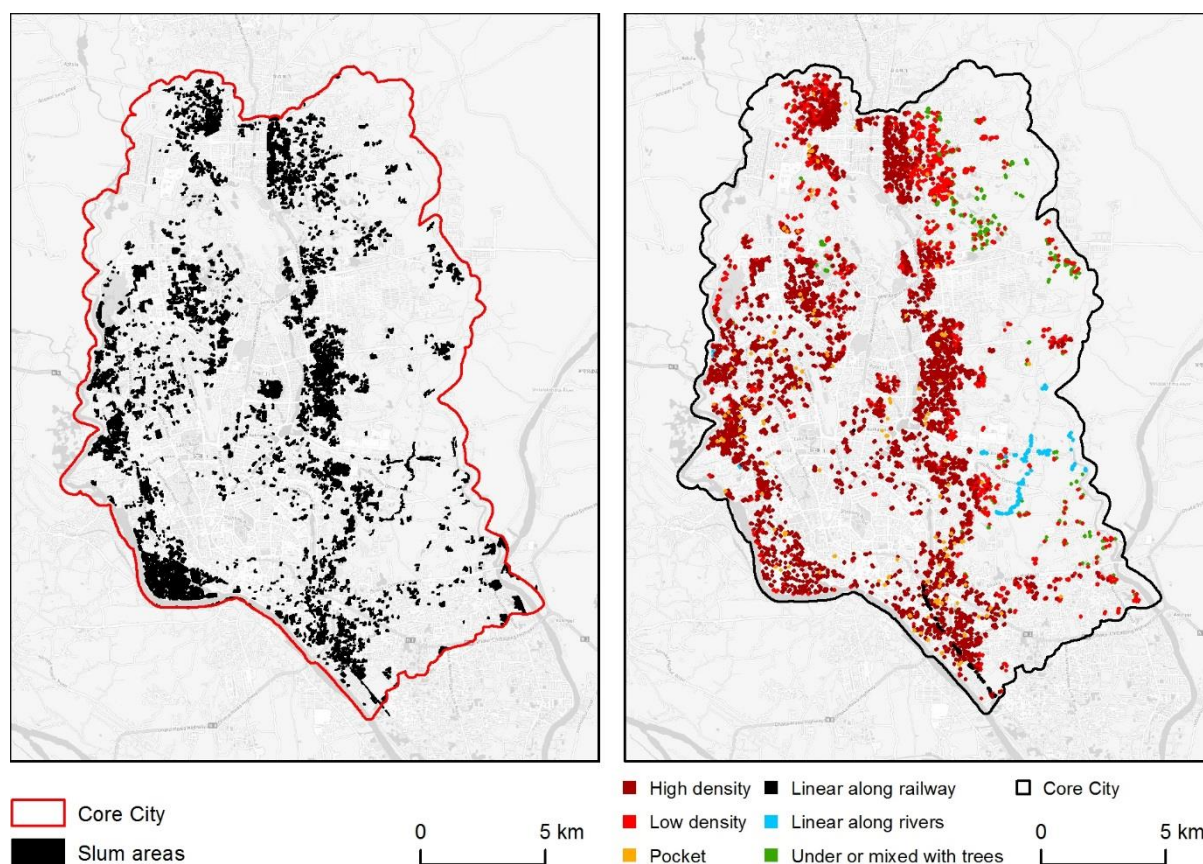


Figure 17: Informal Settlements/Slum Areas (left) and Slum Areas Types represented by points (right) - Spatial Distribution 2017.

From Figure 17 it can be seen that slums are not so often in the very core city, but mainly on the edges. Most of them are high density slums.

The following figure shows two graphs representing the proportion of different informal settlements/ slum areas within the Core City in % (on the left), and the actual area covered by the different informal settlements/ slum areas in the Core City in km² (on the right).

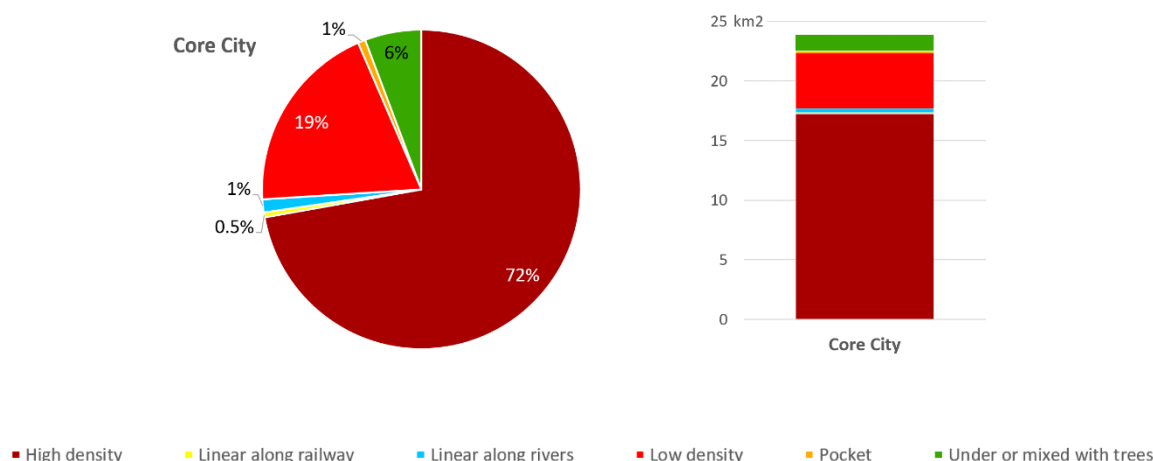


Figure 18: Informal Settlements/Slum Areas distribution per type - overall, in Core City.

The most common type of informal settlements is the *High density* type representing 72 % of all informal settlement types, i.e. area of 17.3 km². The second most common type of informal settlements is the *Low density* type occurring in 19 % of cases and covering 4.7 km². This is followed by the informal settlements *Under the trees or Mixed with trees* type (6 %, i.e. 1.4 km²). *Linear informal settlements along rivers* and *Pocket type* of informal settlements occur in approximately 1 % of cases (0.3 km² and 0.2 km² respectively). Last type of informal settlements, *Linear along railway* was recorded in 0.5 % of cases and covers the area of 0.1 km².

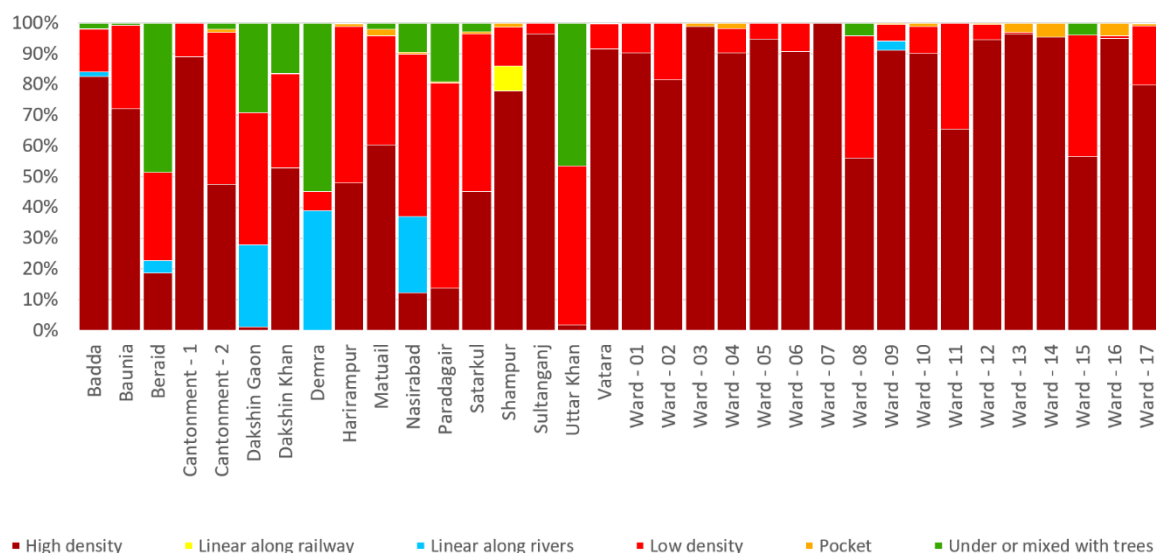
Table 8: Area and Share of different Informal Settlements/Slums Types

INFORMAL SETTLEMENTS / SLUM TYPES	Area (km ²)	%	Share per Urban Fabric (%)
High Density	17,25	72,18	3,84
Low Density	4,66	19,50	1,04
Under or mixed with trees	1,37	5,73	0,31
Linear along rivers	0,32	1,34	0,07
Pocket	0,18	0,75	0,04
Linear along railways	0,12	0,50	0,03
TOTAL	23,90	100	5,32

Table 8 shows area, share and share per urban fabric of different types of Informal settlements/Slums. Most common types are High and Low density Informal settlements/Slums. Total Share per Urban Fabric is 5.32 %.

It must be noted that the distinction between Informal settlements/Slums categories is approximate, some transitional classes yielded lower accuracies and not all of them could be precisely extracted (especially inhomogeneous and incontiguous pocket slums).

Figure 19 displays the proportional distribution of informal settlements/ slum areas types in each district. Considering the large number of districts the graph has been split into three parts.



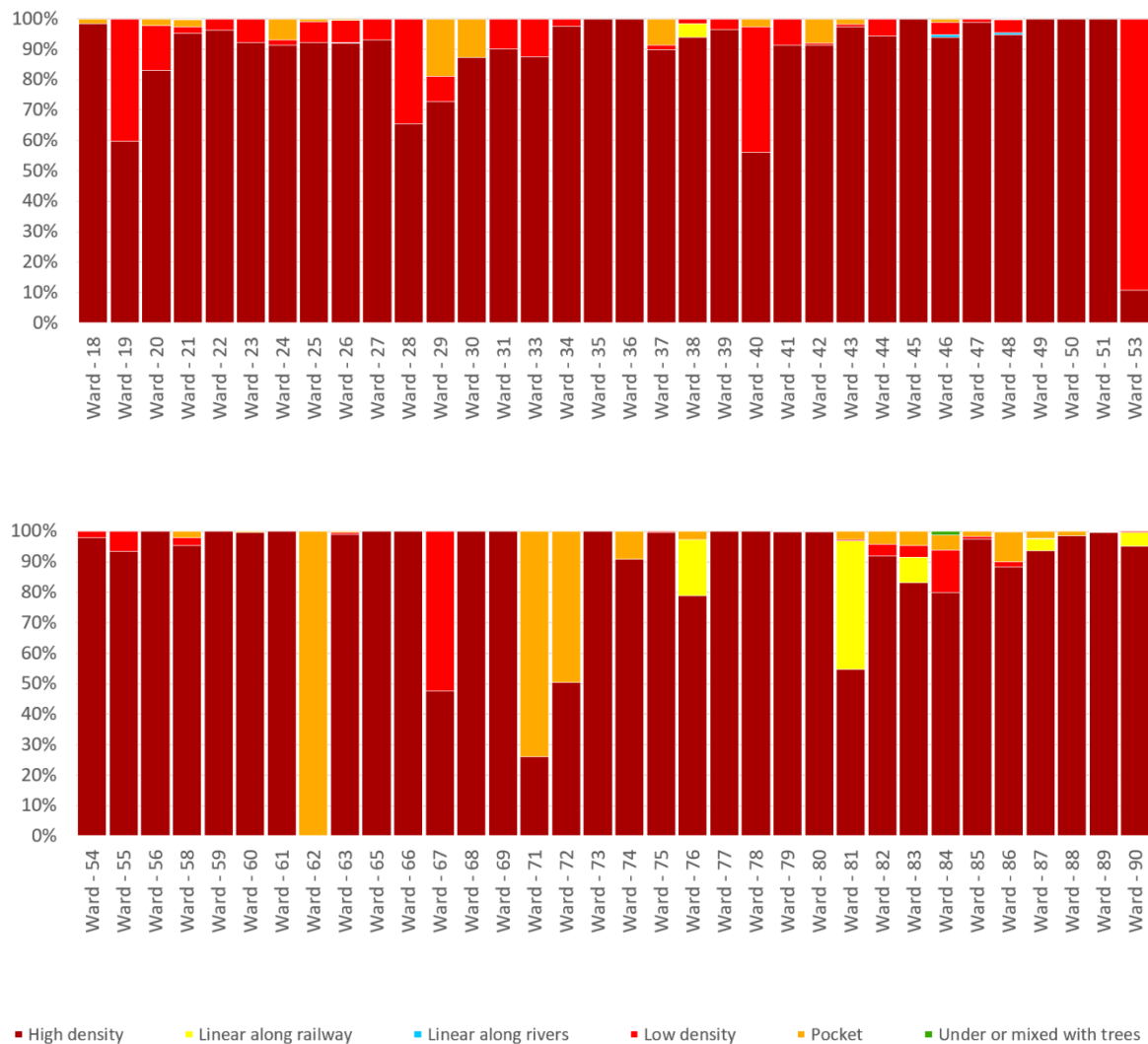
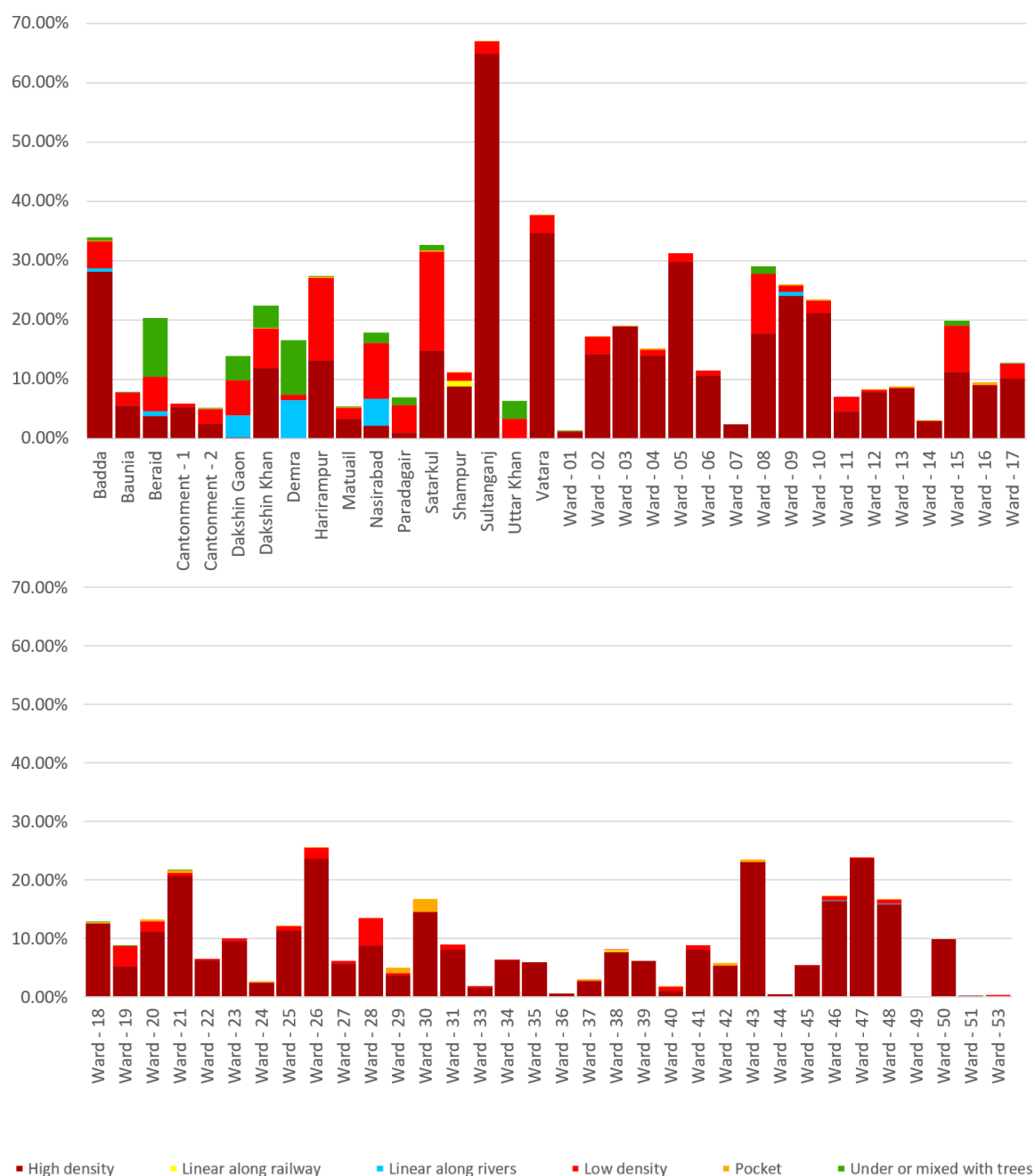


Figure 19: Informal Settlements/Slum Areas distribution per type in % per districts area – status 2017.

The informal settlements are classified into six classes: *High density*, *Linear along railway*, *Linear along rivers*, *Low density*, *Pocket*, and *Under or mixed with trees*. The graph above looks into the proportionate distribution of each of the informal settlement type within each district.

Regarding the *High density* type of informal settlements this class represents 100 % of all informal settlements types in many districts. *Linear along railway* type covers the proportionally largest area in Ward-81 district where it presents 42 % of all informal settlements, whereas *Linear along rivers* type covers the proportionally largest area in Demra district where it presents 39 % of all informal settlements types. Ward-53 is the district where the largest proportion of *Low density* type can be found covering 89 % of all informal settlements area. *Pocket* type covers the proportionally largest area in Ward-62 district where it presents 100 % of all informal settlements types and *Under or mixed with trees* type presents the largest portion of informal settlements types in Demra district covering 55 % of the informal settlements area.

Figure 20 displays the proportional distribution of informal settlements/ slum area types in relation to the urban fabric area within each district. Considering the large number of districts the graph has been split into three parts.



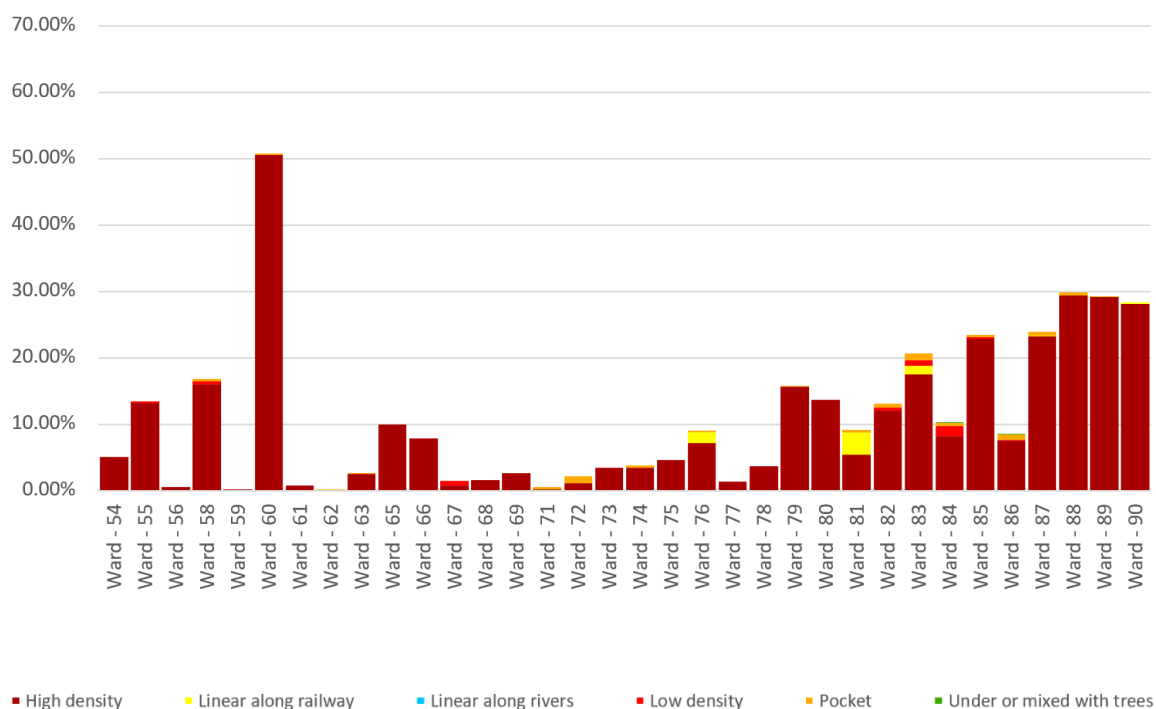


Figure 20: Informal Settlements/Slum Areas distribution per type in % per total urban fabric area in each district – status 2017.

The largest area of *High density* type of informal settlement was recorded in the Sultanganj (Kamrangirchar) district where the area covers 2.3 km² representing 64.9 % of the whole urban fabric area of the district. *Linear along railway* type covers the largest area in the Shampur district (0.06 km²) and *Linear along rivers* type covers the largest area in Demra district (0.13 km²) which presents 6.4 % of the district's urban fabric area. In the Harirampur district the largest area of both *Low density* type and *Pocket* type of informal settlements was recorded covering 0.88 km² and 0.01 km² respectively. As for the *Under or mixed with trees* type of informal settlements the largest area recorded is in the Dakshin Khan district covering 0.38 km².

Next figure presents two views on spatial distribution of slums. On the map on the left a portion of slums per district area is displayed and on the map on the right a portion of slums per urban fabric area is shown.

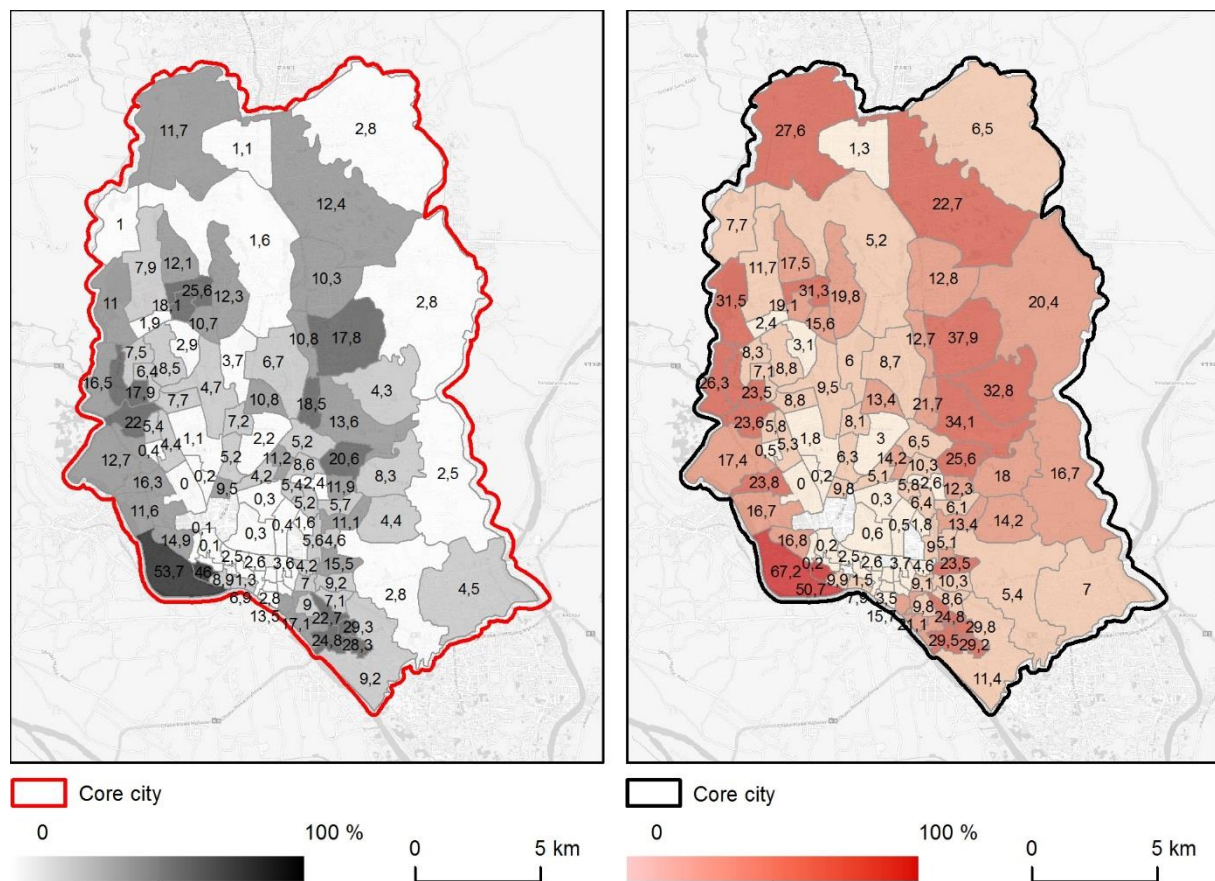


Figure 21: Informal Settlements/Slum Areas proportion in % of district area (left) and Informal Settlements/Slum Areas proportion in % of total urban fabric area in each district (right) – status 2017.

Districts in south-west part of core city are mainly affected by slums. Significant shares of slums per district/urban fabric are located on the edge of core city.

Figure 22 shows different statistics of slums. Figure on the left presents change rate of slums (calculated as ratio of slum area as of 2017 on slum area as of 2006). Figure on the right presents changes in informal settlements/slum areas as proportion of total urban fabric area in %.

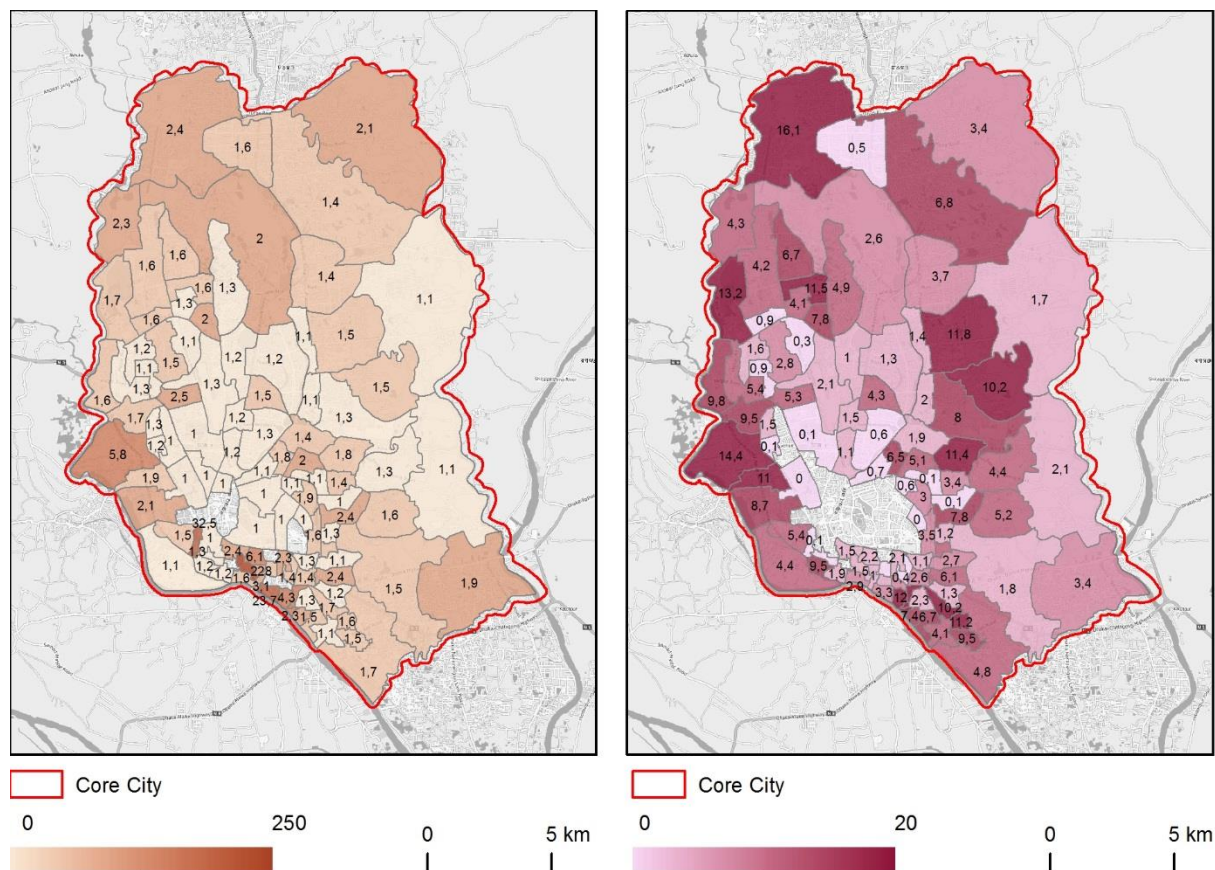


Figure 22: Informal Settlements/Slum Areas development: change rate of Informal Settlements/Slum Areas (calculated as ratio of slum area as of 2017 on slum area as of 2006) (left) and changes in Informal Settlements/Slum Areas proportion in % of total urban fabric area in each district (right) – period 2006-2017.

Above presented Figure shows districts affected by development of informal settlements. Increasing of proportion of Informal settlements per urban fabric is visible in districts on the edge of city center. In very Core city there is not much space for further expansion of informal settlements.

Figure 23 displays the proportions of informal settlements/ slum area of the Dhaka Core City which are in high/ medium/ low risk or no risk of flooding (left graph). The same is displayed in km² on the right.

It should be noted, that following statistics have been obtained by overlaying informal settlements and flood risk products. Flood risk product is based on combination of flood hazard represented by recorded (detected) extents of historic flood inundations (over the last decade) and flood exposure represented by scaled vulnerability based on 2017 LULC product. While detection of floods from HR satellite data is typically very accurate in open spaces, it may be subject to errors of omission in urban areas, generally higher in denser built-up areas. Similarly, flood history assessment at this precision level does not take into account current flood prevention measures (dams, embankments etc.) and derived risk level may be exaggerated with respect to current situation. As a result, following statistical figures should be taken only as indicative and verified with local knowledge.

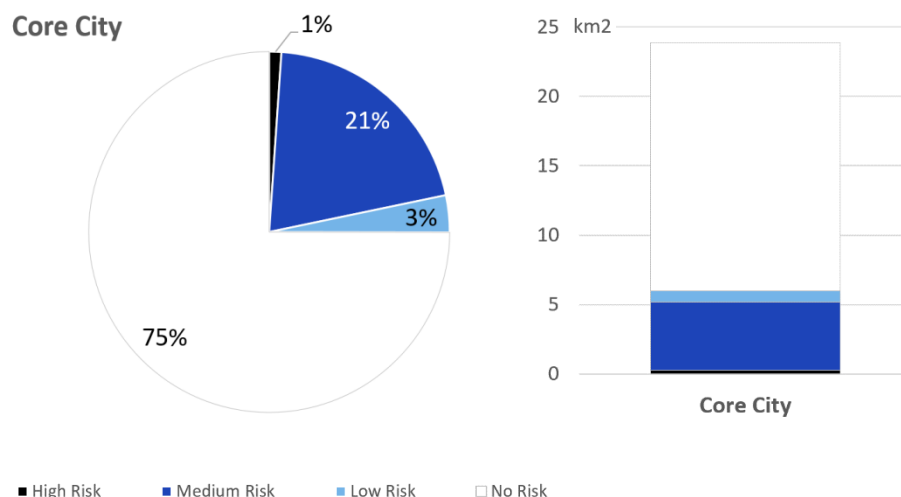
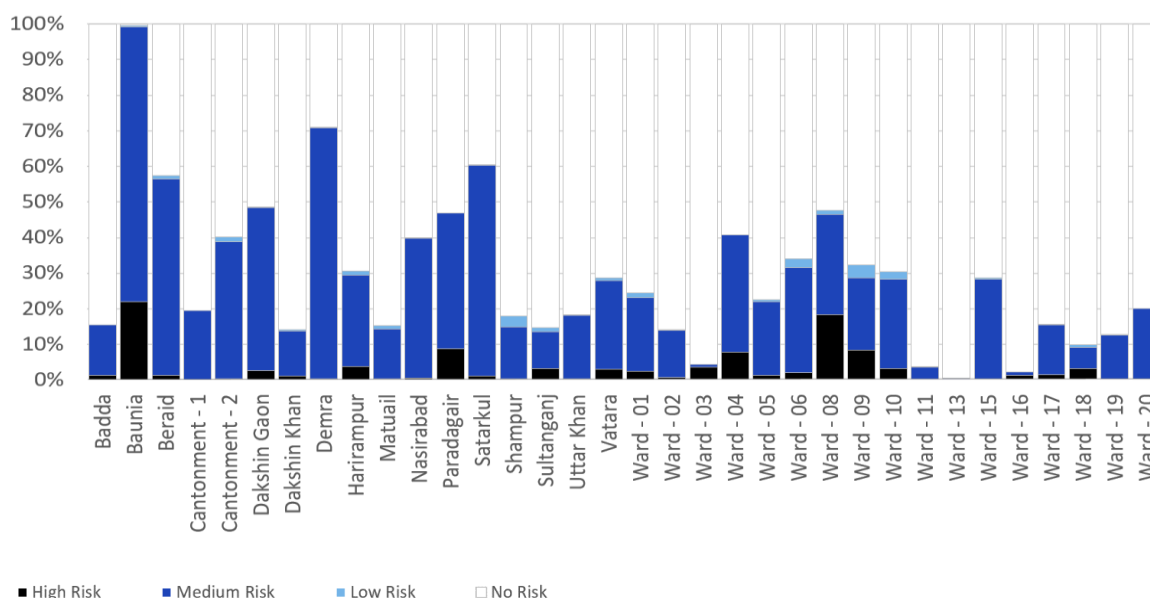


Figure 23: Proportion of Informal Settlements/Slum Areas in flood prone areas (high/low risk) – overall, in Core City in % of slum area (left) and in km2 (right) – status 2017.

The total area of Dhaka Core City covered by the Informal Settlements/ Slum areas detected from 2017 imagery is 23.9 km², out of which 75 % (17.9 km²) of the area is in no risk of flooding zone. A quarter of the area of Informal settlements/ Slum areas in the Core City is situated in the flood prone areas- 1 % (0.3 km²) is located in the high risk, 21 % (4.9 km²) in the medium risk, and 3 % (0.8 km²) of the informal settlement area is situated in the low risk of flooding area zone.

The next figure shows the proportion of each Informal settlement/ Slum area within each district located in either flood prone areas (high, medium, low risk of flooding) or in the no risk of flooding zones. Considering the high number of districts the graph is divided into two parts. Only districts where informal settlements/ slum areas are situated are displayed.



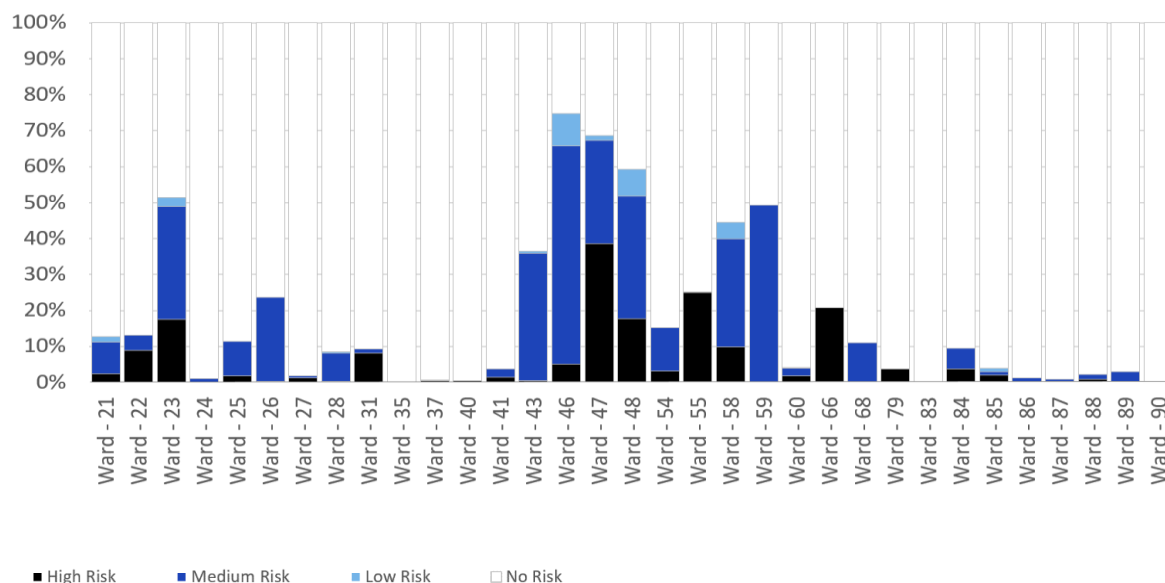


Figure 24: Proportion of Informal Settlements/Slum Areas in flood prone areas (high/low risk) in % of slum area per districts – status 2017.

The district of which the largest proportion of informal settlements/ slum areas is located within flood prone zone is the Baunia district (99.5 %), leaving only 0.5 % of the informal settlements/ slum areas of this district to be situated in the no risk of flooding zone. On the other hand Ward-35 is the district in which all the informal settlements/ slum areas are located in the no risk of flood zone.

The largest proportion of informal settlements/ slum areas located in the high risk of flood zone is located in the Ward-46 district (9 %) where this area represents 0.06 km². The largest proportion of informal settlements/ slum areas located in the medium risk of flood zone is located in the Baunia district (77.1 %), whereas the largest area located in the medium risk of flood is situated in the Harirampur district (0.44 km²). The largest proportion of informal settlements/ slum areas located in the low risk of flood zone is located in the Ward-47 district (38.6 %) where this area represents 0.10 km².

In addition to standard Informal Settlements/Slum Areas products, more detailed service has been provided for Dhaka city in order to identify main (spatially-driven) factors and patterns that can predict levels of water and sanitation coverage in the slums. Modelling exercise based on Slum Severity Index (SSI) utilizes number of spatial automatically generated slums characteristics e.g. neighbourhood locational, neighbourhood accessibility, morphological characteristics, LULC proportional characteristics, internal structure and dwelling characteristics (see list below on Figure 25) as well as local information from in-site surveys (e.g. WB WASH-P surveys). Few examples of different slums and slum neighborhoods can be seen on Figure 26.

Indicator Group / Indicator	Indicator type	Input Data	Indicator Group / Indicator	Indicator type	Input Data
Neighbourhood locational			Neighbourhood shape morphological characteristics		
Distance to paved road	Quantitative	OSM, LULC	Area	Quantitative	GIS
Distance to railroad	Quantitative	OSM, LULC	Perimeter	Quantitative	GIS
Distance to centre/CBD	Quantitative	LULC	Shape compactness	Quantitative	GIS
Distance to nearest important connectivity node	Quantitative	OSM, LULC	Neighbourhood LULC proportional characteristics		
Distance to (heavy) industry	Quantitative	LULC	LULC structure in surroundings	Quantitative	GIS
Distance to shoreline (river, canal, lake or sea)	Quantitative	LULC	Neighbourhood internal structure		
Neighbourhood accessibility			Built-up homogeneity	Quantitative	OBIA
Distance to arterial (capacity) road	Quantitative	OSM, LULC	Built-up density	Quantitative	OBIA
Distance to (selected) public services	Quantitative	OSM / GPS	Open spaces density	Quantitative	OBIA
Occurrence of feature within X metres	Qualitative	OSM, LULC	Greenness density	Quantitative	OBIA
Road network "winding index"	Quantitative	OSM	Neighbourhood dwelling characteristics		
Density of road network	Quantitative	OSM	Mean dwelling size	Quantitative	Imagery
Structure of road network typology	Qualitative	OSM	Mean dwelling separation	Quantitative	Imagery
Road connectivity - end point nodes	Quantitative	OSM	Mean estimated dwelling height	Quantitative	Imagery
Road connectivity - junction point node weights	Quantitative	OSM	Roof heterogeneity	Quantitative	Imagery
Road connectivity - junction point nodes	Quantitative	OSM	Roof & masonry material	Qualitative	Imagery
			Estimated dwelling age	Qualitative	LULC

Figure 25: List of spatial characteristics automatically generated for slums and slum neighborhoods.

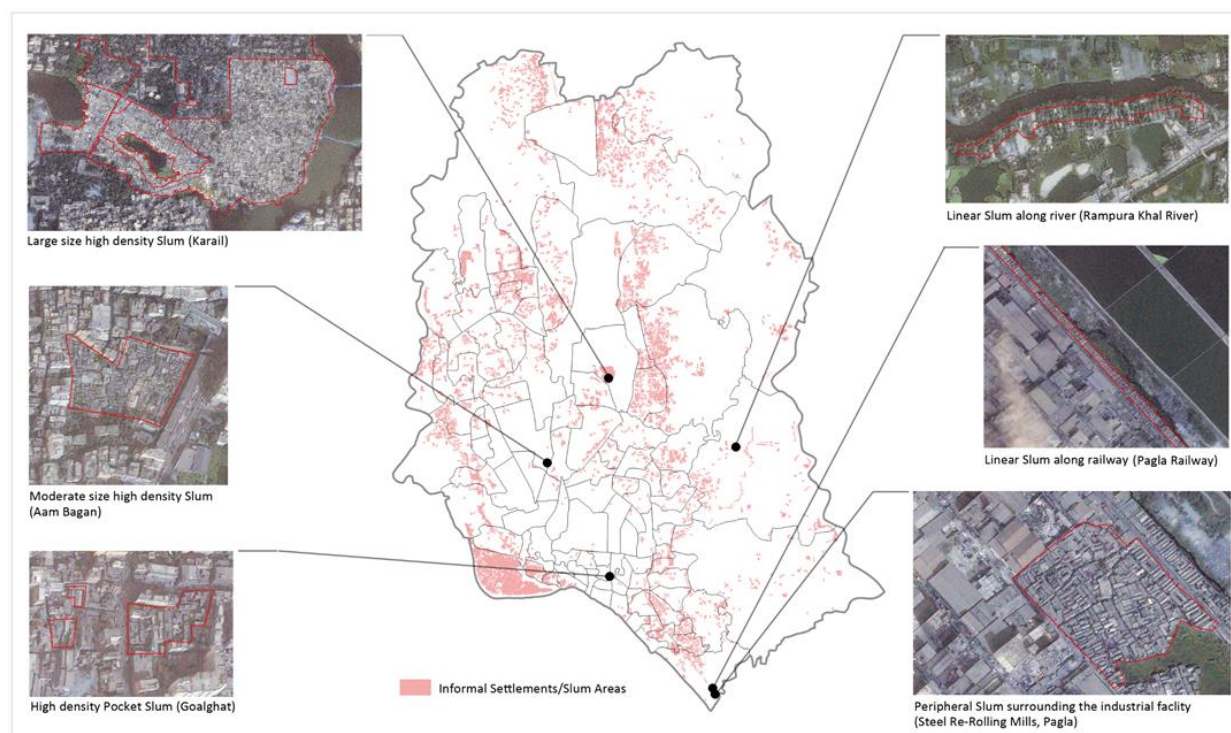


Figure 26: Examples of Informal Settlements/Slum Areas Types

4.5 Urban Green Areas and Open Spaces

This chapter provides insight into spatial distribution of Green Areas and Open Spaces elements within the *Core city*: green areas share, greenness index and its evolution, proximity, detailed situation mapping, fragmentation/connectivity and accessibility. Furthermore, statistics about development of green areas and greenness in general have been aggregated at level of administrative units (wards) or at level of urban fabric blocks (polygons delimited by bounding linear features – mostly roads).

In addition to standard Urban Green Areas product being part of EO4SD's products portfolio and consisting of two classes, advanced service with extended detailed typology is provided: **Open and green spaces (OGS)**. It was derived by interpretation of VHR imagery as of 2017. Public urban spaces such as streets, open spaces, green areas, parks, and public buildings are a part of cities that is often overlooked. Inadequate, poorly designed, or privatized public spaces often lead to exclusion, marginalization, and degradation of liveability in the urban environment. That is why the importance of green areas and open spaces are now embedded within the Sustainable Development Goals, particularly Goal 11.7 on universal access to safe, inclusive and accessible green and public spaces.

The service showcases how the use of geospatial information derived from very high resolution (VHR) satellite imagery can contribute to development of a framework to examine current situation and future potential of urban spaces to transform urban environments by promoting inclusive green growth and enhancing liveability in megacities and, furthermore, how such framework, supported by satellite image based information, can be scaled up to other global megacities and large cities in future. Figure 27 shows example of OGS product. More details on Urban Green Areas and Open Spaces typology and elements identified can be found in the Annex 1.

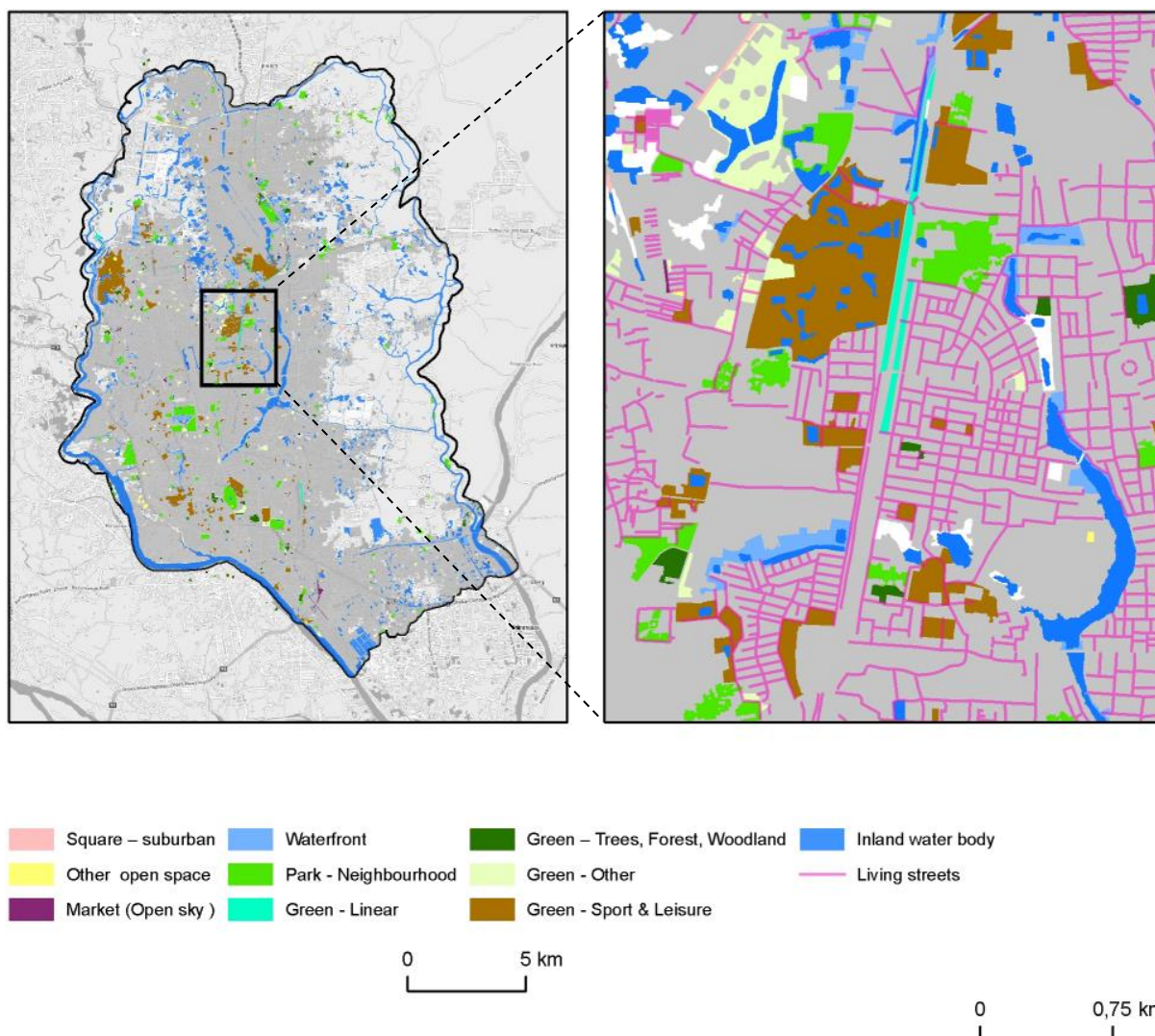


Figure 27: Map of open and green spaces based on extended nomenclature – overview and detail maps

Distribution of Open and green spaces is presented in the Figure 27. Extended typology derived by CAPI interpretation is demonstrated in detail on the right. This product represents maximalist option in terms of spatial and thematic detail for open and green areas, which is feasible to be extracted from VHR optical satellite imagery. It must be noted that extraction at this level of detail for large city like Dhaka is quite costly. Thematic detail (of nomenclature) can be reduced in favor of applicability of semi-automated or fully automated approach. Discussion on such service cost balancing and trade-offs is currently ongoing with WB team support as part Phase 2 activities.

It worth to note that further independent categories of urban green may be derived automatically: distribution and density of green areas defined by presence of low or high vegetation (=land cover) without distinguishing by its type (= land use).

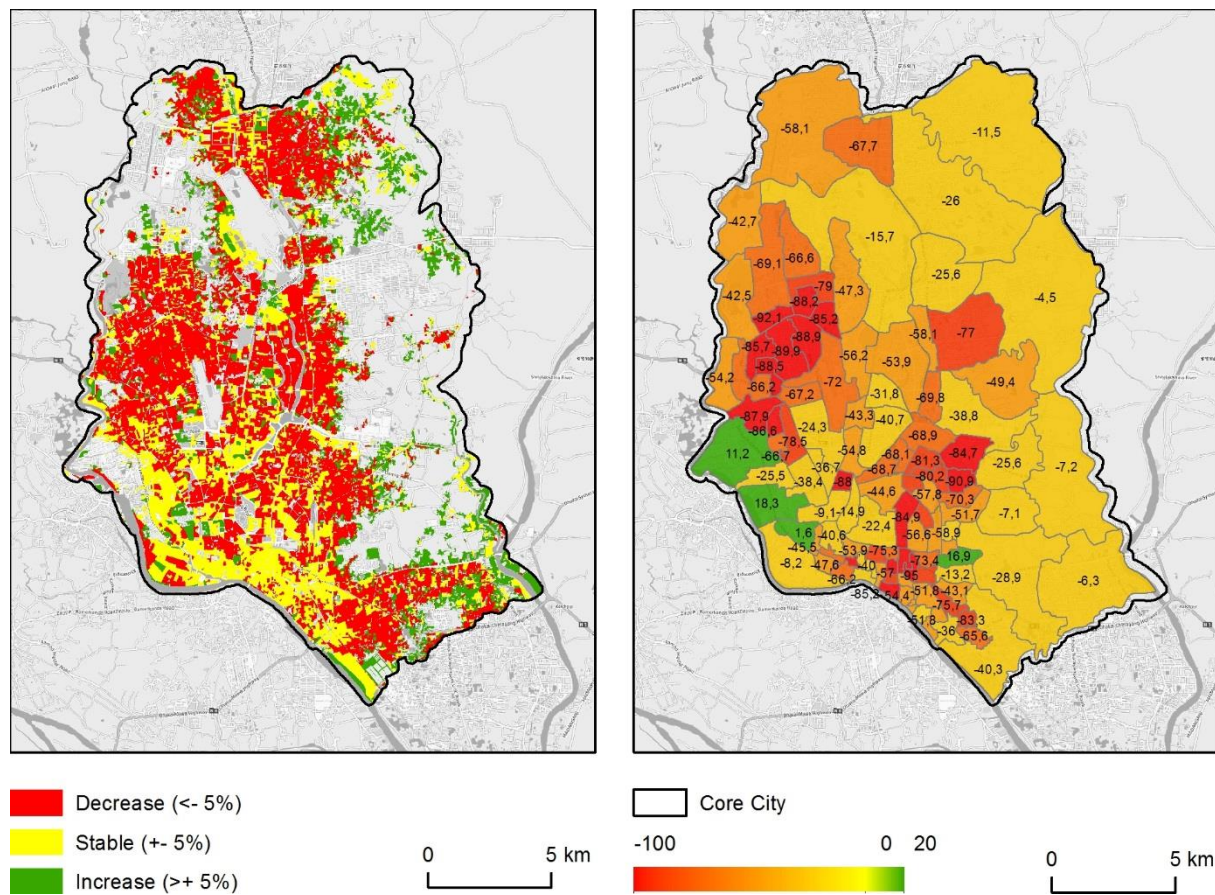


Figure 28: Development of green patterns: change of greenness index (2006-2017) per urban fabric blocks – left; Cumulative change of greenness index (2006-2017) per ward – right.

Greenness index and its change has been derived from VHR imagery by means of automatic detection of green patterns (= green cover type). The index takes into consideration both green areas according to LULC definition and small patches of green areas within urban fabric blocks, which have originally not been delineated as independent polygons of urban green classes due to minimum mapping unit constraint.

A pattern of greenness index development at detail level of urban fabric blocks is shown in the left figure. Figure on the right presents change between years 2006 and 2017 aggregated at district level. The most districts and blocks suffered from reduction of greenness between 2006 and 2017 due to internal urban fabric development – densification of built-up. This development goes typically at expense of open and vacant land within the city and inside urban fabric blocks. Almost 50% of urban blocks faced decrease of content of their “greenness” (represented by index as a sum of all green patches) during the observed period.

Furthermore, development of new urban green, sport & leisure spaces has not kept the pace of rapid annual urban form extent (1.9%) and urban population (4 - 5%) growth rates. Average annual growth rate of urban green areas estimated from LULC product (2006-2017) is by far much lower in the Core area (0.8%). Moreover, some originally green spaces were consumed by urban development.

The maps and graphs presented below have been designed to support **spatial analysis of sustainable public spaces network** in the Dhaka by describing following characteristics:

- *Availability*: share of public spaces out of a unit area
- *Accessibility*: distance from a public space to nearest road
- *Connectivity*: distance from a public space to the nearest neighboring public space
- *Inclusivity*: share of population living within 400m distance

Calculation of following statistics was conducted on following subset of classes from OGS product that best adhere to definition of spaces with potential “public” function: Parks (downtown, neighborhood, waterfronts), Squares (central, neighborhood), Market.

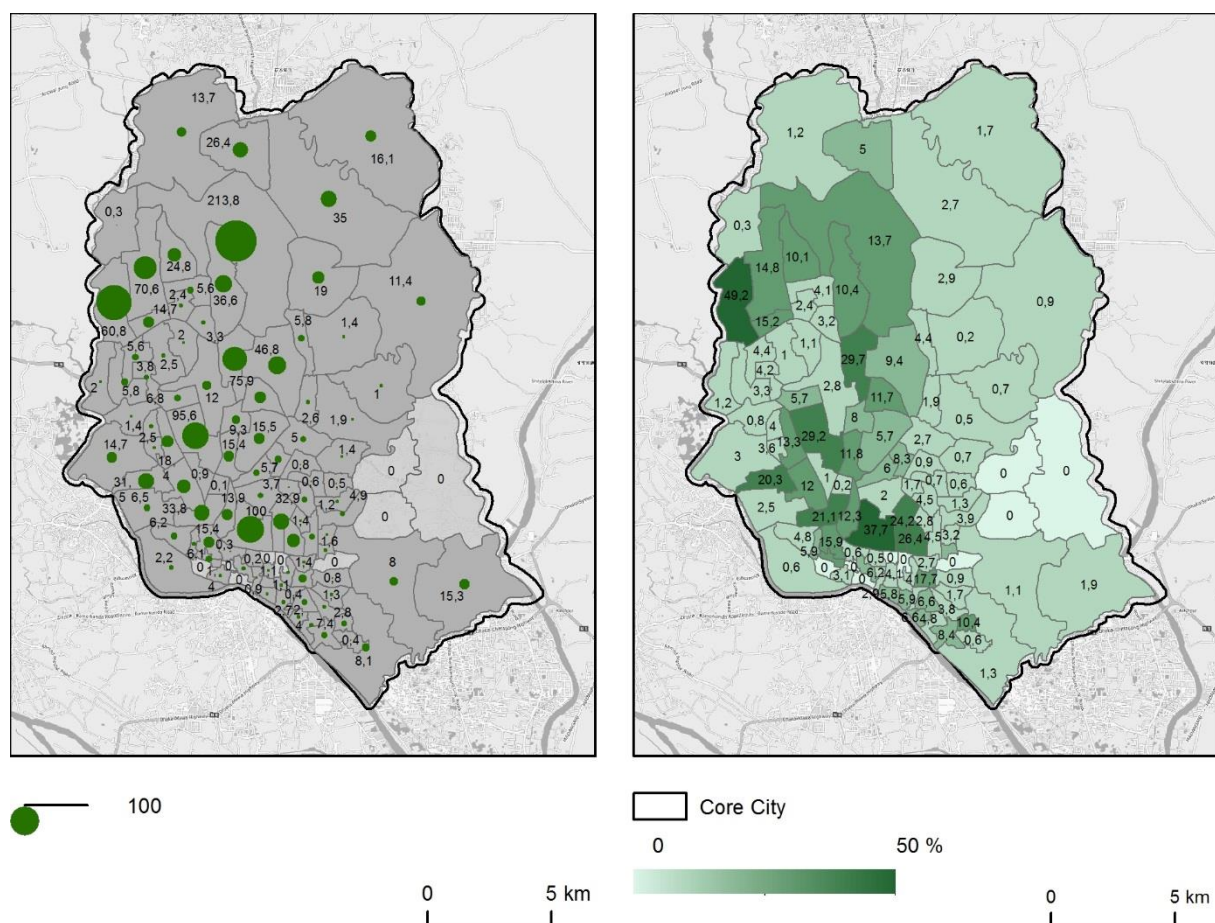


Figure 29: Share of OGS (potential public spaces) Total area of Open and green spaces per ward (ha) – left; Share of Open and green spaces on Urban fabric per ward (%) – right

Figure 29 shows the total area of urban green and open spaces areas (ha) per each ward. Urban green areas used for this statistics were aggregated from LULC product and Open Spaces were selected from Open Green classes. . Figure on the right shows Mean ratio of Urban Green and Open Spaces areas per Urban Fabric areas in ward. In addition Urban Green and Open Spaces area per capita could be also calculated, when detailed population statistics available per ward.

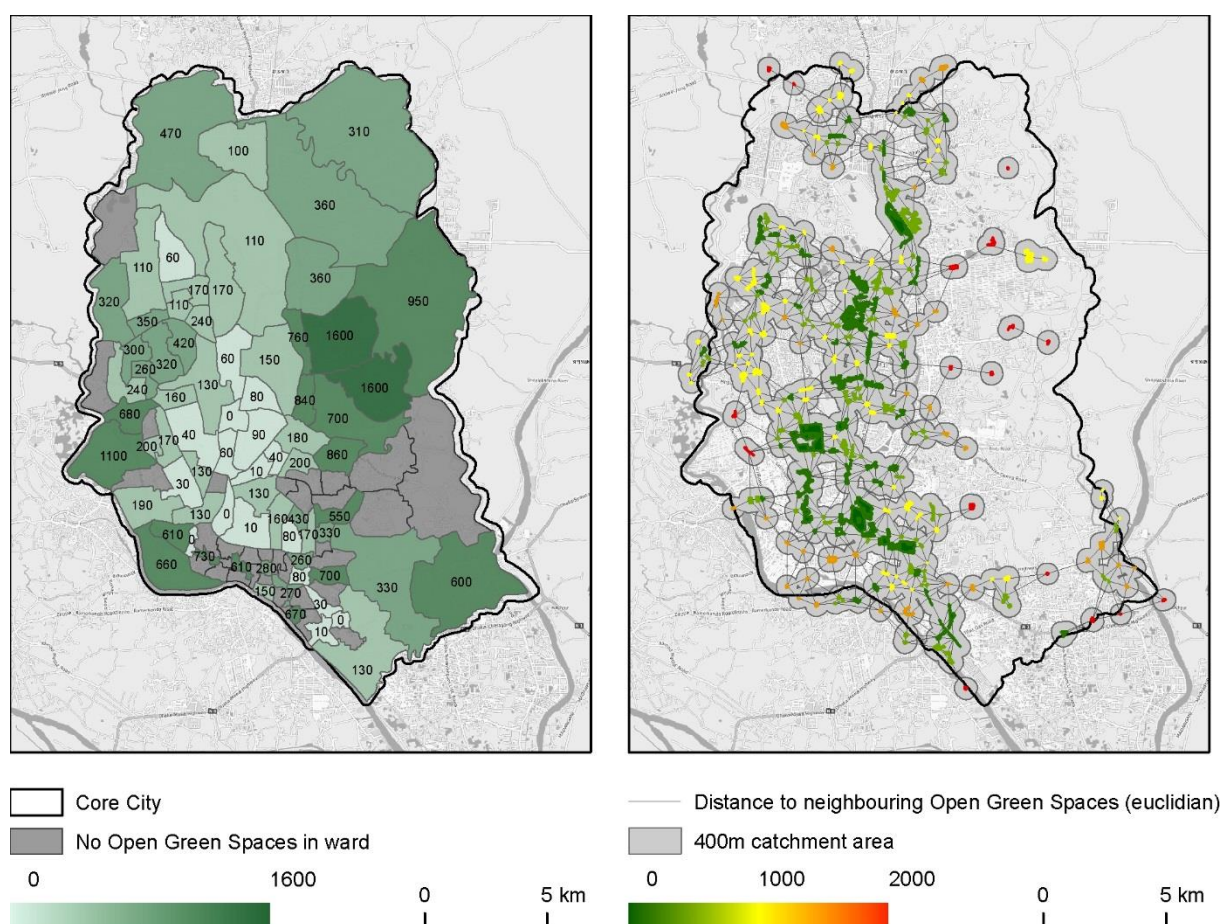


Figure 30: Open Green Spaces Connectivity: Mean distance between nearest Open Green Spaces in ward (m) – left; Euclidian distance to the nearest Open Green Spaces calculated for each Open Green Space - right

Figure 30 shows another insight into distribution of connectivity of the Open and green spaces. The figure on the left represents Mean distance between nearest Open Green Spaces (OGS) per ward.

Figure on the right represents mutual distance relationships between Open Green Spaces themselves - each OGS is classified by Euclidian distance to its nearest OGS with indication of linear connection links and their densities. This can support interpretation of fragmentation of selected OGS areas and potential axes to support better connectivity – OGS infrastructure network within city.

Table 9: Inclusivity: Percentage of population living within 400m catchment areas

	Estimation of total population in AOI (aggregated from GPWv4 1000m GRID)	Estimation of total population within 400 m catchment area	Share of estimated population within 400m catchment area on estimated total per AOI (%)
Dhaka – Core City	10.636.285	7.997.645	74 %

Table 9 shows population number calculated as disaggregation of GPWv4 (1km grid) population data. Table shows total population within AOI, within 400m catchment area around Open Green Spaces and their ratio. It must be noted that these numbers are approximate due to coarse resolution of population data. For detailed information about population around each of input Open Green Space refer to the table 1-2 in Annex 1.

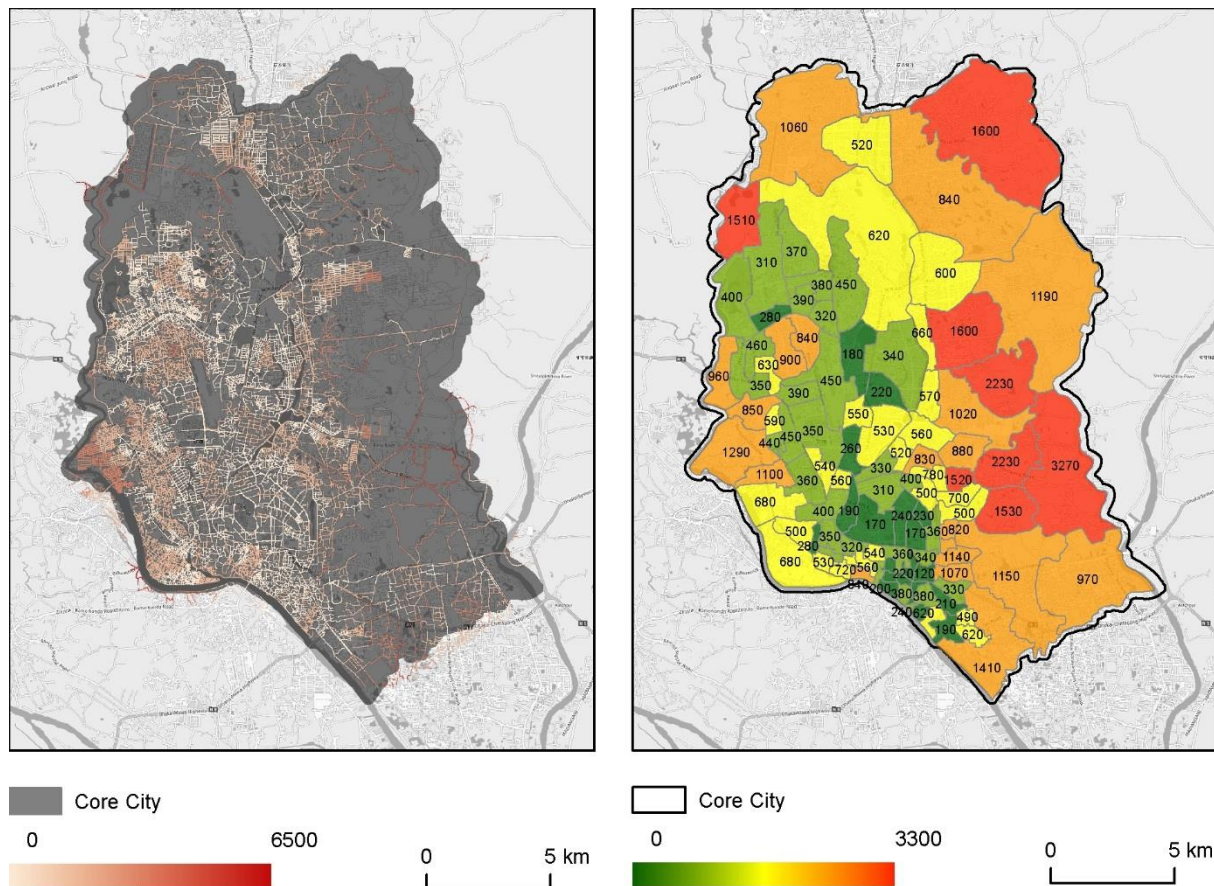


Figure 31: Open Green Spaces Accessibility. Street path distance to the nearest Open Green Space (m) – left; Average street path distance to the nearest Open Green Space (m) per ward – right

Both maps relate to Open and Green Spaces accessibility. Left figure shows distance to the nearest OGS (via road network) – distance was calculated for each point in the network assuming all roads and streets in the city while excluding highways. The figure on the right denotes average street path distance to the nearest OGS aggregated from roads in each ward.

As seen in the Figure 30 and Figure 31, there is high density of both large and smaller OGS being at close distance in the city's central parts. Accessibility of OGS measured by distance to the nearest OGS decreases when moving to the peripheral areas. However, even in relatively central parts of the city there are wards with high level of built-up density and inferior accessibility to OGS as well as low proportion of OGS in general.

Spatial aspects of OGS distribution within the city are further analyzed using following maps and graphs. They show relationship of selected variables on distances from the central business district (CBD) and directional axis defined by 30° transects. In the Dhaka, CBD point was defined on the Kwan Bazar circle.

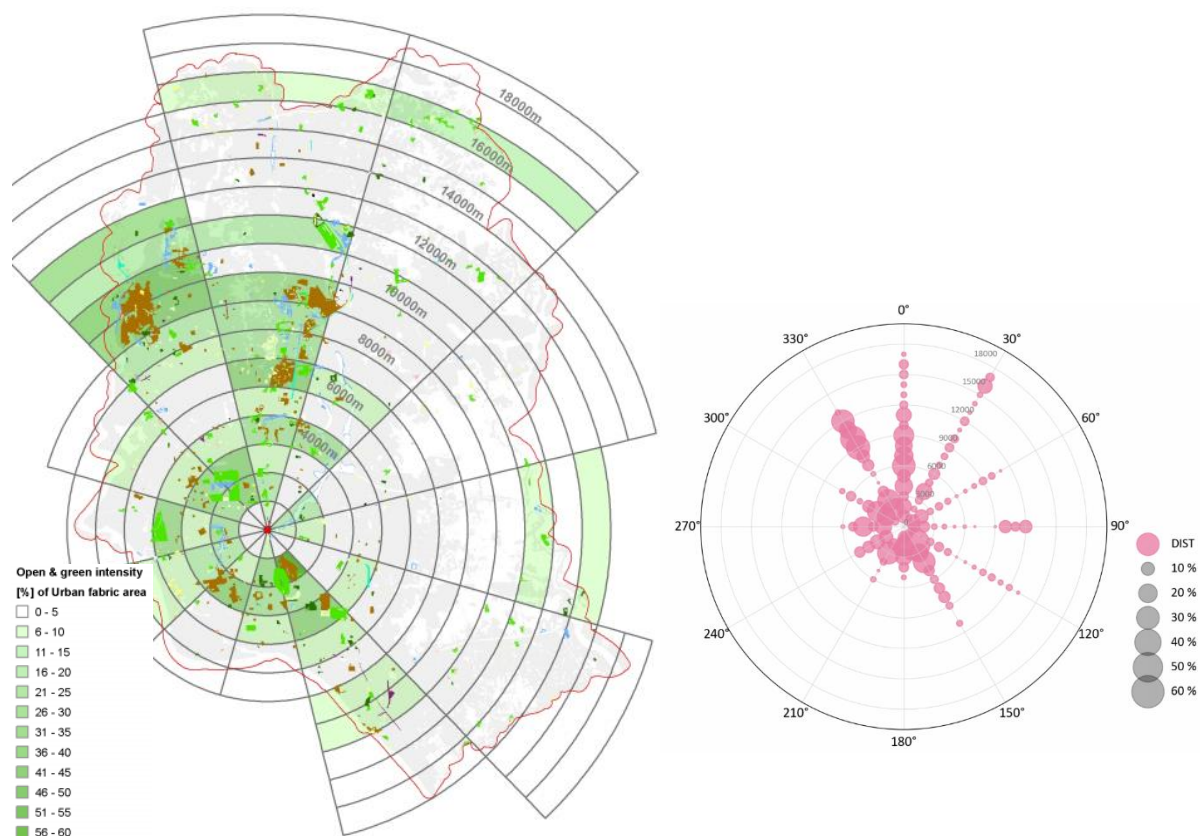


Figure 32: Directional distribution and density (as % of Urban Fabric) of selected OGS as a function of distance from the CBD

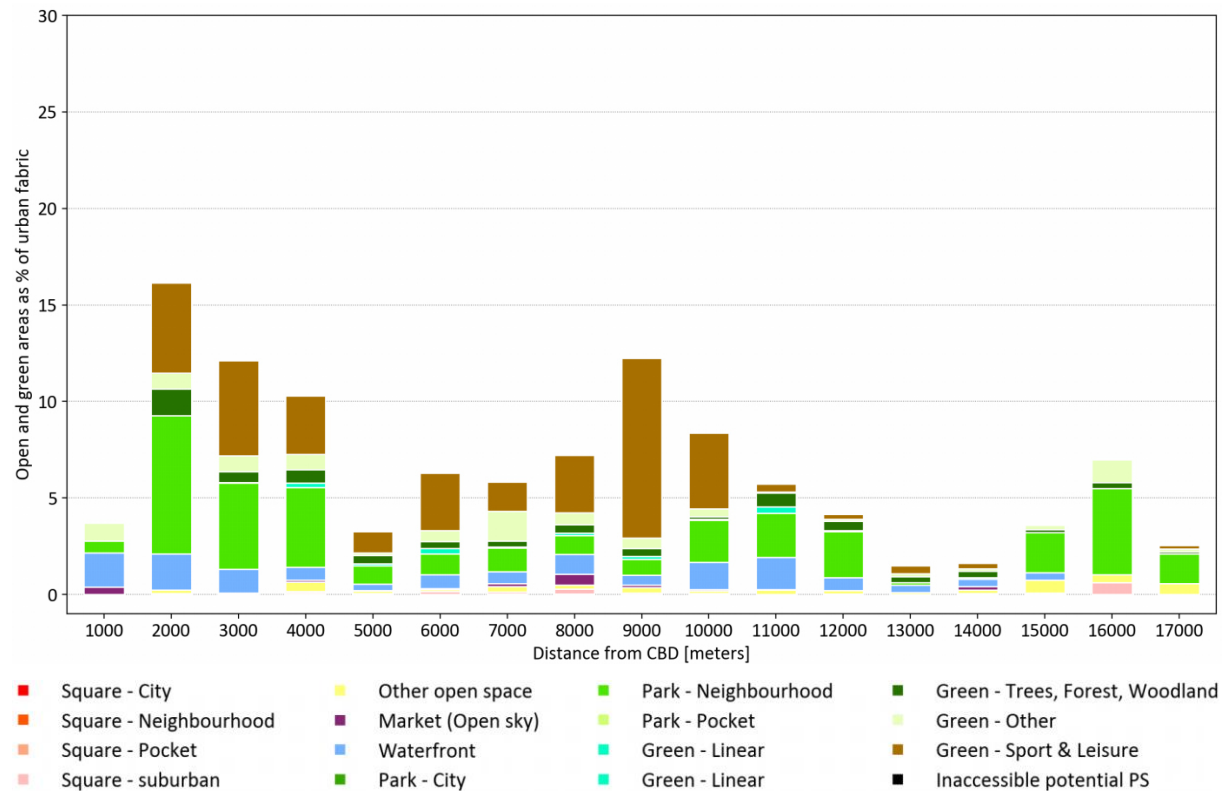


Figure 33: Structure of OGS as a function of distance from the CBD

4.6 Transportation network – Status

Transportation network has been derived for the core city from Open Street Maps. Typology of road hierarchy has been primarily adopted from Open Street Map. Where necessary, typology has been updates assuming situation visible from the most up-to date satellite imagery (as of 2017) in order to maintain consistent classification of transportation axes especially at primary and secondary hierarchical level.

Transportation network has been widely used to support analytical assessment in Slum and Green products.

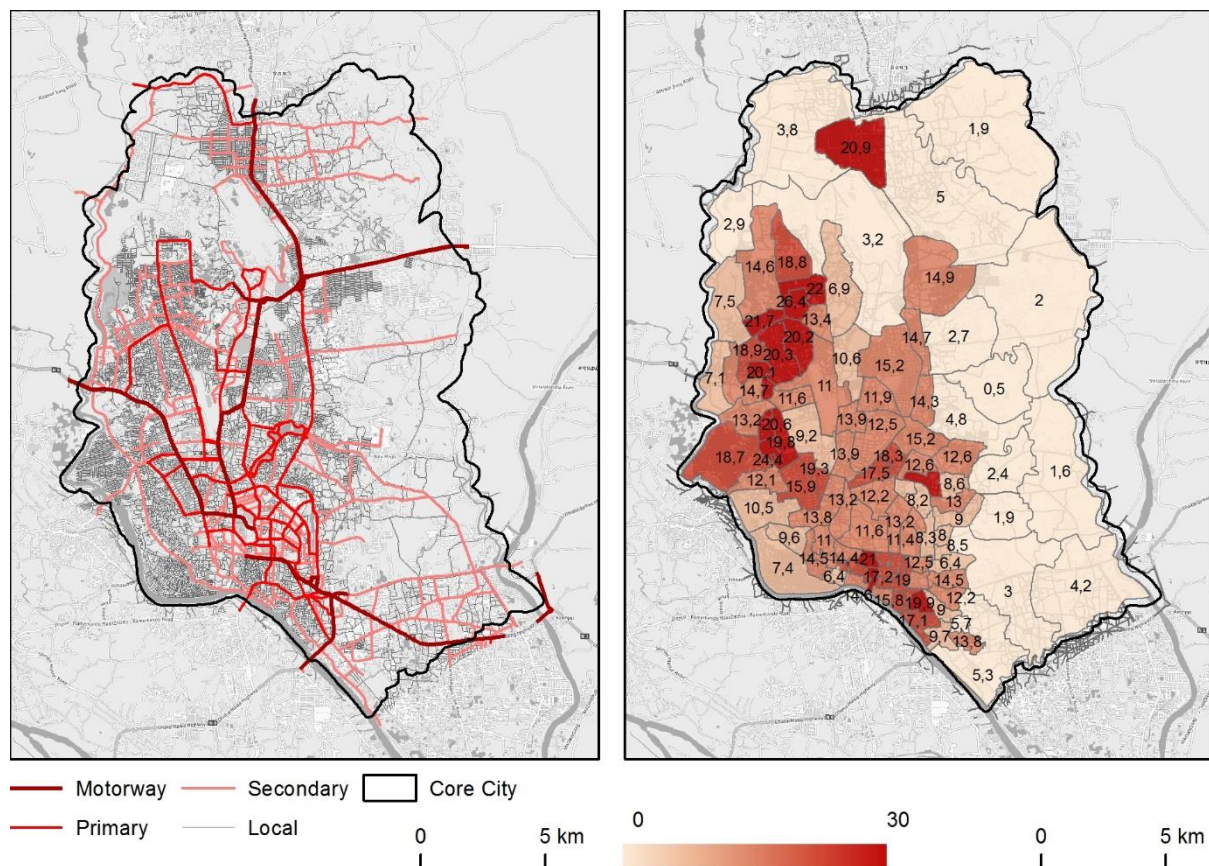


Figure 34: Current status of road network (left) and its density - mean length per area unit per ward (km/km²)

Figure 34 shows the transportation network spatial distribution and density in Core City of Dhaka. Right figure denotes length of all roads in kilometers per square kilometer of district area. High densities indicate mostly presence of formal residential fabric with dense grid composed of local roads.

In Dhaka mean length of roads per sq.km of urban fabric is 12.65 and mean number of intersection per sq.km is 69.2 of intersections per sq.km of urban fabric, which is 31% less than UN-HABITAT standard of 100.

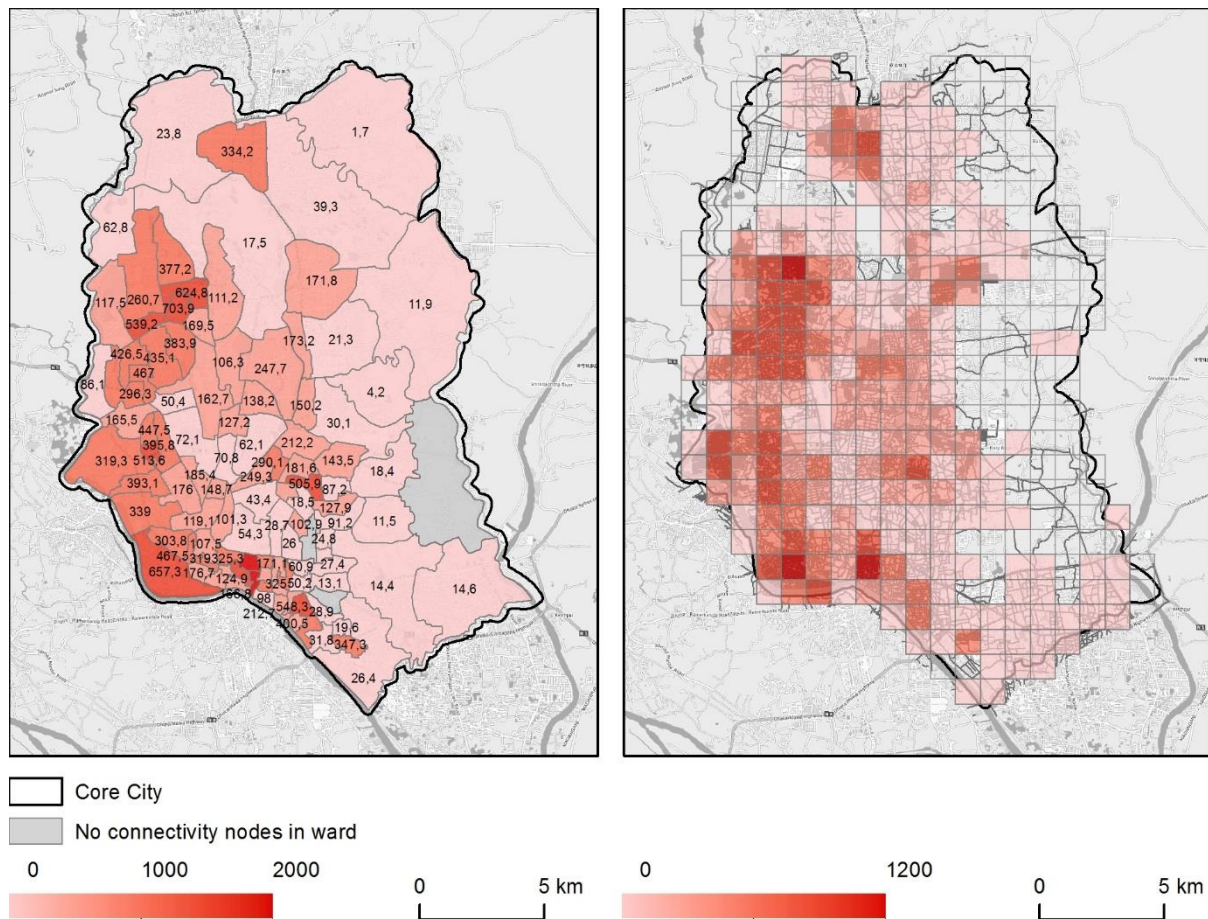


Figure 35: Density of street intersections: Number of intersections per 1 km² of urban fabric per ward (left) and intersections per 1 km² grid (right)

Figure 35 presents Street Intersection Density. Left figure shows number of intersections per 1 km² of urban fabric. Intersection is supposed to be a street network node with a minimum of 3 converging roads (i.e. road dead ends are excluded).

The figure on right presents intersections density per 1km² visualized in the regular grid.

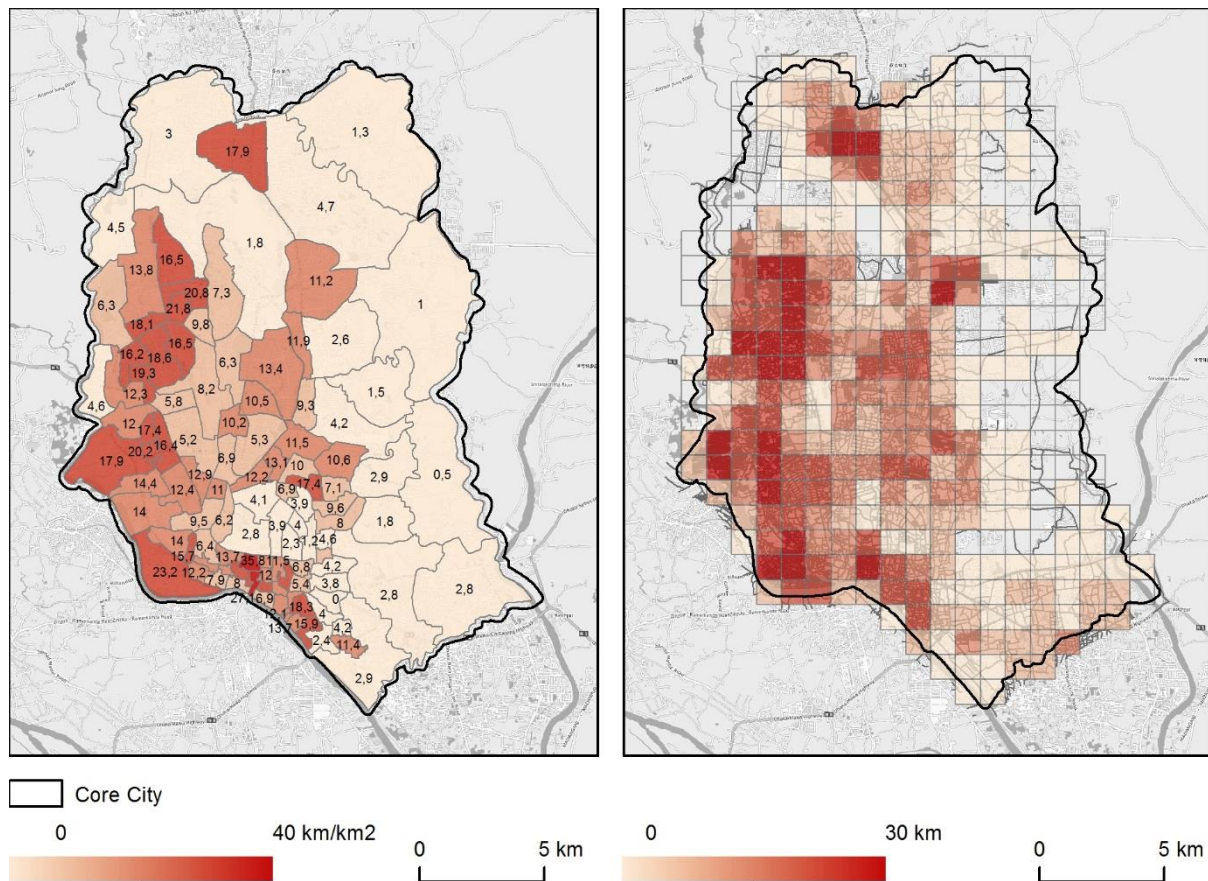


Figure 36: Urban Street Density – Street Length in km per 1km² per ward (left) and Street Length per 1km² grid (right)

Figure on the left displays density of street network calculated as length of streets per 1 km² of urban fabric per ward.

This figure on the right present, same as left one, Urban street density in Dhaka region. In this case are the values recalculated per 1 km² grid to provide results in a granular form.

4.7 Flood History, Hazard and Risk

Figure 37 shows proportion of flood risks in each district.

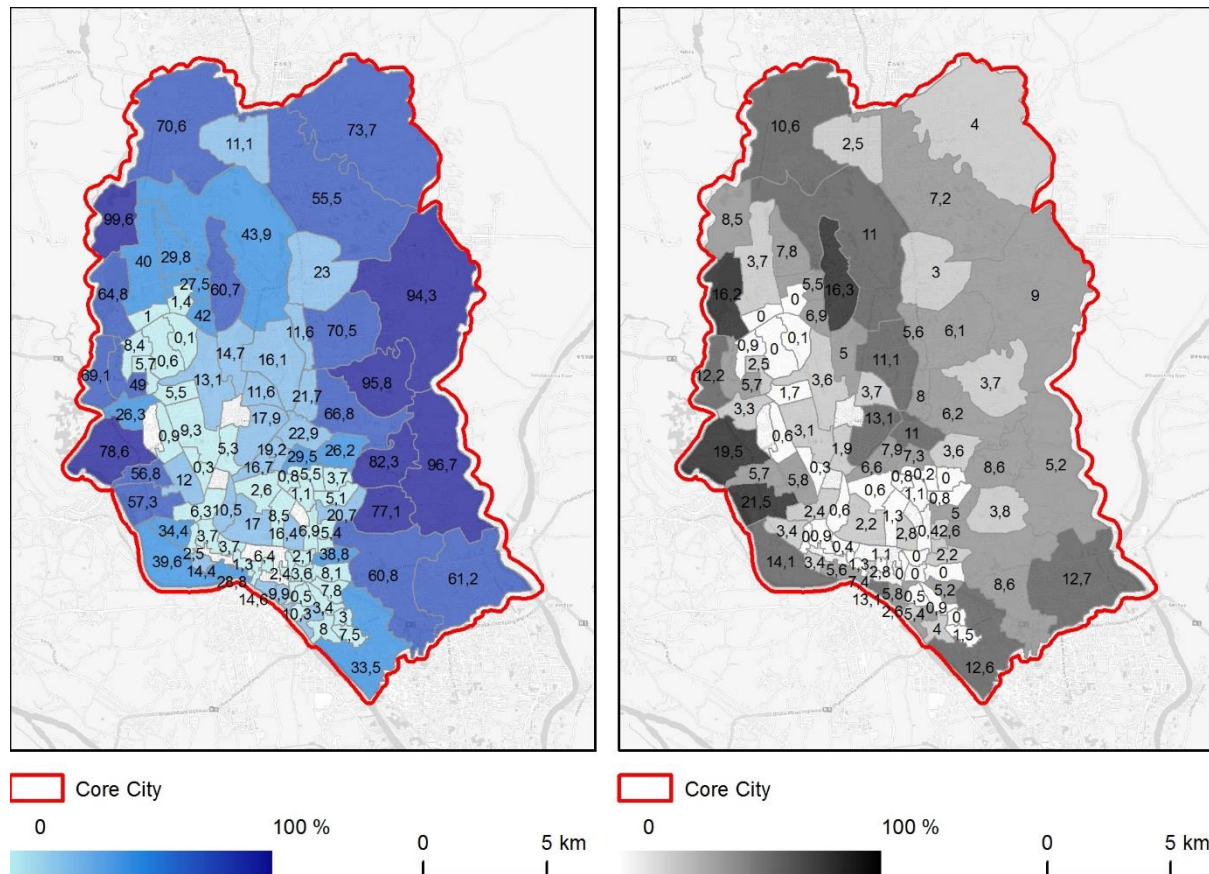


Figure 37: Proportion of (all, aggregated) flood risk classes per district (left) and Proportion of Flood High Risk per district (right) in % as of 2017

The left figure shows proportion of all flood risk (high + medium + low risk) in districts. The highest risks are located of course near rivers, on the other hand high risk of floods are not as often as it would seem.

It must be mentioned that detection of floods in highly urbanized areas is limited because of narrow streets or acquisition angle of images. Because of the results for very Core center could be underestimated.

In general, slums in Dhaka are not located in flood prone with high risk (except two districts on the west), On the other hand many slums are located in low risk flood prone zones in the arc on the east. If we consider that slums are often poorly built, even low flood risk could be fatal.

For city Dhaka, it is one of the most flooded cities in Bangladesh. The city has been flooded 4 times in the past 10 year. During the past decade, it has been flooded quite frequently. Major floods occurred in 2004, 2007, 2014 and 2016 caused by tropical cyclones.

Regarding flooding, three products have been derived based on flood inventory during past 10 years. The year 2004 flooding event is the worst in recent decades, so it was also mapped as a reference for comparing the significance of recent flooding events. To give an overview, three products were derived:

1. Product 1: Flood extent maps
2. Product 2: Flood hazard maps
3. Product 3: Flood risk maps

1) Flood Extent

The flood extent was derived from historical optical satellite imagery of 30 meter resolution (Figure 38). According to the flood extent maps, the flooding situation Dhaka is extreme in terms of both temporal and spatial dimensions. As shown in Figure 38, in the year 2004, almost half of the city was suffering from flooding. The majority affected lands are agriculture land and some low density residential area. 2014 flooding event was national wide and catastrophic, the extent was comparable to that in 2004. The event in 2014 was also significant with boarder of the peri-urban area totally flooded. The other flooding event (2007, 2016) was in less magnitude compared with these two major floods.

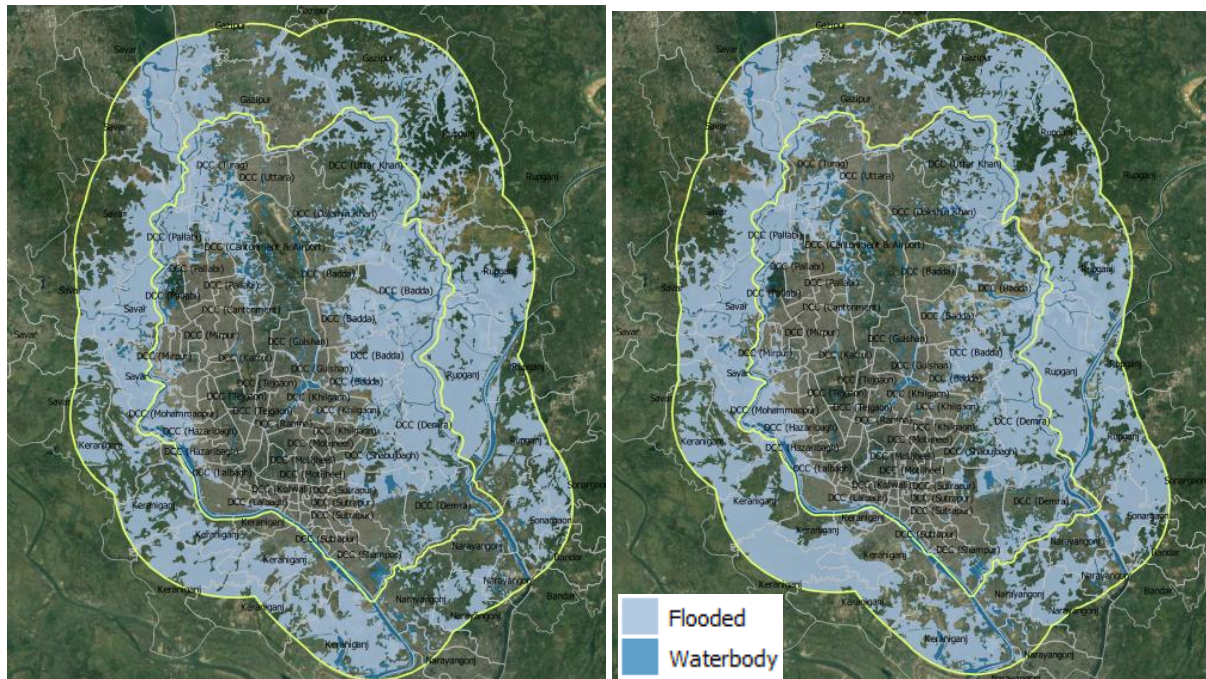


Figure 38: Flooded areas within AOI for the historical remarkable flooding event in 2004 (left) and 2014 (right).

As there is flood occurring in Bangladesh currently, extra effort was made to derive current flooding extent using Sentinel 1 imagery acquired on 10th and 24th of August, 2017 (see on Figure 39). Radar imagery is of added value for flood damage evaluation. It is valuable as it is a good compensate of cloudy optical imagery covering AOI. The affected regions are in the west and east areas that are close to the river. The flood remains in the affected areas after 14 days.

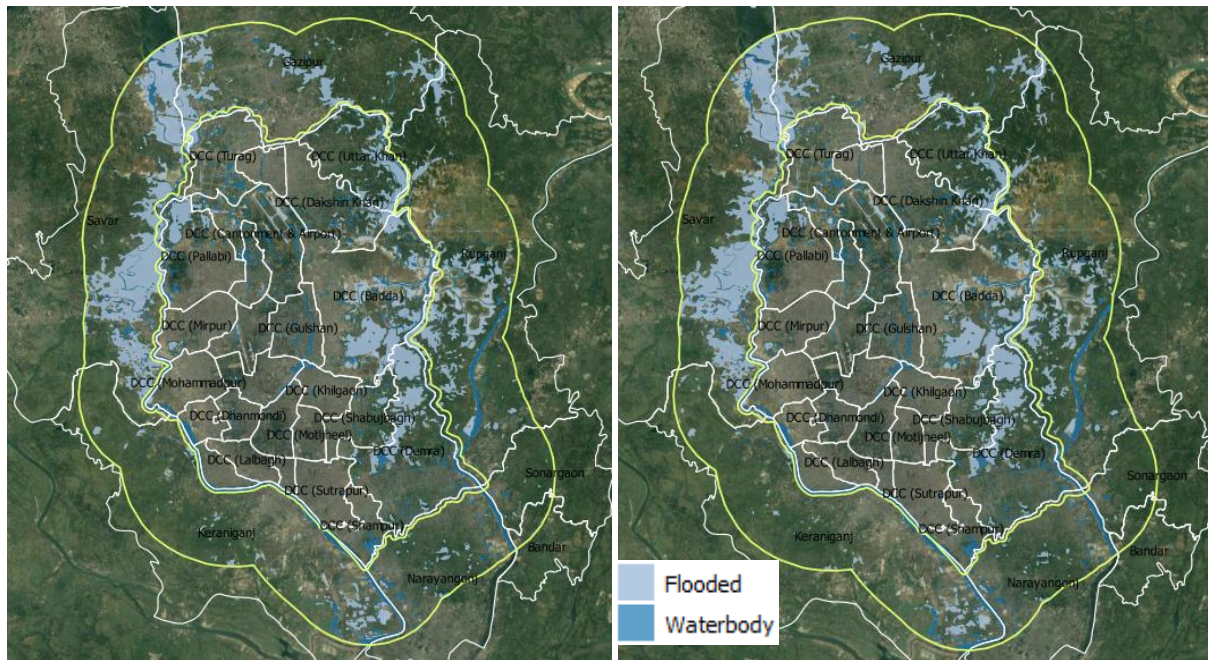
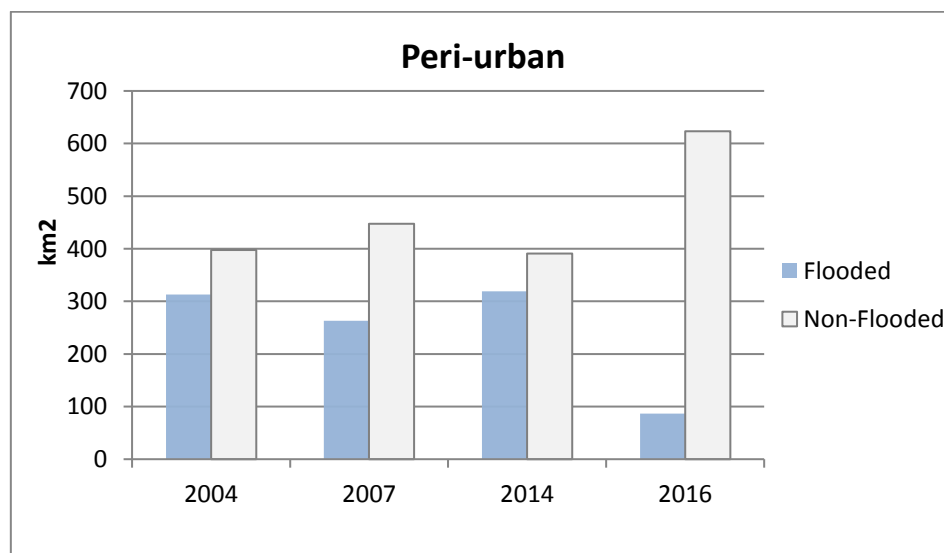


Figure 39: Flooded areas within AOI for the flooding event on August 10th, 2017 (left) and August 24th, 2017 (right).

As shown in Figure 40, during the two remarkable flood events in 2004 and 2014, 44.0% and 44.9% of the city was flooded within peri-urban area, and 35.7% and 30.8% in core urban area. In 2007, the flooding affected 37.0% and 30.8% of peri-urban and urban areas respectively. Regarding the flooding year 2016, the extent decreased significant to around 12.2% and 9.9% for peri-urban area and core urban area.

In general, 16 upazilas were affected by the flooding events, among Badda, Shabujbagh, Demra, Rugganj and west part of Savar were more frequently flooded.



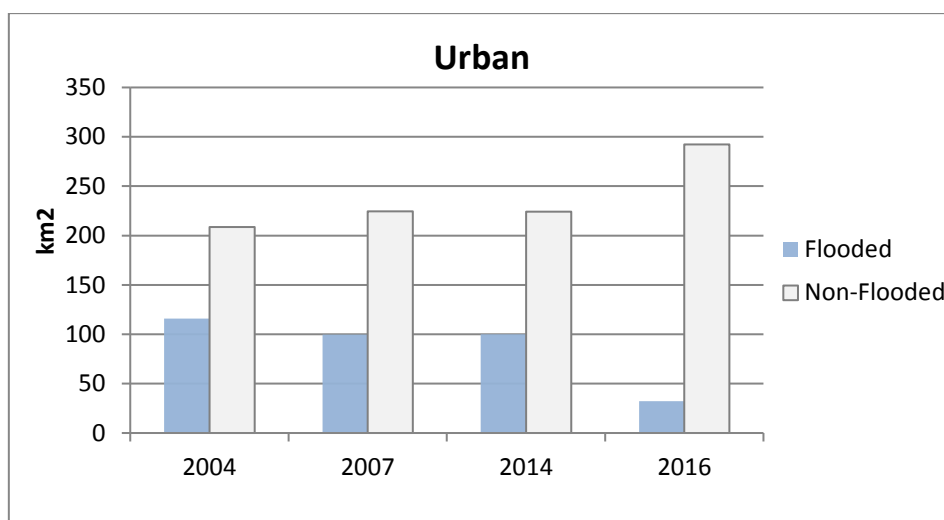


Figure 40: Flooded areas of AOI for peri-urban (top) and core-urban area (below) in km² for the year 2004, 2007, 2014 and 2016. The total area of peri-urban is 710.73km² and core urban area is 324.34km².

2) Flood Hazard

The flood hazard map was generated based on the occurrence of flooding events during the past 10 years, combined with the remarkable historical flood event that occurred in the year 2004. The map aims to give an idea about the flood presence in terms of both frequency and extent in the city, and illustrates which part is in general flooded more often than other areas (Figure 41). As the core urban is located in high elevated area, it is not as much severely flooded as the rest area.

Of all flooded area within peri-urban AOI, majority area was flooded once or twice during the past decade as illustrated in Figure 31 making up 41.6% of the total area (295km²). The area that has been flooded 3 times is around 78km², which is 11.1% of the whole AOI area. Around 37.4% was not flooded at all, and this area is around core urban area in the city (Figure 41).

In general, Savar is the most flooded upazila. This upazila located on the western part close to Turag River. Other susceptible upazilas are Badda, Rupganj and Demra all located in the east of Dhaka.

Regarding the core urban area, the proportion of different flood hazard differs slightly from peri-urban area. 8.9% of the land (29km²) was flooded 3 times, and 30.6 % was flooded 1 or 2 times. And the non-flooded area increase to 48.36% percent. Badda and Shabujbagh was the two most flooded upazilas in core urban area. According to the Figure 41, in the year 2004, some residential area next to the farm lands was flooded as well.

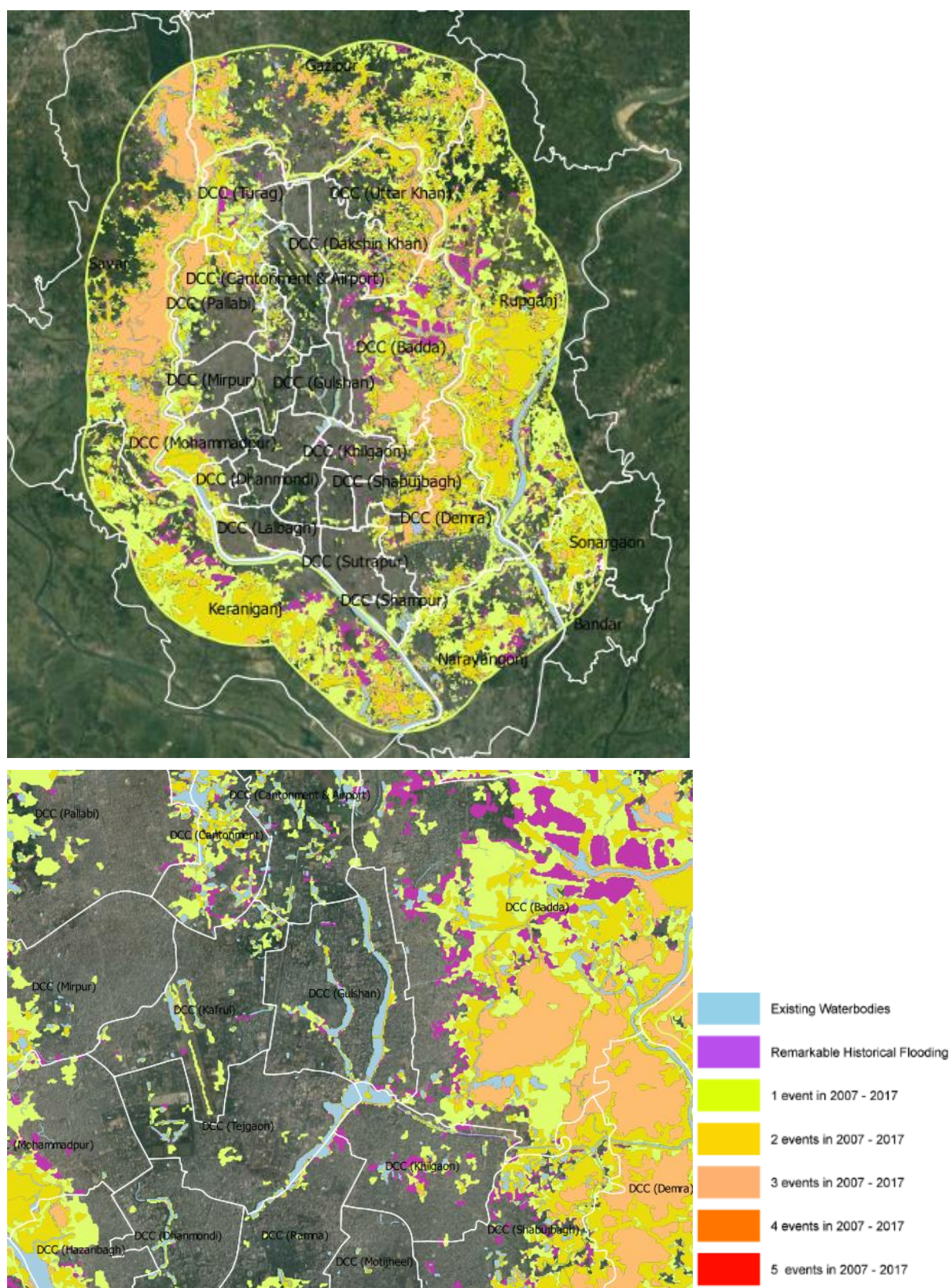


Figure 41: Flood hazard within AOI (top) and core urban area (bottom) based on flooding events from 2007-2017.

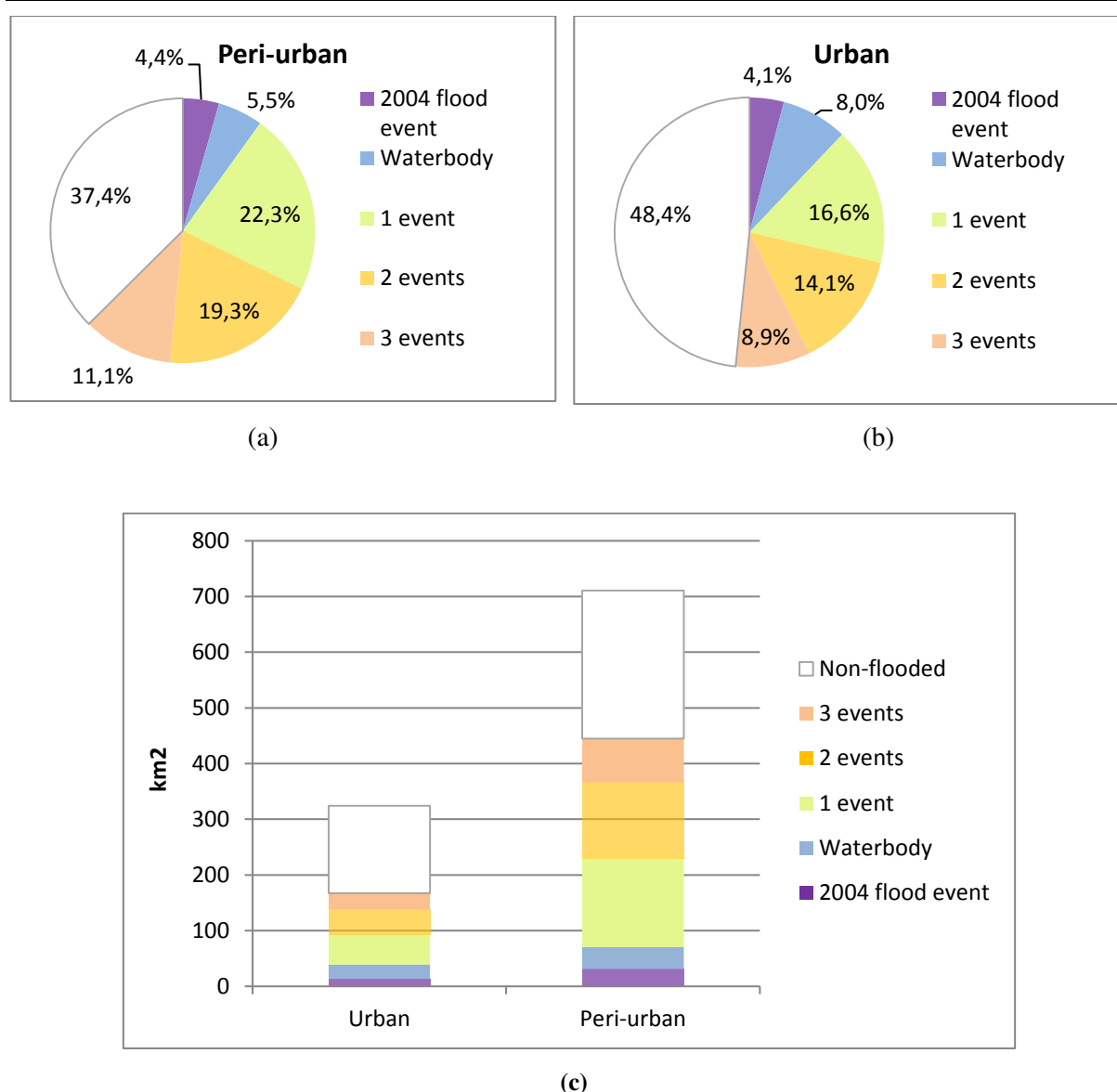


Figure 42: Flood hazard statistics for peri-urban (a) and core urban (b) area proportion in % and in km² (c).

3) Flood risk

The flood risk maps and statistics (Figure 43 and Figure 44) show that majority part has a flooded risk level MediumRisk (88.9 km² and 277.8 km² for core urban and peri-urban area), which means that it has been flooded more than twice and the damage on land use is equal or above B level (please refer to risk matrix in appendix for more information). This medium level is explained by the land use type being agriculture and frequent floods events. In total, the medium risk area is 27.4% (88.9km²) for urban and 39.1% (277.8km²) for peri-urban area.

Some high risk level areas are seen along the main river that passes along the border of the city (Figure 43). Mainly in western and eastern upazilas which are Savar, Rupganj, Badda and Demra.

However, it is important to know that this risk level is determined based on the risk matrix described above, and it still involves some degree of human interpretation. So it is recommended to treat the risk level here as relative metrics rather than absolute metrics.

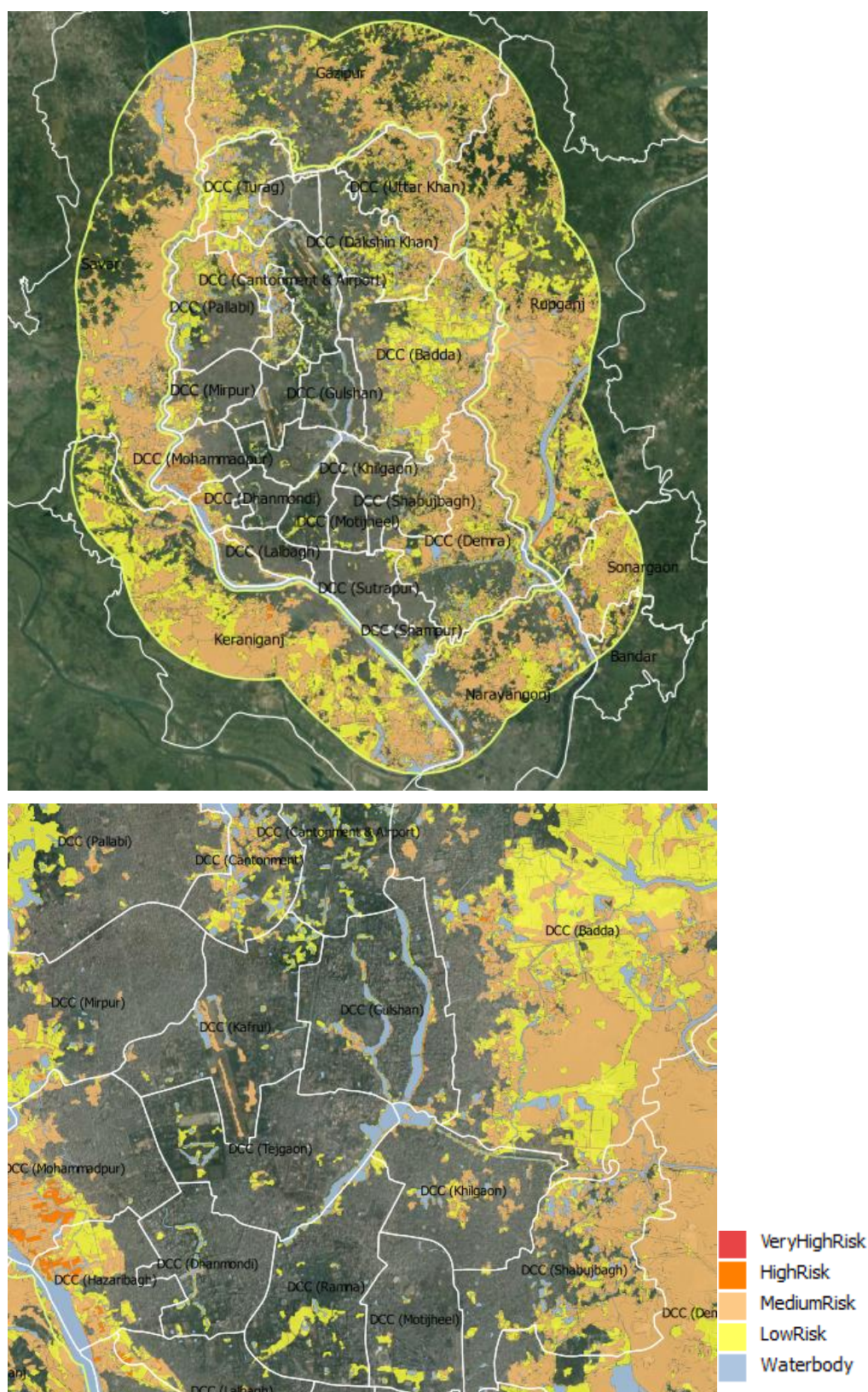


Figure 43: Flood risk within AOI of peri-urban (top) and core urban area (bottom) based on flooding events from 2007-2017.

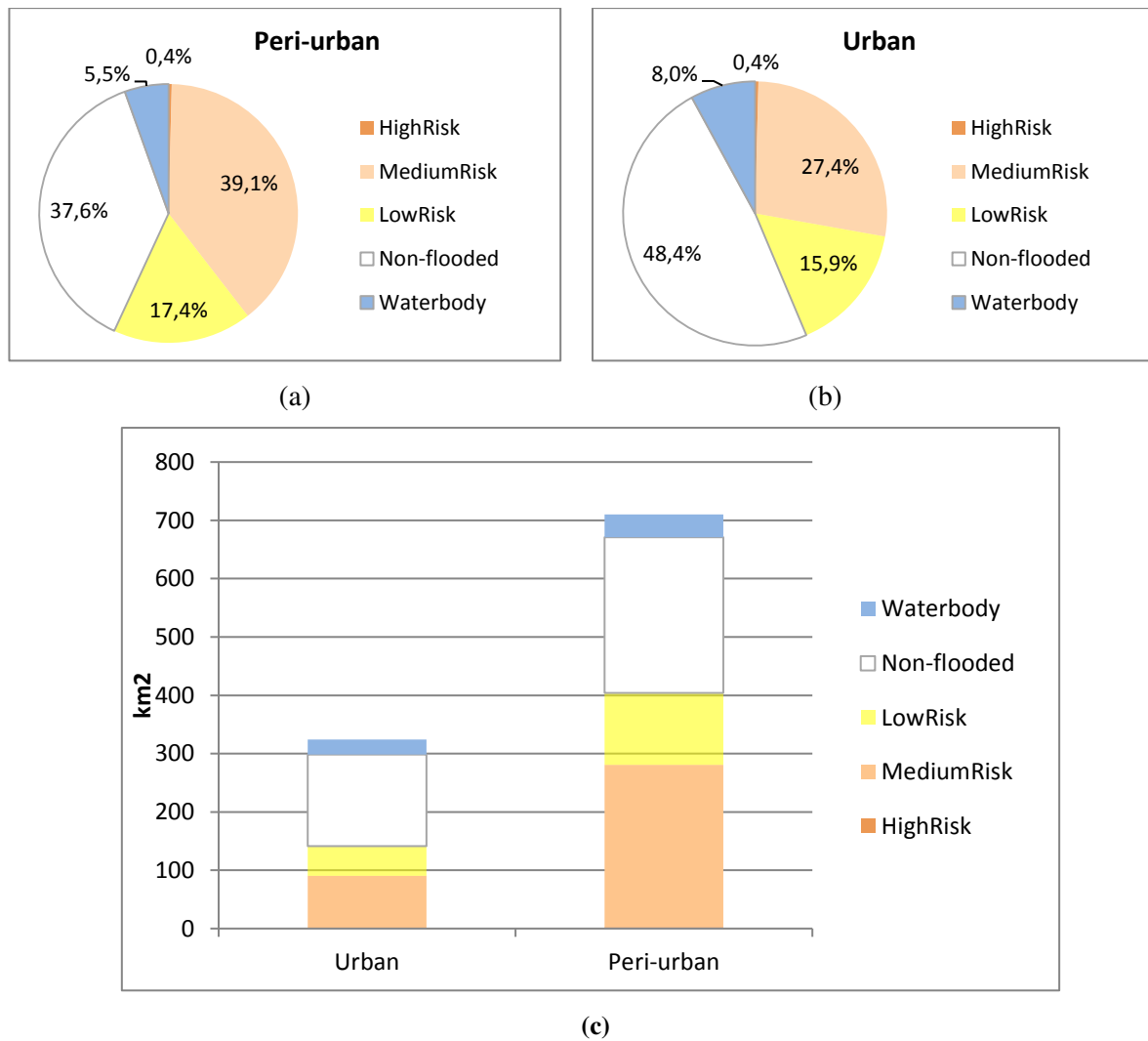
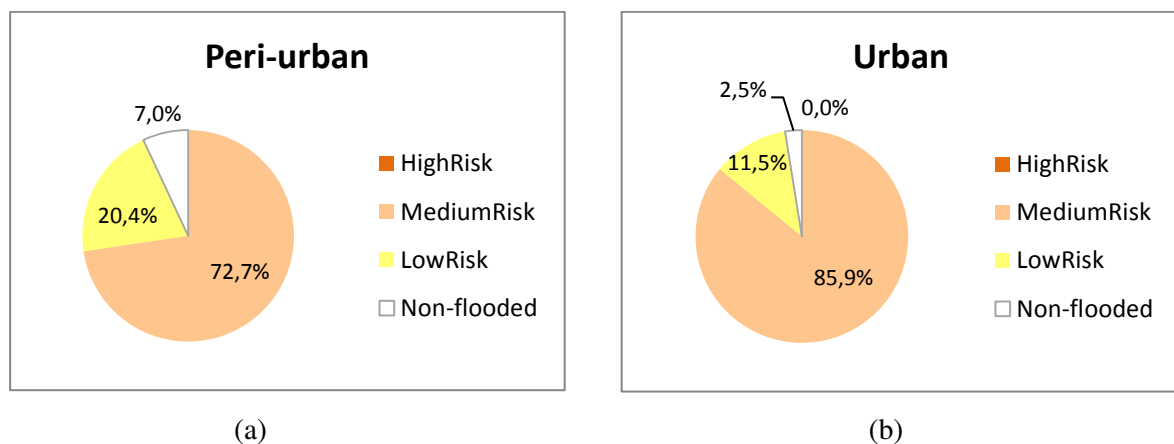


Figure 44: Flood risk for peri-urban (a) and core urban area (b) within AOI in area proportion % and km² (c)

Flood Risk Statistics per LULC class

All floods generally have a big impact on agriculture area as shown in Figure 45. 93.1% (97.4% of core urban) of all agricultural lands have risk. Majority of the agriculture area are exposed to medium risk level, 72.7% and 85.9% for peri-urban and urban area, which are around 195.2km² and 40.8km² respectively.



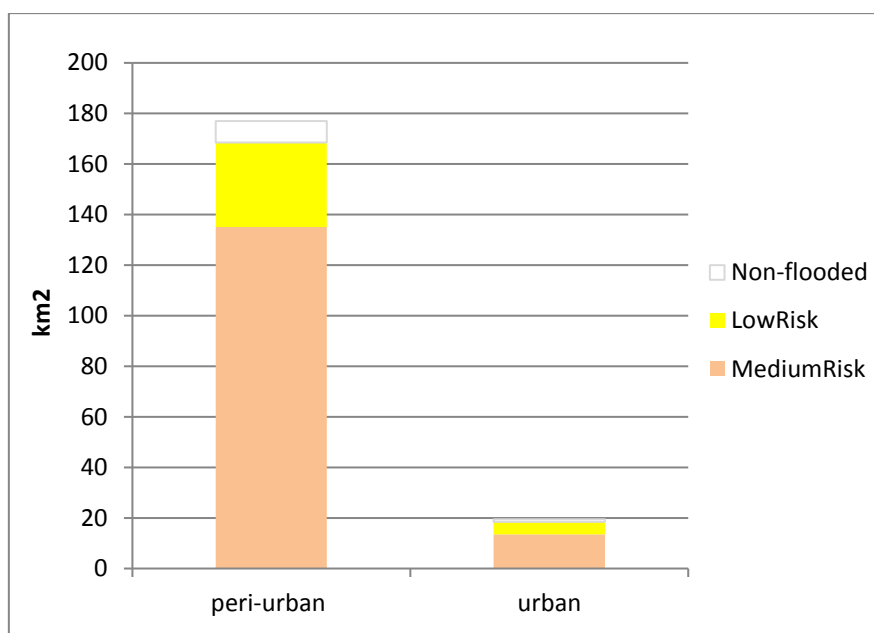


Figure 45: Flood risk for agriculture land for peri-urban (a) and core urban area (b) in area proportion % and area km2 (c).

Regarding core urban region, the high risk level is found in residential area as the damage level to economy and society is high. 10.8% (8.4 km²) and 35.2% (8.7km²) of high density continuous and high density discontinuous residential area experience floods during the past decades, 48.7% (8.3km²) of low density discontinuous residential area had floods in the past decades. Majority of the area have a risk level of medium (Figure 46 (a) and (b)). The medium risk areas are 9.1% (7.1km²) for continuous residential area and 35.2% (6.9km²) for discontinuous residential area, whereas for low density residential area, it grows up to 48.7% (8.3km²). Only in high density continuous residential area has 1.7% (1.3km²) continuous residential area is high risky.

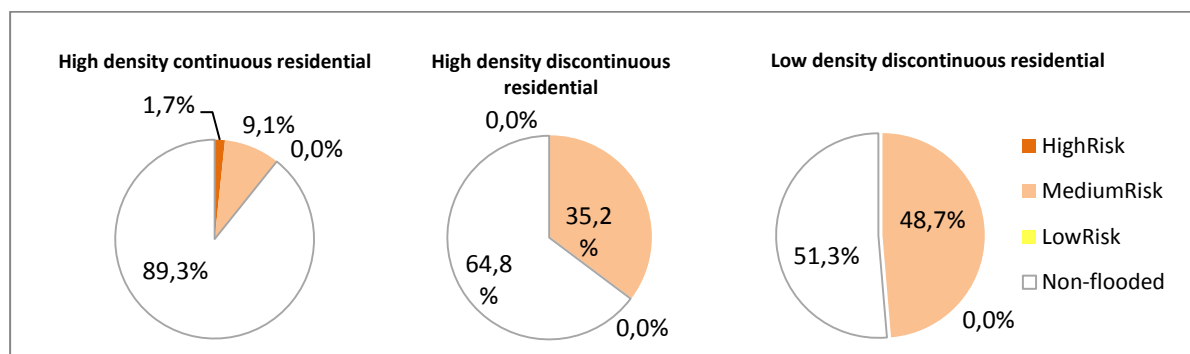


Figure 46: Flood risk level for high density continuous and discontinuous residential areas (left and middle) and low density residential area (right) in area proportion %.

Figure 47 then shows absolute figures (in km²) for areas under particular flood risk level risk for the major land use classes within urban area. As seen majorities are non-flooded or have a small proportion exposed to medium or low risk level. Village settlement, however, is the opposite, more than half of village settlement area is at medium flood risk level.

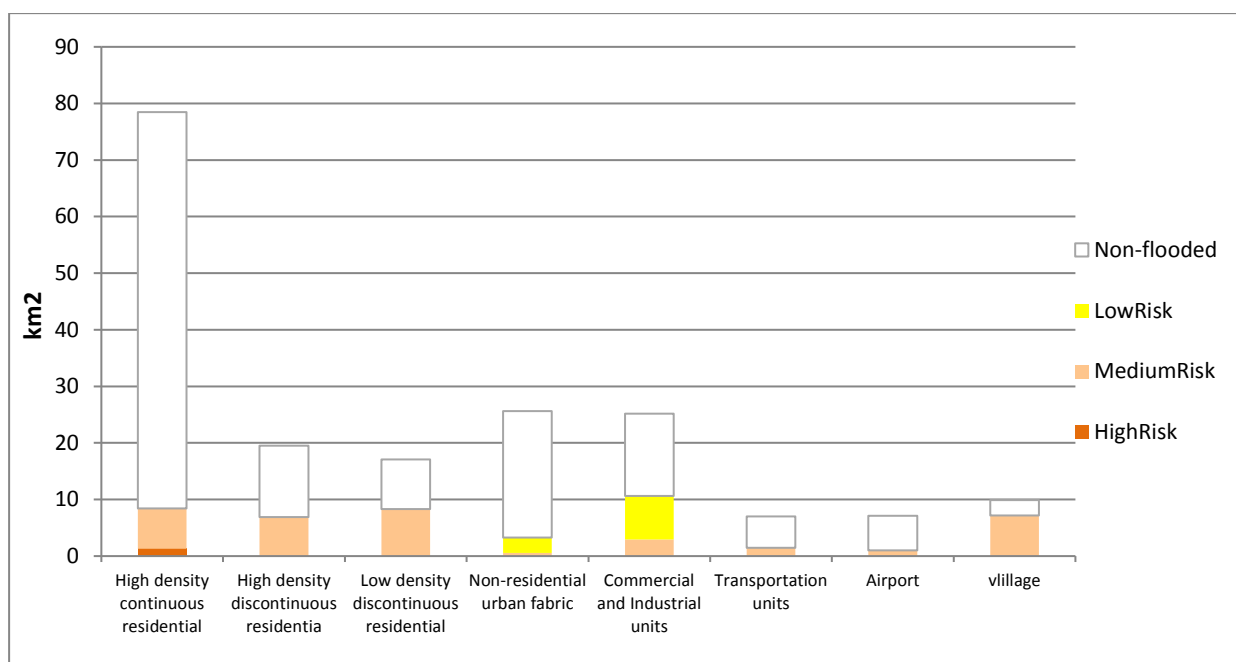


Figure 47: Flood risk levels for different land use classes within urban region are demonstrated in km².

Annex 1 – Processing Methods for EO4SD-Urban Products

Summary of Processing Methods

Global Urban Products – Settlement Extent and Imperviousness

Global Settlement Extent Product Description

The 2014-2015 settlement extent product is generated by properly exploiting multitemporal Sentinel-1 (S1) radar and Landsat-8 optical data intersecting the investigated Area of Interest (AOI) in the period 2014-2015. Specifically, the rationale of the adopted methodology is that given a series of satellite images for the AOI, the temporal dynamics of human settlements are sensibly different than those of all other land-cover classes (e.g., vegetated and cultivated areas are prone to multiple changes over 1/2-year timeframe, whereas this generally does not occur for built-up structures). Thus global Settlement Extent product can be effectively generated in fully-automated way using rich high resolution temporal (both radar and optical) image stack as outlined in following paragraphs. There is a little refinement for a particular area, but the Global Urban Products are focused on first cost-effective global screening and benchmarking due to high data comparability and availability globally. Identified hot-spots identified using the Global Urban Products can be then further explored and analysed using other more detailed and tailored products from the EO4SD portfolio.

As regards radar data, Ground Range Detected S1 scenes acquired at high resolution in Interferometric Wide Swath Mode (IW GRDH) are used. Each scene is pre-processed by means of the SNAP software available from ESA; specifically, this task includes: orbit correction, thermal noise removal, radiometric calibration, Range-Doppler terrain correction and conversion to dB values. Scenes acquired with ascending and descending pass are processed separately due to the strong influence of the viewing angle in the backscattering of built-up areas. Moreover, the joint employment of VV/VH imagery did not provide any sensible improvement with respect to the solely use of VV data; accordingly, VH data are excluded. As a means for characterizing the behavior over time, the backscattering temporal maximum, minimum, mean, standard deviation and mean slope is derived for each pixel. Texture information is also extracted to ease the identification of lower-density residential areas; in particular, the coefficient of variation (COV) of the temporal mean backscattering is computed, which is defined for each pixel as the ratio between the local standard deviation and the local mean calculated over a $N \times N$ spatial neighborhood. In the light of the 10m spatial resolution of the considered S1 data, a neighborhood of 5x5 pixels proved to be an effective choice.

Concerning optical data, only Landsat-8 scenes with cloud cover lower than 60% are taken into consideration (indeed, further raising this threshold often results in accounting for images with non-negligible misregistration error). Data are calibrated and atmospherically corrected using the LEDAPS tool available from USGS and the CFMASK software is applied for removing pixels affected by cloud-cover and cloud-shadow. Next, a series of 6 spectral indices suitable for an effective delineation of settlements (identified through extensive experimental analysis) are extracted; these include – among others – the Normalized Difference Built-Up Index (NDBI), the Modified Normalized Difference Water Index (MNDWI) and the Normalized Difference Vegetation Index (NDVI). For all of them, the same set of 5 key temporal statistics used in the case of S1 data are generated for each pixel in the AOI. Moreover, to improve the detection of suburban areas, for each of the 6 temporal mean indices also the corresponding COV is computed in a neighborhood of 3x3 pixels, which proved being the most effective choice in the light of the 30m spatial resolution of Landsat data. For matching the spatial resolution of Sentinel data, the whole stack of Landsat-based features is finally resampled to 10m spatial resolution.

To identify reliable training points for the settlement and non-settlement class, a strategy has been designed which jointly exploits the temporal statistics computed for both S1 and Landsat data, along with additional ancillary information. In the case of optical data, in general the most of settlement pixels can be effectively outlined by properly jointly thresholding the corresponding NDBI, NDVI, and MNDWI temporal mean; likewise, this holds also for non-settlement pixels. Nevertheless, being all 3 spectral indices affected by the presence of vegetation, absolute threshold values are not universally effective since vegetation strongly varies depending on climate. To overcome this drawback, by

accounting for the well-established updated Köppen Geiger climate classification, for each zone specific thresholds have been determined for outlining both candidate settlement and non-settlement training samples. Furthermore – in the reasonable hypothesis that the higher is the number of cloud/cloud-shadow free acquisitions, the more robust are the corresponding temporal statistics –all pixels whose number of Landsat-8 clear-sky acquisitions is lower than 5 are excluded.

Regarding radar data, it generally occurs that the temporal mean backscattering of most settlement samples is sensibly higher than that of all other non-settlement classes. Accordingly, samples whose temporal mean backscattering (either in the case of data acquired in ascending and descending pass) computed from more than 4 scenes is: i) lower than -7 dB are not eligible to be labelled as settlement training samples; and ii) greater than -11 dB are not eligible to be labelled as non-settlement training samples. Finally, it is worth noting that in complex topography regions: i) radar data show high backscattering comparable to that of urban areas; and ii) bare rocks are present, which often exhibit a behaviour similar to that of settlements in the Landsat-based temporal statistics. Accordingly, to exclude these from the analysis, all pixels are masked whose slope - computed based on SRTM 30m DEM for latitudes between -60° and +60° and the ASTER DEM elsewhere - is higher than 10 degrees.

Support Vector Machines (SVM) with Radial Basis Function (RBF) Gaussian Kernel are used in the classification process. However, as the criteria defined above for outlining training samples might results in a high number of candidate points, for AOIs up to the a size of ~10000 km² the most effective choice proved extracting 1000 samples for both the settlement and non-settlement class. However, since results might vary depending on the specific selected training points, as a means for further improving the final performances and obtain more robust classification maps, 20 different training sets are randomly generated and given as input to an ensemble of as many SVM classifiers. Then, a majority voting is applied and each pixel is finally associated with the settlement class only in the case it is labeled as settlement in at least 11 over 20 of them.

It worth noting that the stacks of Landsat-8-based and S1-based temporal features are classified separately as this proved more effective than performing a single classification on their merger.

In both cases, a grid search with a 5-fold cross validation approach is employed to identify for each training set the optimal values for the learning. The values resulting in the highest cross-validation overall accuracy is selected and used for classifying the corresponding AOI. In particular, this is carried out by employing the largely employed open source C++ library libSVM.

A final post-classification phase is dedicated to properly combining the Landsat- and S1-based classification maps and automatically identifying and deleting potential false alarms. To this purpose, an updated version of the post-editing object-based approach adopted in the production of the GUF2012 has been used. Specifically, it consists of two phases. First, segmentation is performed for categorizing each cluster of connected pixels in the two classification maps as individual image objects; then, a ruleset is employed for selecting whether: i) to combine the S1-based and the Landsat-based objects; ii) to keep just one; or iii) to discard both of them. For instance, objects are excluded if their shape is most likely not compatible with real settlements (generally roads falsely detected in the Landsat-based classification) or if they exhibit high average temporal maximum NDVI (i.e., mostly the case of false detection in the S1-based classification in correspondence of specific types of dense forests). The final classification map is given by the merger of the objects preserved in the Landsat- and S1-based classification maps.

The above-described methodology has been further adapted for outlining the settlement extent in the past solely based on Landsat-5/7 imagery available since 1984; indeed, no long-term SAR data archive at comparable spatial resolution is freely accessible for the same timeframe (ESA ERS-1/2 data are available from 1991 without systematic world coverage and often proved too complicated to pre-process). In particular, for the given target period and AOI, all available Landsat imagery with cloud cover lower than 60% is pre-processed in the same fashion as described in the previous paragraphs and the same set of temporal statistics and texture features are extracted. Based on the hypothesis that settlement growth occurred over time (meaning that a pixel cannot be marked as settlement at an earlier time if it has been defined as non-settlement at a later time), all pixels categorized as non-settlement in the 2014-2015 extent map are excluded from the analysis. Then, training samples are derived by thresholding the temporal mean NDBI, MNDWI and NDVI; specifically, a dedicated strategy has been

implemented for automatically determining the thresholds for the 3 indices by comparing their cumulative distribution function (CDF) for the target period with that exhibited for the period 2014-2015. Also in this case, an ensemble of 20 SVMs is used, each one trained on a different subset of 2000 samples (i.e., 1000 for the settlement and 1000 for the non-settlement class) and majority voting is then employed for generating the final map. It is worth noting that, when deriving the past settlement extent for multiple times (e.g., 1999-2000, 2009-2010), both the masking and threshold adaptation are performed on the basis of the results derived for the next target period (e.g., in the case of 2 target periods 2004-2005 and 2009-2010, first the 2009-2010 settlement extent is derived by exploiting the 2014-2015 settlement extent and its corresponding temporal statistics; then the 2004-2005 settlement extent is derived by exploiting the 2009-2010 settlement extent and its corresponding temporal statistics and so on).

Global Imperviousness Product Description

Urban growth is associated not only to the construction of new buildings, but – more in general – to a consistent increase of all the impervious surfaces (hence also including roads, parking lots, squares, pavements or railroads), which do not allow water to penetrate, forcing it to run off. To effectively map the extent of all such areas is then of high importance being it related to the risk of urban floods, the urban heat island phenomenon as well as the reduction of ecological productivity. Moreover, monitoring the change in the PIS over time is of great support for understanding, together with information about the temporal evolution of the extent of urban areas, also more details about the type of urbanization occurred (e.g., if areas with sparse buildings have been replaced by highly impervious densely built-up areas or vice-versa).

To this purpose an imperviousness product is generated by properly exploiting Landsat-4/5/7/8 multitemporal imagery acquired over the study area within a given time interval of interest in which no relevant changes are expected to occur (typically a time period of 1-2 years allows to obtain very accurate results). For all the considered scenes, cloud masking and, optionally, atmospheric correction are performed. Next, the normalized different vegetation index (NDVI) is extracted for each image. Here, being it inversely correlated with the amount of impervious areas (i.e., the higher the NDVI, the higher the expected presence of vegetation, hence the lower the corresponding imperviousness) the core idea is to compute per each pixel its temporal maximum which depicts the status at the peak of the phenological cycle. It is worth noting that for different pixels in the study area, different number of scenes might be available. However, in the hypothesis of a sufficient minimum number of acquisitions for computing consistent statistics, this does not represent an issue. Moreover, in this framework it is also possible to obtain spatially consistent datasets to be employed for the desired analyses even when investigating large areas. Areas associated with impervious surfaces are then extracted at high spatial resolution [e.g., by photointerpretation of VHR imagery (e.g., from Google Earth), the analysis of OpenStreetMap layers or information derived from in-situ campaigns] in various parts of the study region and then rasterized and aggregated at the Landsat 30m spatial resolution. A support vector regression SVR module is then employed for properly correlating the resulting training information with the temporal maximum NDVI to finally derive the PIS for the entire area of interest.

Summary of Processing Methods

Urban and Peri-Urban Land Use/Land Cover and Change

The input data include Very High Resolution (VHR) and High Resolution (HR) satellite imagery from different sensors acquired at different times for covering both Core Urban and Peri-Urban Areas. The data are pre-processed to ensure a high level of geometric and radiometric quality (ortho-rectification, radiometric calibration, pan-sharpening).

Core Urban Area of AOI is covered by VHR satellite imagery to ensure the ability to extract LULC information within the artificial areas at Level 3-4 of the nomenclature. The complexity when dealing with VHR images comes from the internal variability of the information for a single land-use. For instance, urban area is represented by a high number of heterogeneous pixel values hampering the use of automated pixel-based classification techniques.

Nevertheless, it is possible from VHR images to identify textures (or pattern) inside an entity such as an agricultural parcel or an urban lot. In other words, whereas pixel-based techniques focus on the local information of each single pixel (including intensity / DN value), texture analysis provides global information in a group of neighbouring pixels (including distribution of a group intensity / DN values but also spatial arrangement of these values). Texture and spectral information are combined with a segmentation algorithm in an Object Based Image Analysis (OBIA) approach to reach a high degree of automation for most of the peri-urban rural classes. However, within urban areas, land use information is often difficult to extract from the imagery alone and ancillary/in situ data needs to be used. The heterogeneity and format of these data mean that another information extraction method based on Computer Aided Photo-Interpretation techniques (CAPI) needs to be used to fully characterise the LULC classes in urban areas. Therefore, a mix of automated (OBIA) and CAPI are used to optimise the cost/quality ratio to extract the LULC/LULCC information from VHR EO data.

Although LULC information at the most detailed level (3-4) within artificial areas is extracted by means of visual interpretation, it is also relevant to mention that the distinction between the different classes defined for residential urban fabric units is performed using an automated approach based on the additional Imperviousness product (dedicated section about the processing method in this annex). Indeed, the average value of Imperviousness Degree (IMD) or Soil Sealing (SL) per urban fabric unit well characterizes the density of built-up area and allows to associate each residential unit to the relevant classes well defined in the LULC nomenclature.

Peri-Urban Area of AOI is covered by HR satellite imagery only. As Level-1 nomenclature is applied, it is quite easy to get the LULC information using automated pixel-based classification algorithm outside the urban mask extracted by visual interpretation beforehand.

Manual enhancement is the final post-processing step of the production framework. It aims to validate the detected thematic classes and refine the geometry of polygon features if necessary, especially for ensuring that the correct MMU has been respected. Finally, a thorough completeness and logical consistency check is applied to ensure the topological integrity and coherence of the product which is delivered in vector format. That makes easier its integration in GIS and for subsequent analysis.

Change detection - Four important aspects must be considered to monitor land use/land cover change effectively with remote sensing images:

(1) detecting that changes have occurred, (2) identifying the nature of the change, (3) characterising the surface extent of the change and (4) assessing the spatial pattern of the change.

Apart from visual comparison, the change detection layer can be based on automated image-to-image classification approach given that the same sensor is used for both reference dates. An original and efficient image processing chain is promoted consisting in the comparison of images acquired at two different dates providing multi-labelled changes. The approach mainly relies on texture analysis, which has the benefits to deal easily with heterogeneous data and VHR images.

Summary of Processing Methods

Urban Extent and Change

For each reference date, the product is directly and easily derived from the baseline LULC dataset by extracting and merging all polygon features from the classes related to artificial areas (1xxxx).

Change information layer is basically derived from the geometric intersection of the historic and current Urban Extent layers by means of GIS operation. This resulting product finally provides information about permanent artificial areas, converted ones to another land use category and new ones. Quality control and accuracy assessment tasks have been performed through the LULC product generation.

Summary of Processing Methods

Urban Imperviousness

The Imperviousness product is intended to represent the impervious surfaces because of urban development, layers of completely or partly impermeable artificial material (asphalt, concrete, etc.) and infrastructure construction. Therefore, the Imperviousness Degree (IMD) or Soil Sealing (SL) information can be produced relatively easily based on the Urban Extent derived from the baseline LULC information product and the linear model between impervious areas and vegetation presence that can be determined and characterized from Landsat or Sentinel-2 NDVI time series.

More precisely, the raster product is generated at 10m - 30m spatial resolution by properly exploiting Landsat-4/5/7/8 or Sentinel-2 multitemporal imagery acquired over the study area within a given time interval of interest in which no relevant changes are expected to occur (typically a time period of 1-2 years allows to get very accurate results). Each acquired EO data is pre-processed (ortho-rectification, radiometric calibration, pansharpener, cloud-masking). Then, the Normalized Difference Vegetation Index (NDVI) is extracted for each image within the urban mask (corresponding to Urban Extent product). NDVI is inversely correlated with the amount of impervious areas, i.e. the higher the NDVI is, the higher the expected presence of vegetation, hence the lower the corresponding imperviousness degree. The core idea is to compute per each pixel its temporal maximum which depicts the status at the peak of the phenological cycle.

It is worth noting that for different pixels in the study area, different number of scenes might be available. However, in the hypothesis of sufficient minimum number of acquisitions unavailable for computing consistent statistics, this does not represent an issue. Indeed, in this framework, it is also possible to get spatially consistent datasets useful for the desired analyses, even when investigating large territories. Areas associated with different levels of impervious surfaces are then extracted by visual interpretation from data sources with higher spatial resolution (e.g. VHR imagery, Google Earth imagery). OSM layers or information derived from in-situ campaigns are other auxiliary data sources which can also be used for this purpose. At the end, reference data are extracted in various parts of the study region and then rasterized and aggregated at the spatial resolution of input EO data.

A support vector regression SVR module is then used for properly correlating the resulting training information with the temporal maximum NDVI to finally derive the Percentage of Impervious Surface (PIS) or Imperviousness Degree (IMD) for the entire AOI. Specifically, 8bit integer values from the raster product range from 0 (no impervious surface in the given pixel) to 100 (completely impervious surface in the given pixel).

Processing Methods

Informal Settlements and Change

Generic methodology

Informal Settlements Extent and Change product describe spatial location, extent, typology and development of building agglomerations that can be classified as informal settlement/slum areas.

According to widely used definition of UN-HABITAT, informal settlements (sometimes referred as slums) are characterized by sub-standard housing and lacking tenure security. They are supposed to be attributed to (any) combination of various features including inadequate access to safe water, sanitation and other basic infrastructure including public services, poor structural quality of housing and design arrangements, overcrowding and insecure residential status. According to World Bank slums are generally considered to be a residential units in the cities where the poor are concentrated in substandard conditions. Despite plenty of definitions of informal settlements or slums, there is no standard slum specification or typology harmonized globally across different cities. Slums differ in their both de facto and de jure characteristics from country to country, from city to city and in terms of their physical characteristics they may substantially differ even within single city.

Methodology of delineation and assessment of informal settlements with remote sensing focuses on identification of physical characteristics related to typical morphological features (Kuffer et al, 2016), mostly using very high resolution optical satellite images. Other characteristics are not discernible from imagery, however there may exist links between some non-physical and physical characteristics. As a consequence, output product derived from remote sensing shall be supposed as a map of “probable informal settlements” localising areas with visible physical characteristics typical to slums in the area. In order to translate the product into “confirmed informal settlements” with high confidence, local expert knowledge engagement and field validation are needed.

For deriving of informal settlements neighbourhoods pan-sharpened imagery with best available resolution and lowest inclination angles are needed. Semi-automatic and CAPI visual interpretation consider following visible features:

- Dwelling size and geometry – typically very small, coupled locally determined distinct shapes, low building height (mostly)
- Roofing materials - heterogeneous
- Pattern – organic structural layout of dwellings in the neighbourhood, irregular (dirt) road pattern, narrow roads and no building set-backs
- Density – very high roof coverage and soil sealing, small amount of green and open spaces
- Site – pockets within formal residential fabric or on hazardous locations (prone to floods, slides, proximity to industry, heavy traffic lines etc), lack of access roads

Furthermore, using archive very high resolution images, current and historic extent of informal settlements is compared to identify their temporal development – extent growth or reduction.

Resulting product emphasises physical patterns and key characteristics of informal settlements such as growth (in terms of location and area) and type. Multiple indicator metrics based on above features related to physical characteristics can be derived – e.g. spatial metrics such as compactness, patch size and density of buildings, construction material, spatial patterns and location-based indicators such as distance to selected features or accessibility. Furthermore, spatial context of the settlement with respect to modelled hazard or risks, which are derived by classification/interpretation of EO imagery or spatial analysis of additional geospatial products, could be also considered. Individual metrics attributed to detected informal settlement patches or their combination provide means to tailor custom typology.

Specific methodology applied in Dhaka

As indicated in generic methodology, multiple variables can be extracted from EO imagery or by GIS analysis for each detected slum polygon. Conceptual framework and description of each variable is presented in separate Annex 3 (attached to SOR). Most of variables could be derived automatically from baseline VHR imagery by means of object based image analysis or GIS geoprocessing.

Please note that having wide set of variables different typological nomenclatures might be defined and implemented.

- 1) Pocket slum – located in Saddar district && size < 0.25 ha && fully surrounded by urban fabric (geometry completely within respective LULC polygons)
- 2) Linear along rivers: distance to river <10m && linear shape
- 3) Linear along railways: distance to railway < 10m && linear shape
- 4) Low density slum: located in Saddar district && sealing > 0.099 && sealing <= 0.799
- 5) High density: located in Saddar district && sealing > 0.799
- 6) Under Trees: greenness > 0.699

Summary of Processing Methods

Urban Green Areas and Change

Generic methodology

Urban green areas refer to land within and on the edges of a city that is partly or completely covered with grass, trees, shrubs, or other vegetation. This includes public parks, private gardens, cemeteries, forested areas as well as trees, river alignments, hedges etc. The product delivered within EO4SD-Urban project thus provides accurate information (1 m resolution) on the spatial location and extent of the green areas located within the Urban Extent derived from the baseline LULC information product.

Detecting and monitoring urban green coverage needs VHR EO data, which explains the product generation over the Core Urban Area of AOI only. The same images have been logically used as for generating the LULC information product. Consequently, the usual preliminary quality check and pre-processing tasks were already implemented.

Urban Green Areas have been detected using automated non-supervised classification method. More precisely, each single multispectral VHR scene has been classified by specifying the most appropriate algorithm and class number. Then, pixel units from the classes considered as representing green areas have been combined into 1 single class. From this operation results the required binary raster product. At this stage, it only remains necessary to apply some post-processing steps, such as eliminating the pixel groups under the MMU, merging the products resulting from the classification process of each single scene, but also integrating the information provided by the LULC product (class 14100 - *Urban Greenery*). In case of reasonable data volume, the raster layer can be converted in vector format and smoothing algorithm can be applied before delivery to users.

This processing method is applied for each reference date for which the product is required. Then, change information layer is basically derived from the geometric intersection of the historic and current Urban Green Areas layers by means of GIS operation. This resulting product finally provides information about permanent vegetation, loss of urban green areas and new ones. Quality control and accuracy assessment tasks are performed by means of visual interpretation and considering the LULC dataset.

Specific methodology applied in Dhaka

In Dhaka, extended typology for Urban Green areas was requested. While LULC at level 3 provides only delineation of urban green, extended typology aims to stratify areas according to tailored categorization addressing urban planning and management needs. Such typology includes also concept of open spaces. Most of classes are derived within geometric outline of original LULC classes, but not exclusively due to different application of land “use” concept in some cases (e.g. square may lay over multiple land cover / land use classes).

Nomenclature implementing the typology which is derivable from EO or ancillary data is detailed in the following table.

Table 1-1: Open and Green areas nomenclature

Class	Categorization	Included land use types	Not included
Squares	Central square	CS	
	Other (neighbourhood) square	NS	
Open sky markets		CS, NS, large streets with concentration of stalls	
Living streets		Roads surrounded predominantly by urban residential and non-residential fabric	Capacity roads Highways Overpasses Underpasses Road links
Waterfronts	Beach	Beach	

	Waterfront park	Green waterfront areas. Could include amusement parks	Mangrove forests
	Sealed waterfront surfaces	Berthing and boardwalk areas	Industrial areas, port areas except berthing places for "public" river transport
Green urban areas	Central park / Downtown park	Large parks in central and down-town areas. Horticultural management is apparent	
	Neighbourhood park	Smaller parks in neighbourhood areas. Horticulture management is visible or likely	
	Mini pocket park	Small green area surrounded by buildings (originally not classified as green urban area in LULC product). Only where high vegetation (trees)	
	City woodland	Forest within in city outline. No horticulture management apparent	
	Sub-urban forest	Forest in peri-urban area. No horticultural management is apparent	
	Green alley	Elongated linear forest structures in the streets (of boulevard type)	
	<i>Sport & leisure facilities</i>	Green sport areas incl. Buildings and associated land, playgrounds, amusement parks, golf courses etc	

Methodology for deriving extended typology depends on utility of both VHR optical satellite data and ancillary data from free sources – Open Street Map.

Green areas

Inputs: VHR Imagery (pan-sharpened multispectral), LULC product , Open Street Map (OSM)

Green areas polygons from LULC product are taken as candidate polygons. Their geometry is revised by interpretation of satellite image and assuming geometry of respective classes from OSM. Polygons are further supplemented with small green area spot polygons which were originally derived by automated approach but fell below the MMU threshold applicable for LULC product and now are above custom MMU applicable for green areas product.

Soil Sealing, Greenness, openness and size attributes are automatically calculated at a LULC block level to supplement decision tree implemented in later stage.

Polygons are classified according to extended typology definition by expert interpretation. Information visible in satellite imagery is considered as well as information from OSM. Public areas typically cannot be distinguished from non-public with confidence as this information is not always obvious from the imagery.

Living streets and connectivity

Inputs: LULC product, Transportation network product (based on OSM)

Living streets are identified by geoprocessing operations in GIS environment on top of road network line layer and LULC layer. Layer comes either directly from OSM or could be further adapted by visual interpretation (adding missing segments, re-classification, removing of surplus segments) before geoprocessing. Living streets have been defined streets with a potential for being "public spaces" and space for economic activities. Applicable GIS definition followed a concept of identification of street segments that may fulfil the above generic conditions. As such following attributes have been considered in the workflow:

- Roads were split into segments between junctions (crossings)
- Segments out of urban area (based on LULC mask) were rejected
- Segments being (1st level) capacity roads (highways, motorways) were rejected
- Segments being (2nd level) capacity roads may be optionally rejected (they need to be first identified by interpretation assuming:
 - o Visible presence of huge traffic on wide (multi-lane) roads
 - o Mostly recruiting from Primary / Secondary roads
- Remaining segments are filtered out depending on condition of being surrounded by residential and non-residential urban fabric blocks from LULC).

Resulting segments shall be considered as *potential* living streets.

Depending on user definition, road segments may be aggregated into capacity and non-capacity, with capacity roads having higher potential for being served by public or other transportation services. In Karachi, capacity roads have been selected as highways, primary and secondary roads according to OSM typology. Transport connectivity nodes are generated by network analysis at junctions of aggregated capacity segments.

Further network, distance, connectivity or risk indicators could be derived similar to extended slum product to any set of points or polygons (green areas, open spaces, public buildings etc).

Open spaces

Inputs: VHR Imagery , OSM, Living streets product

Open spaces such as squares and waterfronts are further delineated by visual interpretation of VHR imagery. The condition of accessibility to public mostly cannot be interpreted from satellite imagery.

Public buildings and institutions can be derived by subsetting of related OSM layer. However, spatial and thematic consistency cannot be assured and fully depends on quality of OSM data in the area of interest.

Table 1-2: Detail inside into population distribution within each Open Green Space

Open Green Space ID	Population		
1	4296	43	9031
2	7184	44	14312
3	18791	45	11570
4	7763	46	83510
5	7411	47	30666
6	14439	48	29857
7	12651	49	12631
8	9325	50	12928
9	4488	51	13043
10	17685	52	37523
11	10307	53	35960
12	6745	54	44649
13	7138	55	50213
14	9569	56	3162
15	17190	57	4051
16	18593	58	21669
17	19562	59	20312
18	7252	60	24296
19	21938	61	23923
20	8482	62	18986
21	9234	63	3021
22	7072	64	2728
23	4119	65	38427
24	5230	66	56063
25	3686	67	31769
26	5812	68	48626
27	5079	69	47747
28	8728	70	49245
29	7555	71	50906
30	7924	72	19281
31	6605	73	25190
32	9020	74	44756
33	14457	75	39507
34	9688	76	30952
35	5620	77	86826
36	23800	78	52871
37	5349	79	48423
38	5250	80	35119
39	12396	81	80703
40	4515	82	48615
41	2743	83	64803
42	2452	84	58112
		85	48667
		86	7890

87	55933	132	25503
88	3259	133	61253
89	52104	134	4530
90	33199	135	32337
91	9159	136	27136
92	18948	137	10648
93	11833	138	41463
94	49303	139	43073
95	25526	140	7782
96	81216	141	5983
97	37480	142	23921
98	47548	143	154212
99	71399	144	132443
100	136272	145	116622
101	39351	146	62778
102	31709	147	83792
103	31435	148	54386
104	24412	149	46707
105	30448	150	44645
106	11981	151	61652
107	12020	152	3621
108	6311	153	10521
109	3296	154	36386
110	3396	155	36497
111	3476	156	102586
112	4264	157	12408
113	86047	158	34530
114	58063	159	30878
115	71465	160	4547
116	82711	161	45409
117	20124	162	10433
118	10756	163	11166
119	11024	164	43360
120	55597	165	16378
121	102323	166	89859
122	55080	167	8451
123	61571	168	9362
124	72818	169	36354
125	76165	170	1155
126	156254	171	4261
127	7068	172	3462
128	12015	173	1477
129	21332	174	3807
130	20877	175	12338
131	18802	176	31608

177	73552
178	83340
179	30127
180	6781
181	38568
182	3382
183	8122
184	52461
185	3274
186	2302
187	3415
188	38591
189	32073
190	34694
191	32418
192	49093
193	29068
194	38575
195	67309
196	15314
197	11215
198	60271
199	37562
200	58270
201	56620
202	44479
203	10403
204	36363
205	13167
206	14799
207	6338
208	83319
209	3985
210	3808
211	52930
212	39406
213	32986
214	46670
215	34966
216	9733
217	38758
218	3558
219	43353
220	36310
221	41961

222	54131
223	49980
224	56617
225	3108
226	2520
227	3312
228	3699
229	38691
230	2841
231	3256
232	904
233	60343
234	63744
235	68621
236	8489
237	26801
238	55314
239	31020
240	38674
241	19869
242	24698
243	5282
244	4423
245	52709
246	38530
247	47640
248	71863
249	49884
250	79051
251	67748
252	68868
253	39603
254	16173
255	20346
256	54512
257	81871
258	44472
259	67311
260	81451
261	6932
262	35699
263	30646
264	28213
265	32603
266	36585

267	45659	312	17120
268	18177	313	3771
269	7934	314	10884
270	27238	315	7507
271	20620	316	9713
272	19758	317	16678
273	3688	318	17609
274	36109	319	958
275	29413	320	1091
276	25683	321	30687
277	3431	322	55997
278	7167	323	39196
279	6520	324	43029
280	8607	325	27957
281	22618	326	36829
282	75176	327	42214
283	49431	328	41028
284	9305	329	33184
285	42139	330	33660
286	8661	331	41251
287	10658	332	56110
288	61626	333	37597
289	10937	334	43107
290	11041	335	41636
291	9169	336	43268
292	18565	337	21790
293	1412	338	41320
294	726	339	35941
295	5942	340	49151
296	1582	341	67298
297	764	342	45755
298	35657	343	49078
299	48404	344	63022
300	55368	345	54632
301	44589	346	15796
302	45416	347	5652
303	9200	348	5234
304	13203	349	54826
305	6180	350	6681
306	13290	351	40159
307	33555	352	6551
308	5090	353	17818
309	26730	354	47349
310	19223	355	6345
311	38238	356	60333

357	64586
358	48750
359	8896
360	53256
361	59457
362	50064
363	52190
364	63924
365	70779
366	57872
367	46904
368	3563
369	20565
370	17419
371	35657
372	59705
373	45104
374	18857
375	50149
376	58059
377	61444
378	58781
379	45343
380	57230
381	56556
382	57855
383	59182
384	66604
385	65368
386	56433
387	67171
388	53496
389	59665
390	53463
391	13769
392	106670
393	69977
394	73704
395	30867
396	10871
397	99125
398	31619
399	85436
400	61833
401	78990

402	4096
403	18748
404	23399
405	36461
406	35938
407	46433
408	41592
409	62404
410	69946
411	56777
412	46198
413	43218
414	41955
415	47031
416	73818
417	159111
418	42158
419	41089
420	21784
421	41320
422	20181
423	53004
424	21899
425	29471
426	34299
427	22143
428	37452
429	1454
430	7420
431	41306
432	44887
433	36210
434	29463
435	18176
436	58762
437	33138
438	32143
439	48127
440	72138
441	55971
442	5633
443	10240
444	34170
445	36876
446	30932

447	18386
448	16924
449	33409
450	33156
451	29774
452	48301
453	37070
454	69368
455	21611
456	78970
457	65868

Summary of Processing Methods

Flood History and Risk

Products based on flood history monitoring or modelling provide information about spatial location and intensity of flood hazard. Thus, they represent key input into identification of exposed assets, infrastructure and population and assessment of flood-related risk. Hazard, exposure and risk levels are estimated through numeric indexes for describing the severity of phenomena. The service consists in the provision of flood-related geospatial information and is composed of three products which are all delivered in vector format:

- Product 1: Flood Extent (inventory)
- Product 2: Flood Hazard (susceptibility)
- Product 3: Flood Risk (considering vulnerability)

The flood events retrieval was based on literature review and social media release. The major platforms used for this purpose are:

- <http://floodobservatory.colorado.edu/Archives/index.html>
- Copernicus Flood list: <http://floodlist.com/tag/bangladesh>

Available open datasets, including Sentinel-1/2 imagery, Landsat-5/7/8 imagery, ENVISAT ASAR, MODIS daily reflectance product MOD09GA and SRTM Digital Elevation Model (DEM) data, are considered and if matching with flood event chronology, selected and downloaded to generate the required products. In Dhaka, the use of ENVISAT data has been investigated, but either no data or proper data mode is available during the flood periods. The downloaded data cover the entire time span of each flood event and even 1- 2 months longer in some cases to solve the issue caused by the presence of clouds and shadows. All selected HR EO data were visually checked to be assured that their quality fits well for the processing steps to be performed

Product 1: Flood Extent

Flood extent products are derived to show the flooded area at the peak of historical event. This type of product has thus been generated for each event for which HR optical data was available. It is worth to remind that 2017 HR imagery was only used as reference data for QC.

Flood extent is derived from selected historical HR satellite imagery using automated classification approach. Normalized Difference Water Index (NDWI) and Modified Normalized Difference Water Index (MNDWI) were obtained from the imagery subject to the availability of spectral bands in the different scenes. A threshold method was used to derive water extent. The two indices are proved to be the most effective water extent extraction approach by many researches, and show good performance in this case. During the application of the threshold method, some scattered small water extent pixels were considered as false flooded area and were filtered out (based on size). Reference water course and bodies were extracted from the baseline LULC information layer. The flood extent was finally derived by subtracting the reference water bodies from the water extent automatically extracted.

For this product 1, STRM DEM (30m) and 2017 cloud-free reference images (Landsat-8 & VHR) were used for the validation purpose. For the area where clouds were present on HR images, flood extent was manually adjusted based on the previous flood event extents and local DEM variations together.

Product 2: Flood Hazard

Flood hazard intensity over Area of Interest relies on the occurrence of flood events during the past 10 years, optionally combined with the remarkable historical flood events that occurred in the year 2004 and 2014. The flood hazard mapping product aims at giving accurate indication about the flood presence in terms of both frequency and extent in the city, and illustrates which area is in general flooded more often than others.

The flood extent (Product 1) resulting layers from the different study years are easily stacked together to generate the Flood Hazard product. For every area feature, the occurrence of flood events within the

last decade were counted and assigned to *Hazard Code* attribute field. 0 is the value assigned to areas corresponding to reference water bodies, while -999 is the one indicated for the areas only affected by 2004/14 significant historical event (to be considered as additional indicator).

Product 3: Flood Risk

Flood Risk product considers both the flood hazard intensity level (susceptibility) and the exposure level related to population and assets regarding such natural event (vulnerability). Flood Hazard product only shows the potential extent and frequency of flood events based on a series of significant historical ones, demonstrating the importance of this type of natural hazard. By introducing the damage cost on the different land use categories as exposure indicator, the actual impact of flood hazard on human society through different aspects is also taken into account. Consequently, Flood Risk product carries two-fold meaning in both perspectives.

Flood Hazard level

Flood Hazard intensity level is calculated based on the frequency of the flood occurrences in the past 10 years (see Product 2) which is further classified into 3 levels with *Hazard Code* as described in the table below.

Table 1-3: Flood Hazard level classification

Flood Hazard Code	Frequency of Occurrence	Flood hazard Level
-999	Flood in 2000 only	1
0	Water course / body	1
1	Once in past 10 years	1
2-3	2 or 3 times in past 10 years	2
>=4	More than 4 times in past 10 years	3

Exposure level - land use damage cost

Damage cost on land use indicates the exposure level of population and assets regarding flood hazard. This vulnerability indicator is assessed on 4 aspects: economic, social, physical and flood severity (the number of days the flood event lasted), which are well described in the table below.

Table 1-4: Damage aspects

Damage aspect	Explanation
Economic costs	Income, employment, city budget/subsidy, household assets
Social damage	Population, health, social capital
Physical damage	Electricity, water, sanitation, road accessibility
Flood severity	Persistent damage after 3 days of flooding

For each LULC class, the damage cost on each aspect is estimated on a scale from 0 to 2. The damage cost level from all aspects is then summed up for further evaluation. Finally, the total damage cost values are assigned to 4 different categories of increasing exposure level: A, B, C and D. The evaluation results are provided in the table below.

Table 1-5: Evaluation results of land use damage cost

LULC class	Damage cost level per aspect				Total	Level
	Economic Costs 0-2	Social Damage 0-2	Physical Damage 0-2	Flood Presence 0-2		
Urban / artificial areas	1,5	1,5	1	0,5	4,5	C
Formal high density residential (Sealing level: 50%-100%)	1,5	1,5	2	1,5	6,5	D
Formal low density residential (Sealing level: 10%-50%)	1,5	1	2	1	5,5	C
Village settlements (Sealing level 1-10%)	1,5	1,5	0,5	1,5	5	C
Commercial and industrial units	2	0,5	1	0,5	4	B
Non-residential urban fabric	1	1	0,5	1	3,5	B
Roads and associated land	1,5	1	2	1,5	6	C
Mineral extraction sites	1	0	0,5	1	2,5	B
Waste sites	0	1,5	0,5	0	2	A
Construction sites	1	0,5	0	0	1,5	A
Land without current use	0	0	0	0	0	A
Urban greenery	0,5	0,5	0,5	1	2,5	B
Sports and leisure facilities	0,5	0,5	0	0,5	1,5	A
Agriculture	1,5	0,5	0	1	3	B
Trees (Forests)	0,5	0	0	0	0,5	A
Other natural and semi-natural areas including wetlands	0	0	0	0	0	A
Water bodies	0	0	0	0	0	A

Flood risk matrix

Flood risk matrix is two-dimensional, taking into account both flood hazard level and exposure level (land use damage cost). At the end, the values resulting from the combination of the ones provided by the input indicators are further grouped and classified into 4 flood risk levels: low - medium - high - very high risk. This risk level categorization, however, is relative. The flood risk matrix is finally provided in the table below.

Table 1-6: Flood risk matrix

		Land use damage cost			
		A	B	C	D
Flood hazard level	1	1A	1B	1C	1D
	2	2A	2B	2C	2D
	3	3A	3B	3C	3D
Flood Risk classification					
Low risk	1A 1B 2A				
Medium risk	1C 1D 2B 2C 3A 3B				
High risk	2D 3C				
Very high risk	3D				

Once the values of the input indicators are well estimated and the flood risk matrix duly established, the flood hazard layer (Product 2) with level attribute information and the current LULC layer with exposure level information (damage cost) can be easily stacked together to generate the Flood Risk product. The

flood risk level can then be populated for each polygon thanks to the attribute information from both input layers. The content of Product 3 attribute table is described in the table below.

Table 1-7: Flood Risk product - attribute information

Attribute field	Description
<i>COUNTRY</i>	Country name
<i>CITIES</i>	City name
<i>WaterArea</i>	Water body / flooded area
<i>HazardCode</i>	Number of flood occurrences in the past 10 years 0 – water body -999 – historical flood in the year 2000
<i>Area</i>	Geometry - area
<i>Remark</i>	Explanation given about 2000 remarkable flood event
<i>RiskCode_H</i>	Flood hazard level
<i>RiskCode_L</i>	Exposure level - Land use damage cost
<i>ACTUAL</i>	Current LULC classification code
<i>ITEM_ACTU</i>	Current LULC classification name
<i>IMGACTUSRC</i>	EO data used to generate LULC product
<i>COMMENT</i>	Comment on land use classification
<i>RiskLevel</i>	Flood risk level

Annex 2 – Filled Quality Control Sheets

This page is intentionally left blank!

Earth Observation for Sustainable Development - Urban

Quality Assurance and Quality Control Sheets

These QA/QC Templates were prepared by GAF AG compliant with ISO 9001:2008 Quality Management System standards and can only be used by Partners in the current EO4SD-Urban Project.

Project Title:	EO4SD-Urban		
Project Leader:	GAF AG		
Service Provider:	GISAT, s.r.o.	Editor:	Václav Stonáček
Client:	WB	Date:	15.9.2017

<i>Overview of QC-Sheets and Processing Steps</i>	<i>Sheet used</i>	<i>Sheet filled in</i>
---	-------------------	------------------------

Requirements

0.1 Requirements

Yes	Yes
-----	-----

Specifications of Input Data

1.1 List of EO Data

Yes	Yes
-----	-----

1.2 List of In-situ Data

Yes	No
-----	----

1.3 List of Ancillary Data

Yes	Yes
-----	-----

Data Quality Checks

2.1 EO Data Quality

Yes	Yes
-----	-----

2.2 In-situ Data Quality

Yes	No
-----	----

2.3 Ancillary Data Quality

Yes	Yes
-----	-----

Pre-Processing of EO Data

3.1 Geometric Correction

Yes	Yes
-----	-----

3.1.1 Data Fusion

Yes	Yes
-----	-----

3.2 Data Processing

Yes	Yes
-----	-----

Thematic Processing

4.1 Classification

Yes	Yes
-----	-----

Accuracy Assessment

5.1 Thematic Accuracy

Yes	Yes
-----	-----

5.2 Error Matrices

Yes	Yes
-----	-----

Delivery Checks / Delivery

6.1 Completeness

Yes	Yes
-----	-----

6.2 Compliancy

Yes	Yes
-----	-----

Glossary (index numbers in the QA/QC tables refer to the glossary at the end of this document)

Further QC-relevant Documents:

Comments / Characteristics:

o.1 Requirements		
Product 1 ⁽²⁸⁾	Urban Land Use/ Land Cover 2006 and 2017	
Abstract		
The Land Use/Land Cover product contains spatial explicit information on different land use and land cover in both Core and Peri-Urban areas of the city of Dhaka (Bangladesh) for the years of 2006 and 2017. The Core area has detailed LU/LC nomenclature whereas the Peri-Urban area LU/LC nomenclature is at an aggregated Level. The input data for the Core area was the Very High Resolution data (Pleiades for 2017, Quickbird-2 for 2006) and the input data for the Peri-Urban area was Sentinel (2017) and Landsat (2006) data. The LU/LC product is the Baseline Product from which various derived products (such as Green Areas and Informal Settlements) are produced.		
Service / Product Specifications		
Area Coverage		
Country:	Bangladesh	
City:	Dhaka	
Area km²	Core Urban:	324,34 km²
	Peri-Urban:	710 km²
Time Period - Update Frequency		
A) Baseline Year(s): 2006, 2017		B) Update Frequency 2 time horizons, no future update
Comments:.		
Geographic Reference System		
EPSG: 32646. UTM 46N / WGS84		
Mapping Classes and Definitions		
Formal high density continuous residential (Sealing level > 80%)	Urban fabric where formal housing (in the form of individual houses or apartment blocks) dominates. Formal houses expected to be organized in a relatively regular spatial pattern with clearly visible roads. 0 %-20 % of the urban fabric consists of non-sealed or vegetated surface.	
Formal high density discontinuous residential (Sealing level: 50% - 80%)	Urban fabric where formal housing (in the form of individual houses or apartment blocks) dominates. Formal houses expected to be organized in a relatively regular spatial pattern with clearly visible roads. 20 %-50 % of the urban fabric consists of non-sealed or vegetated surface.	
Formal low density discontinu (Sealing level: 10% - 50%)	Urban fabric consist of individual houses with little evidence of apartment blocks. Plot size are relatively large compare to formal high density residential. 50 % - 90 % of the urban fabric consists of non-sealed or vegetated surfaces.	
Village settlement (Sealing level < 10%)	Subclass of low density residential class. Village settlements were labelled if located outside of urban perimeter.	
Commercial and industrial units	Factories, warehouses, kilns, shopping malls and all associated facilities and land includes medium to large scale compound units.	
Non-residential urban fabric	Governmental, Medical, Religious, Military, Educational areas including associated areas.	
Roads and associated land	Motorways, primary and secondary roads.	
Airports	Administrative area of airports, mostly fenced. Included are all airports installations: runways, buildings and associated land.	
Construction sites	Areas with on-going building/infrastructure construction activity or areas obviously prepared for construction.	
Sand-filled area	Sub-class of construction site.	
Land without current use	Vacant land for which there is no evidence of on-going building/infrastructure construction activity.	

Urban greenery, sport and leisure facilities	Urban green area, recreational use as gardens, zoos, parks including sport and leisure facilities, sport fields.	
Agriculture	Cultivated areas non-irrigated or permanently irrigated including rice fields. Arable land (annual crops), permanent crops, orchards.	
Trees	High woody vegetation in natural forests including bushes and shrubs at the fringe of the forest.	
Other Natural and Semi-natural areas incl. wetlands	Shrubs and/or herbaceous vegetation incl. transitional woodland and wetlands.	
Bare land	Natural areas where there is no or very little evidence of vegetation and does not serve as construction site.	
Water bodies	Visible water surface areas.	
Cloud and Cloud Shadow Detection and Removal		
EO data without clouds or shadows.		
Spatial Resolution		
n.a. (Product provided as Shapefile)		
Minimum Mapping Unit (MMU)		
Minimum Mapping Unit is 0.25 ha for Core City, 0.5 ha for Peri-urban areas.		
Data Type & Format		
Vector files (ESRI File Geodatabase, Shapefile), Maps (JPEG, TIFF, PDF, KMZ)		
Bit Depth		
N/A		
Class Coding		
Class Code	Class Name	RGB Code
11100	Formal high density continuous residential (Sealing level > 80%)	(RGB: 115-0-0)
11200	Formal high density discontinuous residential (Sealing level: 50% - 80%)	(RGB: 255-0-0)
11300	Formal low density discontinuous residential (Sealing level: 10% - 50%)	(RGB: 255-190-190)
11500	Village settlement (Sealing level < 10%)	(RGB: 168-112-0)
12000	Commercial and industrial units	(RGB: 197-0-255)
12100	Non-residential urban fabric	(RGB: 232-190-255)
12200	Roads and associated land	(RGB: 78-78-78)
12300	Airports	(RGB: 205-102-102)
13000	Construction sites	(RGB: 255-0-197)
13100	Sand-filled area	(RGB: 158-170-215)
13200	Land without current use	(RGB: 190-232-255)
14000	Urban greenery, sport and leisure facilities	(RGB: 85-255-0)
20000	Agriculture	(RGB: 255-235-175)
31000	Forest	(RGB: 38-115-0)
32000	Other Natural and Semi-natural areas incl. wetlands	(RGB: 180-215-158)
33000	Bare land	(RGB: 204-204-204)
50000	Water bodies	(RGB: 0-112-255)
Metadata		
Provided as INSPIRE conformant *.xml data set, covering at least the mandatory elements.		
Service / Product Quality		
Thematic Accuracy		
Overall Accuracy: 92%		
Positional Accuracy		
RMSE < 30 m.		

Delivery Procedure		
Service Provision		
Online via FTP		
Delivery Date		
End of June 2017		
o.1 Requirements		
Product 2 ⁽²⁸⁾	Urban Land Use/ Land Cover Change	
Abstract		
<i>The Land Use/Land Cover Change product contains spatial explicit information on different land use and land cover change in both Core and Peri-Urban areas of the city of Dhaka (Bangladesh) between the years of 2006 and 2017. The Core area has detailed LU/LC nomenclature whereas the Peri-Urban area LU/LC nomenclature is at an aggregated Level. The input data for the Core area was the Very High Resolution data (Pleiades for 2017, Quickbird-2 for 2006) and the input data for the Peri-Urban area was Sentinel (2017) and Landsat (2006) data. The LU/LC product is the Baseline Product from which various derived products (such as Green Areas and Informal Settlements) are produced.</i>		
Service / Product Specifications		
Area Coverage		
Country:	Bangladesh	
City:	Dhaka	
Area km ²	Core Urban:	324,34 km ²
	Peri-Urban:	385,73 km ²
Time Period - Update Frequency		
A) Baseline Year(s): 2006, 2017		B) Update Frequency 2 time horizons, no future update
Comments:.		
Geographic Reference System		
EPSG: 32646. UTM 46N / WGS84		
Mapping Classes and Definitions		
Urban extension: extension of residential urban fabric	Formation of new residential urban fabric over non-artificial land over non-artificial land	
Urban extension: extension of commercial & industrial units	Formation of new commercial & industrial fabric over non-artificial land	
Urban extension: extension of non-residential urban fabric	Formation of new non-residential urban fabric over non-artificial land over non-artificial land	
Urban extension: extension of construction sites	Formation of new construction sites over non-artificial land	
Internal urban development	Internal conversion between artificial surfaces.	
Agriculture abandonment	Abandonment of agricultural land in favor of various types of natural and semi-natural land	
Natural areas development: consumption of urban fabric	Conversion of various types of urban fabric into natural areas	
Water bodies development: consumption of agriculture	Conversion of agriculture into water body (related mostly to riverbed development)	
Other changes	Other changes	

Cloud and Cloud Shadow Detection and Removal		
EO data without clouds or shadows.		
Spatial Resolution		
n.a. (Product provided as Shapefile)		
Minimum Mapping Unit (MMU)		
Minimum Mapping Unit is 0.25 ha for Core City, 0.5 ha for Peri-urban areas.		
Data Type & Format		
Vector files (ESRI File Geodatabase, Shapefile), Maps (JPEG, TIFF, PDF, KMZ)		
Bit Depth		
N/A		
Class Coding		
Class Code	Class Name	RGB Code
LCF11	Urban extension: extension of residential urban fabric	(RGB: 255-0-0)
LCF12	Urban extension: extension of commercial & industrial units	(RGB: 197-0-255)
LCF14	Urban extension: extension of non-residential urban fabric	(RGB: 232-190-255)
LCF16	Urban extension: extension of construction sites	(RGB: 255-0-197)
LCF20	Internal urban development	(RGB: 255-127-127)
LCF32	Agriculture abandonment	(RGB: 255-235-175)
LCF62	Natural areas development: consumption of urban fabric	(RGB: 180-215-158)
LCF52	Water bodies development: consumption of agriculture	(RGB: 0-112-255)
XXX	Other changes	(RGB: 156-156-156)
Provided as INSPIRE conformant *.xml data set, covering at least the mandatory elements.		
Service / Product Quality		
Thematic Accuracy		
Overall Accuracy > 80%		
Positional Accuracy		
RMSE < 30 m.		
Delivery Procedure		
Service Provision		
Online via FTP		
Delivery Date		
End of June 2017		

1.1 List of EO Data

Sensors ⁽⁸⁾ QuickBird, Pleiades, Landsat5, Sentinel-2		Incoming Date	Acquisitio n Date	Proc. Level	Path / Row	AOI - City / Region / Countr y	Spatial Res.	No. of Bands	Cloud Cover (Data Provider)	Projection / Spheroid ⁽¹⁶⁾	Data Forma t ⁽³⁾	Bit Depth ⁽⁵⁾	Header / Metada ta ⁽²⁾
File Name													
QuickBird													
1.	o6JAN22045500-M2AS-056358854080_01_P002	03.04.2017	22.01.2006	LV2A	N/A	Dhaka	MUL:2.4 m PAN: 0.6 m	4	0%	UTM46N / WGS84	*.tif	16 Bit u	*.IMD / *.XML
2.	o6DEC07050109-M2AS-056358854080_01_P003	03.04.2017	07.12.2006	LV2A	N/A	Dhaka	MUL:2.4 m PAN: 0.6 m	4	0,4%	UTM46N / WGS84	*.tif	16 Bit u	*.IMD / *.XML
3.	o6DEC20045602-M2AS-056358854080_01_P001	03.04.2017	20.12.2006	LV2A	N/A	Dhaka	MUL:2.4 m PAN: 0.6 m	4	0%	UTM46N / WGS84	*.tif	16 Bit u	*.IMD / *.XML
Pleiades													
4.	PHR1B_MS_201701120450161_SEN_2226845101-002	23.03.2017	12.01.2017	Primary	N/A	Dhaka	MUL:2 m PAN: 0.5 m	4	0%	UTM46N / WGS84	*.tif	16 Bit u	*.XML
Landsat-5													
1.	LT05_L1TP_137043_20061221_20161118_01_T1	14.03.2017	21.12.2006	L1TP	137 / 43	Dhaka	30 m	7	0%	UTM46N / WGS84	*.tif	8 Bit u	*MTL.t xt
2.	LT05_L1TP_137044_20061221_20161118_01_T1	14.03.2017	21.12.2006	L1TP	137 / 44	Dhaka	30 m	7	0%	UTM46N / WGS84	*.tif	8 Bit u	*MTL.t xt
Sentinel-2													
1.	S2A_MSIL1C_20170201T043011_No204_R133_T45QZG_20170201T043825	14.03.2017	01.02.2017	L1C	T45/ OZG	Dhaka	10 m	4	0%	UTM46N / WGS84	*.jp2	16 Bit u	*.safe / *.xml

1.2 List of In-situ Data

N/A

1.3 List of Ancillary Data

Dataset 1 (1.4)	DEM	Incoming Date	Acquisition Date	AOI - City / Region / Country	Data Type (4)	Data Format (3)	Projection / Spheroid (1.6)	Positional Accuracy (1.1)	No. of Classes	Area Coverage	Header / Metadata (2)
SRTM1arcsec		23.03.2017	11.02.2000	Dhaka / Bangladesh	Raster	*.tif	WGS84	Horizontal: CE90 <20m Vertical: CE90 <16m	N/A	100%	*.mta
Lineage:		The Shuttle Radar Topography Mission (SRTM) was flown aboard the space shuttle Endeavour on February 11 - 22, 2000. The National Aeronautics and Space Administration (NASA) and the National Geospatial-Intelligence Agency (NGA) participated in an international project to acquire radar data which were used to create detailed topographic maps. SRTM data are intended for scientific use with a Geographic Information System (GIS) or other special application software. Endeavour orbited Earth 16 times each day during the 11-day mission completing 176 orbits. SRTM successfully collected radar data over 80% of the Earth's land surface between 60° north and 56° south latitude with data points posted every 1 arc-second (approximately 30 meters).									
Source:		USGS Earth Explorer https://earthexplorer.usgs.gov/									

Dataset 2 (1.4)	OSM	Incoming Date	Acquisition Date	AOI - City / Region / Country	Data Type (4)	Data Format (3)	Projection / Spheroid (1.6)	Positional Accuracy (1.1)	No. of Classes	Area Coverage	Header / Metadata (2)
OSM roads/railways		27.02.2017	N/A	Dhaka / Bangladesh	Vector	*.shp	UTM46N / WGS84	N/A	N/A	100%	N/A
Lineage:		Up-to-date data extracts from the OpenStreetMap project. OSM data is crowd sourced data with contribution from volunteers and freely available.									

Source:	Http://www.download.geofabrik.de

2.1 EO Data Quality

Sensors ⁽⁸⁾		QuickBird, Pleiades, Landsat5, Sentinel-2		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Band Specifications			Projection / Spheroid ⁽¹⁶⁾	Scene Location	Completeness of Additional Data	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Cloud Cover (internal check)	Radiometry	Topography	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
File Name [e.g yymmdd; tbd...]		No. of Bands	Band Registration				Spectral/Spatial Resolution													
QuickBird																				
1.	06JAN22045500-M2AS-056358854080_01_P002	Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - V	Y - V	Y - V Q	Y - Q	Y - Q	Y - V Q	Y - V Q	Y - V	N - V Q	Yes	none		
2.	06DEC07050109-M2AS-056358854080_01_P003	Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - V	Y - V	Y - V Q	Y - Q	Y - Q	Y - V Q	Y - V Q	Y - V	N - V Q	Yes	none		
3.	06DEC20045602-M2AS-056358854080_01_P001	Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - V	Y - V	Y - V Q	Y - Q	Y - Q	Y - V Q	Y - V Q	Y - V	N - V Q	Yes	none		
Pleiades																				
4.	PHR1B_MS_201701120450161_SEN_2226845101-002	Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - V	Y - V	Y - V Q	Y - Q	Y - Q	Y - V Q	Y - V Q	Y - V	N - V Q	Yes	none		
5.	PHR1B_P_201701120450161_SEN_2226845101-001	Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - V	Y - V	Y - V Q	Y - Q	Y - Q	Y - V Q	Y - V Q	Y - V	N - V Q	Yes	none		
Landsat5																				
6.	LT05_L1TP_137043_20061221_20161118_01_T1	Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - V	Y - V	Y - V Q	Y - Q	Y - Q	Y - V Q	Y - V Q	Y - V	N - V Q	Yes	none		
7.	LT05_L1TP_137044_20061221_20161118_01_T1	Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - V	Y - V	Y - V Q	Y - Q	Y - Q	Y - V Q	Y - V Q	Y - V	N - V Q	Yes	none		
Sentinel-2																				
8.	S2A_MSIL1C_20170201T043011_N0204_R133_T45QZG_20170201T043825	Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - V	Y - V	Y - V Q	Y - Q	Y - Q	Y - V Q	Y - V Q	Y - V	N - V Q	Yes	none		

2.2 In-situ Data Quality

N/A

2.3 Ancillary Data Quality

Dataset 1 ⁽¹⁴⁾	DEM															
File Name		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Extent	Projection / Spheroid ^(4,6)	Spatial Resolution	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Location	Completeness	Geom. Misalignment	Plausibility	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
SRTM1arcsec		Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - Q	Y - Q	Y - V	Y - V	N - V	Y - V	N - V	Yes	none

Dataset 2 ⁽¹⁴⁾	OSM															
File Name		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Extent	Projection / Spheroid ^(4,6)	Spatial Resolution	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Location	Completeness	Geom. Misalignment	Plausibility	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
OSM roads/railways		Y - Q	Y - Q	N/A	Y - V Q	Y - V Q	N/A	Y - Q	N/A	Y - V Q	Y - V	N - V	Y - V	N - V	Yes	none

3.1 Geometric Correction

Sensors ⁽⁸⁾		QuickBird, Pleiades, Landsat5, Sentinel-2											
No.	File Name [e.g yymmdd; tbd...]	Processing Date	AOI City / Region / Country	Projection / Spheroid ⁽¹⁶⁾	No. & RMS (m) of GCPs ⁽¹⁷⁾	No. & RMS (m) of TPs ⁽¹⁸⁾	No. & RMS (m) of CPs ⁽¹⁹⁾	Digital Elevation Model (DEM)	Model / Algorithm ⁽¹³⁾	Resampling Method ⁽²⁰⁾	Validation Reports	Acceptance Status	Output File Name [e.g yymmdd; tbd...]
QuickBird													
1.	06JAN22045500-M2AS-056358854080_01_P002	11.04.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	28; X RMSE:0 ,45; Y RMSE: 0,46	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006JAN22045500-M2AS- 056358854080_01_P002
2.	06DEC07050109-M2AS-056358854080_01_P003	11.04.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	28; X RMSE:0 ,45; Y RMSE: 0,46	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006DEC07050109-M2AS- 056358854080_01_P003
3.	06DEC20045602-M2AS-056358854080_01_P001	11.04.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	28; X RMSE:0 ,45; Y RMSE: 0,46	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006DEC20045602-M2AS- 056358854080_01_P001
4.	06JAN22045500-P2AS-056358854080_01_P002	11.04.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	25; X RMSE:0 ,83; Y RMSE: 0,47	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006JAN22045500-P2AS- 056358854080_01_P002
5.	06DEC07050109-P2AS-056358854080_01_P003	11.04.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	25; X RMSE:0 ,83; Y RMSE: 0,47	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006DEC07050109-P2AS- 056358854080_01_P003

6.	o6DEC20045602-P2AS-056358854080_01_P001	11.04.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	25; X RMSE:0 ,83; Y RMSE: 0,47	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	o06DEC20045602-P2AS- 056358854080_01_P001
Pleiades													
7.	PHR1B_MS_201701120450161_SEN_2226845101-002	11.4.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	12; X RMSE:0 ,49; Y RMSE: 0,43	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	oPHR1B_MS_2017011204501 61_SEN_2226845101-002
8.	PHR1B_P_201701120450161_SEN_2226845101-001	11.4.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	9; X RMSE:0 ,32; Y RMSE: 0,08	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	oPHR1B_P_201701120450161 _SEN_2226845101-001
Landsat5													
9.	LT05_L1TP_137043_20061221_20161118_01_T1	22.03.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	N/A	N/A	N/A	N/A	N/A	N/A	N/ A	Ye s	LT05_L1TP_137043_20061221 _20161118_01_T1
10.	LT05_L1TP_137044_20061221_20161118_01_T1	22.03.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	N/A	N/A	N/A	N/A	N/A	N/A	N/ A	Ye s	LT05_L1TP_137044_2006122 1_20161118_01_T1
Sentinel-2													
11.	S2A_MSIL1C_20170201T043011_No204_R133_T45QZG_20170201T043825	22.03.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	N/A	N/A	N/A	N/A	N/A	N/A	N/ A	Ye s	S2A_MSIL1C_20170201T0430 11_No204_R133_T45QZG_20 170201T043825

3.1.1 Data Fusion

Dataset 1 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_QB_20060122_MS		EO4SD_Dhaka_QB_20060122_PAN		EO4SD_Dhaka_QB_20060122_PS2Y			
Sensor	QuickBird		QuickBird		QuickBird			
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)			
Spatial Resolution / (MS/Pan)	2,4m	MS	0,6m	Pan	0,6m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

Dataset 2 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_QB_20061207_MS		EO4SD_Dhaka_QB_20061207_PAN		EO4SD_Dhaka_QB_20061207_PS2Y			
Sensor	QuickBird		QuickBird		QuickBird			
Method	PANSHARP2 (PCI)		PANSHARP2 (PCI)		PANSHARP2 (PCI)			
Spatial Resolution / (MS/Pan)	2,4m	MS	0,6m	Pan	0,6m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

Dataset 3 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_QB_20061220_MS		EO4SD_Dhaka_QB_20061220_PAN		EO4SD_Dhaka_QB_20061220_PS2Y			
Sensor	QuickBird		QuickBird		QuickBird			
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)			
Spatial Resolution / (MS/Pan)	2,4m	MS	0,6m	Pan	0,6m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

Dataset 4 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_PLE_20170112_MS		EO4SD_Dhaka_PLE_20170112_PAN		EO4SD_Dhaka_PLE_20170112_PS2Y			
Sensor	Pleiades		Pleiades		Pleiades			
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)			
Spatial Resolution / (MS/Pan)	2m	MS	0,5m	Pan	0,5m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

3.2 Data Processing

Sensors ⁽⁸⁾		QuickBird. Pleiades, Landsat5, Sentinel-2		Processing Date	Atmospheric Correction		Radiometric Processing ⁽¹⁵⁾		Topographic Normalisation		OTHER Calculation:		Acceptance Status	Backup	Comments
No.	File Name [e.g yymmdd; tbd...]	processed	Software / Method		processed	Software / Method	processed	Software / Method	processed	Software / Method					
QuickBird															
1.	o6JAN22o455oo-M2AS-056358854o8o_o1_Poo2	11.o4.2o17	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none		
2.	o6DECo7o5o1o9-M2AS-056358854o8o_o1_Poo3	11.o4.2o17	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none		
3.	o6DEC2oo456o2-M2AS-056358854o8o_o1_Poo1	11.o4.2o17	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none		
Sentinel-2															

4.	PHR1B_MS_201701120450161_SEN_2226845101-002	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none
Landsat5													
5.	LT05_L1TP_137043_20061221_20161118_01_T1	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none
6.	LT05_L1TP_137044_20061221_20161118_01_T1	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	
Sentinel-2													
7.	S2A_MSIL1C_20170201T043011_N0204_R133_T45QZG_20170201T043825	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none

4.1 Classification

Sensor 1 ⁽⁸⁾		QuickBird	Processin g Date	Cloud Masking		Thematic Classification		Manual Enhancement		Mosaicking		Acceptance Status	Backup	Comment
No.	File Name [e.g yymmdd; tbd...]			processed	Software / Method	processed	Software / Method	processed	Software / Method	processed	Software / Method			
QuickBird														
1.	o6JAN22045500-M2AS-056358854080_01_P002 o6DEC07050109-M2AS-056358854080_01_P003 o6DEC20045602-M2AS-056358854080_01_P001		13.04.2017	N	N/A	Y	eCognition/semi-automatic approach	Y	Webim/GISA T proprietary solution	N	N/A	Y	Y	
Pleiades														
2.	PHR1B_MS_201701120450161_SEN_2226845101-002		13.04.2017	N	N/A	Y	eCognition/semi-automatic approach	Y	Webim/GISA T proprietary solution	N	N/A	Y	Y	
Landsat5														
3.	LT05_L1TP_137043_20061221_20161118_01_T1 LT05_L1TP_137044_20061221_20161118_01_T1		01.06.2017	N	N/A	Y	eCognition/semi-automatic approach	Y	Webim/GISA T proprietary solution	N	N/A	Y	Y	
Sentinel-2														
4.	S2A_MSIL1C_20170201T043011_No204_R133_T45Q_ZG_20170201T043825		01.06.2017	N	N/A	Y	eCognition/semi-automatic approach	Y	Webim/GISA T proprietary solution	N	N/A	Y	Y	

5.1 Thematic Accuracy

No.	Product Name	Processing Date	Area Coverage	No. of Classes	Reference Data	Sample Size	Sampling Unit ⁽²¹⁾	Sampling Design ⁽²²⁾	Sample Exclusion Criteria ⁽²³⁾	Thematic Accuracy ⁽²⁴⁾	Acceptance Status	Comment
1.	LULC 2006	2.6.2017	Dhaka – core city	11	QuickBird	520 points	POINT	Stratified random sampling	None	92,3%	Yes	
2.	LULC 2017	2.6.2017	Dhaka – core city	11	Pleiades	535 points	POINT	Stratified random sampling	None	92,5%	Yes	

5.2 Error Matrices														
Class Name (columns = Mapped Class; Rows = Ground Truth) LandUse/Land Cover 2006		Urban fabric	Village settlements	Commercial and Industrial	Transportation	Construction sites	Urban Greenery	Agriculture	Forest	Natural areas	Bare land	Water		User Accuracy and Confidence Interval at 95% Confidence Level
	Class ID	110	115	120	122	130	140	200	310	320	330	500	Totals	
Urban fabric	110	168	0	3	0	0	0	0	0	0	0	0	171	93.3 % ± 2.1
Village settlements	115	10	39	0	0	0	0	0	0	0	0	0	49	97.5 % ± 1.3
Comm. and Industrial	120	0	0	70	0	1	0	0	0	0	0	0	71	85.4 % ± 2.1
Transportation	122	0	0	4	40	0	0	0	0	0	0	0	44	100 % n.a.
Construction sites	130	1	0	2	0	45	0	0	0	0	0	0	48	84.9 % ± 3.1
Urban Greenery	140	0	0	2	0	0	17	0	0	0	0	0	19	89.5 % ± 2.7
Agriculture	200	1	0	0	0	3	1	50	0	0	4	0	59	100 % n.a.
Forest	310	0	0	0	0	0	1	0	20	0	0	0	21	100 % n.a.
Natural areas	320	0	0	0	0	0	0	0	0	13	1	0	14	100 % n.a.
Bare land	330	0	0	0	0	2	0	0	0	0	8	0	10	61.5 % ± 4.1
Water	500	0	1	1	0	2	0	0	0	0	0	10	14	100 % n.a.
Totals		180	40	82	40	53	19	50	20	13	13	10	520	
Producer Accuracy and Confidence Interval at 95% Confidence Level		98.2 % ± 1.1	79.6 % ± 13.5	98.6 % ± 1	90.9 % ± 2.5	93.8 % ± 2.1	89.5 % ± 2.7	84.7 % ± 3	95.5 % ± 1.9	92.9 % ± 2.3	80 % ± 3.4	71.4 % ± 3.9	Overall Accuracy: 92.3% Confidence Interval 90.02% - 94.6%	

5.2 Error Matrices														
Class Name (columns = Mapped Class; Rows = Ground Truth)		Urban fabric	Village settlements	Commercial and Industrial	Transportation	Construction sites	Urban Greenery	Agriculture	Forest	Natural areas	Bare land	Water		User Accuracy and Confidence Interval at 95% Confidence Level
LandUse/Land Cover 2017														
	Class ID	110	115	120	122	130	140	200	310	320	330	500	Totals	
Urban fabric	110	169	0	9	0	0	0	0	0	1	2	0	181	93.9 % ± 2
Village settlements	115	6	40	0	0	0	0	1	0	0	0	0	47	100 % n.a.
Comm. and Industrial	120	0	0	79	0	0	2	0	0	0	0	0	81	87.8 % ± 2.8
Transportation	122	0	0	0	40	0	0	0	0	0	0	0	40	100 % n.a.
Construction sites	130	2	0	2	0	50	0	0	0	0	2	0	56	90.9 % ± 2.4
Urban Greenery	140	1	0	0	0	0	15	0	0	0	0	0	16	75 % ± 3.7
Agriculture	200	0	0	0	0	0	2	49	0	1	0	0	52	98 % ± 1.2
Forest	310	0	0	0	0	0	0	0	20	0	0	0	20	100 % n.a.
Natural areas	320	0	0	0	0	5	1	0	0	17	0	0	23	85 % ± 3
Bare land	330	0	0	0	0	0	0	0	0	0	6	0	6	60 % ± 4.2
Water	500	2	0	0	0	0	0	0	0	1	0	10	13	100 % n.a.
Totals		180	40	90	40	55	20	50	20	20	10	10	535	
Producer Accuracy and Confidence Interval at 95% Confidence Level		93.4 % ± 2.1	85.1 % ± 3	97.5 % ± 0.7	100 % n.a.	89.3 % ± 2.6	93.8 % ± 2	94.2 % ± 2	100 % n.a.	73.9 % ± 3.7	100 % n.a.	76.9 % ± 3.6	Overall Accuracy: 92.5% Confidence Interval 90.3% - 94.75%	

6.1 Completeness

INPUT DATA									
No .	Item	AOI Coverage [km²]	Area Coverage [km²]	Completeness of Coverage ⁽²⁵⁾	No. of Scenes	Scenes used in Production	Completeness of Verification ⁽²⁷⁾	Metadata	Comments
1.	HR EO Data	710 km²	710 km²	100%	3	3	100%	Yes	none
2.	VHR EO Data		324 km²	100%	4	4	100%	Yes	none
4.	Ancillary Data		N/A	N/A	N/A	N/A	N/A	Yes	none
PRODUCTS									
No .	Product ⁽²⁸⁾	AOI Coverage [km²]	Product Coverage [km²]	Completeness of Coverage ⁽²⁵⁾	Completeness of Classification ⁽²⁶⁾	Unclassifiable Area [%]	Completeness of Verification ⁽²⁷⁾	Metadata	Comments
1.	LULC 2006	710 km²	710 km²	100%	100%	N/A	100%	Yes	none
2.	LULC 2016		710 km²	100%	100%	N/A	100%	Yes	none

6.2 Compliance					
Product 1 ⁽²⁸⁾		EO4SD_DHAKA_LULC_2006_2017			
Abstract					
The Land Use/Land Cover product contains spatial explicit information on different land use and land cover in both Core and Peri-Urban areas of the city of Dhaka (Bangladesh) for the years of 2006 and 2017. The Core area has detailed LU/LC nomenclature whereas the Peri-Urban area LU/LC nomenclature is at an aggregated Level. The input data for the Core area was the Very High Resolution data (Pleiades for 2017, Quickbird-2 for 2006) and the input data for the Peri-Urban area was Sentinel (2017) and Landsat (2006) data. The LU/LC product is the Baseline Product from which various derived products (such as Green Areas and Informal Settlements) are produced.					
Service / Product Specifications					
Area Coverage					
Requirements		Achieved Specifications		Compliance	Comments
Country 1:	Bangladesh	Country 1:	Bangladesh	100 %	
Country 2:	N/A	Country 2:	N/A		
Country 3:	N/A	Country 3:	N/A		
Country 4:	N/A	Country 4:	N/A		
A) Wall-to-wall: Sub-national, Dhaka		A) Wall-to-wall: Sub-national, Dhaka		100 %	
Area of approx. 710 km ² , defined by the user.					
B) Sampling based: N/A		B) Sampling based: N/A			
Time Period - Update Frequency					
Requirements		Achieved Specifications		Compliance	Comments
Baseline Year(s): 2000 2015		Baseline Year(s): 2006 2017		Yes	EO data differs from requirements because of availability of VHR EO data.
B) Update Frequency N/A		B) Update Frequency N/A			
Geographic Reference System					
Requirements		Achieved Specifications		Compliance	Comments
EPSG: 32646. WGS84 / UTM 46N		EPSG: 32646. WGS84 / UTM 46N		Yes	-
Mapping Classes and Definitions (Definitions see REDD+ MRV Design Document)					
Requirements		Achieved Specifications		Compliance	
N/A		All classes were agreed by the user.		Yes	
Cloud and Shadow Detection and Removal					
Requirements		Achieved Specifications		Compliance	Comments
Cloud free.		Cloud free scenes used.		Yes	-

Spatial Resolution			
Requirements	Achieved Specifications	Compliance	Comments
VHR for core	VHR for core	Yes	-
Minimum Mapping Unit (MMU)			
Requirements	Achieved Specifications	Compliance	Comments
0.25 ha for Core City, 0.5 ha for Peri-urban	0.25 ha for Core City, 0.5 ha for Peri-urban	Yes	-
Data Type			
Requirements	Achieved Specifications	Compliance	Comments
GIS-ready format	ESRI Shapefile / Geodatabase	Yes	-
Bit Depth			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	N/A	N/A	-
Data Format			
Requirements	Achieved Specifications	Compliance	Comments
*.shp ESRI Shapefile (open cross-platform format)	ESRI Shapefile	Yes	-
Class Coding			
Requirements	Achieved Specifications	Compliance	
N/A	N/A		
Metadata			
Requirements	Achieved Specifications	Compliance	Comments
INSPIRE compliant	ISO/Inspire compliant	Yes	-
Service / Product Quality			
Thematic Accuracy			
Requirements	Achieved Specifications	Compliance	Comments
Overall Accuracy: 85%	89% - 92%	Yes	-
Positional Accuracy			
Requirements	Achieved Specifications	Compliance	Comments
RMSE < 2 m	RMSE < 2 m	Yes	

			Calculated from adjustment to OSM roads. Technical difficulties to co-registrate and find appropriate GCP/TPs due to extremely high amount of changes in the area
Delivery Procedure			
Service Provision			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	FTP	Yes	
Delivery Date			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	Sep 15 th 2017	Yes	Interim products delivered earlier on demand

Glossary

Quality Checks		
Prefix	Suffix	Explanation
Y (Yes)	- V	Visually checked
N (No)	- Q	Quantitatively/qualitatively checked
N/A	- V Q	Visually and quantitatively/qualitatively checked
	- N/A	Not applicable

Readability ⁽¹⁾	Check readability of all required input data. Can the data be stored again?
Header / Metadata ⁽²⁾	Check Image Header Information and/or Metadata for completeness / distinctive features.
Data Format ⁽³⁾	For digital data, please give file format (e.g. *.tiff, *.shp).
Data Type ⁽⁴⁾	Please specify the type of the data (e.g. raster, vector or analogue).
Bit Depth ⁽⁵⁾	Please give pixel depth and sign of raster data (e.g. 8 bit unsigned integer).
Dynamic Range ⁽⁶⁾	Check dynamic range of all image bands. Visual check of dynamic range should be accompanied by histograms and statistics.
Dropped Lines & Artefacts ⁽⁷⁾	Check Image for dropped lines and other artefacts. If such occurs, please give description of extent and influence in the Comments section.
Sensor 1, 2, ... ⁽⁸⁾	Please delete / add additional sensor sections as necessary.
Purpose ⁽⁹⁾	The purpose and use of the given auxiliary or reference data should be stated: ORHTOrectification, GEOmetric correction, REference, VERification source, POSitional ACCuracy assessment, THEmatic ACCuracy assessment.
Sampling Methodology ⁽¹⁰⁾	Methodology of in situ or reference/auxiliary data sampling scheme should be outlined. In case of sample plots, also state how the plot positions have been determined (e.g. from GPS measurements, topographic maps, terrestrial triangulation, EO data, etc.) and how the sampling grid was established.
Positional Accuracy ⁽¹¹⁾	Positional accuracy of collected in situ data should be given. For reference/auxiliary data it MUST be given. If unknown, the data's use must be explained. For DEM or other data with 3D information please specify both vertical and horizontal Positional Accuracy. For analogue data (e.g. maps) try to give approximate accuracy related to mapping scale.
Completeness ⁽¹²⁾	Data should be checked for spatial/temporal/content gaps.
Model / Algorithm ⁽¹³⁾	Give the name of the software and its version. Specify the software module/algorithm used for: a) geometric correction, e.g., Polynomial and its degree, Rational Functions, Thin Plate Spline, etc. b) classification, e.g., ISODATA, Maximum Likelihood, Neural Networks, etc.
Dataset 1, 2, ... ⁽¹⁴⁾	Please delete / add additional dataset sections as necessary.
Radiometric Processing ⁽¹⁵⁾	State whether (and which) radiometric processing was applied (e.g., contrast enhancement, histogram matching, bundle block adjustment, radiometric normalisation, filtering, etc.).
Projection; Spheroid / Ellipsoid ⁽¹⁶⁾	Always specify completely, i.e. at minimum the Projection (+Zone, if applicable), Spheroid / Ellipsoid, Map Datum. Give additional information if necessary to unambiguously define the reference system.

Ground Control Points (GCPs) ⁽¹⁷⁾	Give the number, distribution and RMS of used Ground Control Points (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of GCPs should be attached as a snapshot, or described in the Comments section.
Tie Points (TPs) ⁽¹⁸⁾	In case of mosaicking, give the number of used Tie Points and their total RMS (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of TP's should be attached as a snapshot, or described in the Comments section.
Check Points (CPs) ⁽¹⁹⁾	The real measure of positional accuracy and the only measure which should be examined as to its Acceptable Range. Give the independent Check Points' total RMS (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of CP's should be attached as a snapshot, or described in the Comments section.
Resampling Method ⁽²⁰⁾	Specify, if / which resampling algorithm has been used (e.g. NN=Nearest Neighbour, BIL=Bilinear Interpolation, CC=Cubic Convolution).
Sampling unit ⁽²¹⁾	Specify sampling unit of Accuracy Assessment as POINT, FRAME, POLYGON and how it is treated (e.g. pixel center, polygon centre, etc.).
Sampling Design ⁽²²⁾	SYST=Systematic, RAND=Random, STRAT=Stratified, SBCLASS=Stratified by class, SBAREA=Stratified by area
Sample exclusion criteria ⁽²³⁾	Describe which <u>criteria</u> you apply for sampling point selection resp. exclusion of certain points. For example, if the point is too close to a class boundary (less than 1 pixel), it is excluded. If the selected sample point is not representative of the class, it is excluded. For these reasons, it is recommended to oversample by 10% to compensate for sample point exclusion.
Thematic Accuracy ⁽²⁴⁾	Provide a detailed description of the Accuracy Assessment results in the form of error matrices showing commission and omission errors, user's, producer's and overall accuracies and other measures of Thematic Accuracy, as the confidence level (usually fixed at 95%) and the respective confidence interval, at least for the overall accuracy. If classification is done in a phased approach, e.g. if Forest Area and subsequently Forest Type are mapped, independent reports have to be produced.
Completeness of Coverage ⁽²⁵⁾	State whether the coverage is limited to a subset, or portion of the final product.
Completeness of Classification ⁽²⁶⁾	State whether classification was constrained to a subset, or portion of the final product.
Completeness of Verification ⁽²⁷⁾	State whether verification applied to lineage, positional, or thematic accuracy is constrained to a subset, or portion of the final product.
Product ⁽²⁸⁾	Please delete / add additional product sections as necessary.

Earth Observation for Sustainable Development - Urban

Quality Assurance and Quality Control Sheets

These QA/QC Templates were prepared by GAF AG compliant with ISO 9001:2008 Quality Management System standards and can only be used by Partners in the current EO4SD-Urban Project.

Project Title:	EO4SD-Urban		
Project Leader:	GAF AG		
Service Provider:	GISAT, s.r.o.	Editor:	Václav Stonáček
Client:	WB	Date:	19.9.2017

<i>Overview of QC-Sheets and Processing Steps</i>	<i>Sheet used</i>	<i>Sheet filled in</i>
---	-------------------	------------------------

Requirements

0.1 Requirements

Yes	Yes
-----	-----

Specifications of Input Data

1.1 List of EO Data

Yes	Yes
-----	-----

1.2 List of In-situ Data

Yes	No
-----	----

1.3 List of Ancillary Data

Yes	Yes
-----	-----

Data Quality Checks

2.1 EO Data Quality

Yes	Yes
-----	-----

2.2 In-situ Data Quality

Yes	No
-----	----

2.3 Ancillary Data Quality

Yes	Yes
-----	-----

Pre-Processing of EO Data

3.1 Geometric Correction

Yes	Yes
-----	-----

3.1.1 Data Fusion

Yes	Yes
-----	-----

3.2 Data Processing

Yes	Yes
-----	-----

Thematic Processing

4.1 Classification

Yes	Yes
-----	-----

Accuracy Assessment

5.1 Thematic Accuracy

Yes	Yes
-----	-----

5.2 Error Matrices

Yes	Yes
-----	-----

Delivery Checks / Delivery

6.1 Completeness

Yes	Yes
-----	-----

6.2 Compliancy

Yes	Yes
-----	-----

Glossary (index numbers in the QA/QC tables refer to the glossary at the end of this document)

Further QC-relevant Documents:

Comments / Characteristics:

o.1 Requirements		
Product 1 ⁽²⁸⁾	Urban Green Areas and Change (2006, 2017)	
Abstract		
Urban green areas refer to land within and on the edges of a city that is partly or completely covered with grass, trees, shrubs, or other vegetation. This includes public parks, private gardens, cemeteries, forested areas as well as trees, river alignments, hedges, etc. The product delivered within EO4SD-Urban project thus provides accurate information (1m resolution) on the spatial location and extent of the green areas located within the Urban Extent derived from baseline LULC information product.		
Service / Product Specifications		
Area Coverage		
Country:	Bangladesh	
City:	Dhaka	
Area km²	Core Urban:	324,34 km²
	Peri-Urban:	N/A
Time Period - Update Frequency		
A) Baseline Year(s): 2006, 2017		B) Update Frequency 2 time horizons, no future update
Comments:.		
Geographic Reference System		
EPSG: 32646. UTM 46N / WGS84		
Mapping Classes and Definitions		
Squares	Central square	
	Other (neighbourhood) square	
Open sky markets		Large streets with concentration of stalls
Living streets		Roads surrounded predominantly by urban residential and non-residential fabric
Waterfronts	Beach	Beach
	Waterfront park	Green waterfront areas. Could include amusement parks.
	Sealed waterfront surfaces	Berthing and boardwalks areas
Green urban areas	Central park / Downtown park	Large parks in central and down-town areas. Horticultural management is visible of likely.
	Neighbourhood park	Smaller parks in neighbourhood areas. Horticultural management is visible of likely.
	Mini pocket park	Small green area surrounded by buildings. Only where high vegetation (trees)
	City woodland	Forest within in city outline. No horticultural management apparent
	Sub-urban forest	Forest in peri-urban area. No horticultural management apparent

	Green alley	Elongated linear forest structure in the streets (of boulevard type)
	Sport & leisure facilities	Green sport areas incl. buildings and associated land, playgrounds, amusement parks, golf courses, etc.
Cloud and Cloud Shadow Detection and Removal		
EO data without clouds or shadows.		
Spatial Resolution		
n.a. (Product provided as Shapefile)		
Minimum Mapping Unit (MMU)		
Minimum Mapping Unit is 0.1 - 0.25 ha for Core City		
Data Type & Format		
Vector files (ESRI File Geodatabase, Shapefile), Maps (JPEG, TIFF, PDF, KMZ)		
Bit Depth		
N/A		
Class Coding		
<i>Class Code</i>	<i>Class Name</i>	<i>RGB Code</i>
N/A	N/A	N/A
Metadata		
Provided as INSPIRE conformant *.xml data set, covering at least the mandatory elements.		
Service / Product Quality		
Thematic Accuracy		
Overall Accuracy: 84 %		
Positional Accuracy		
RMSE < 30 m.		
Delivery Procedure		
Service Provision		
Online via FTP		
Delivery Date		
End of June 2017		

1.1 List of EO Data

Sensors ⁽⁸⁾		QuickBird, Pleiades											
File Name		Incoming Date	Acquisition Date	Proc. Level	Path / Row	AOI - City / Region / Country	Spatial Res.	No. of Bands	Cloud Cover (Data Provider)	Projection / Spheroid ⁽¹⁶⁾	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Header / Metadata ⁽²⁾
QuickBird													
1.	o6JAN22045500-M2AS-056358854080_01_P002	03.04.2017	22.01.2006	LV2A	N/A	Dhaka	MUL:2.4 m PAN: 0.6 m	4	0%	UTM46N / WGS84	*.tif	16 Bit u	*.IMD / *.XML
2.	o6DEC07050109-M2AS-056358854080_01_P003	03.04.2017	07.12.2006	LV2A	N/A	Dhaka	MUL:2.4 m PAN: 0.6 m	4	0,4%	UTM46N / WGS84	*.tif	16 Bit u	*.IMD / *.XML
3.	o6DEC20045602-M2AS-056358854080_01_P001	03.04.2017	20.12.2006	LV2A	N/A	Dhaka	MUL:2.4 m PAN: 0.6 m	4	0%	UTM46N / WGS84	*.tif	16 Bit u	*.IMD / *.XML
Pleiades													
4.	PHR1B_MS_201701120450161_SEN_2226845101-002	23.03.2017	12.01.2017	Primary	N/A	Dhaka	MUL:2 m PAN: 0.5 m	4	0%	UTM46N / WGS84	*.tif	16 Bit u	*.XML

1.2 List of In-situ Data

N/A

1.3 List of Ancillary Data

Dataset 1 (1.4)	DEM	Incoming Date	Acquisition Date	AOI - City / Region / Country	Data Type (4)	Data Format (3)	Projection / Spheroid (1.6)	Positional Accuracy (1.1)	No. of Classes	Area Coverage	Header / Metadata (2)
SRTM1arcsec		23.03.2017	11.02.2000	Dhaka / Bangladesh	Raster	*.tif	WGS84	Horizontal: CE90 <20m Vertical: CE90 <16m	N/A	100%	*.mta
Lineage:		The Shuttle Radar Topography Mission (SRTM) was flown aboard the space shuttle Endeavour on February 11 - 22, 2000. The National Aeronautics and Space Administration (NASA) and the National Geospatial-Intelligence Agency (NGA) participated in an international project to acquire radar data which were used to create detailed topographic maps. SRTM data are intended for scientific use with a Geographic Information System (GIS) or other special application software. Endeavour orbited Earth 16 times each day during the 11-day mission completing 176 orbits. SRTM successfully collected radar data over 80% of the Earth's land surface between 60° north and 56° south latitude with data points posted every 1 arc-second (approximately 30 meters).									
Source:		USGS Earth Explorer https://earthexplorer.usgs.gov/									

Dataset 2 (1.4)	OSM	Incoming Date	Acquisition Date	AOI - City / Region / Country	Data Type (4)	Data Format (3)	Projection / Spheroid (1.6)	Positional Accuracy (1.1)	No. of Classes	Area Coverage	Header / Metadata (2)
OSM roads/railways		27.02.2017	N/A	Dhaka / Bangladesh	Vector	*.shp	UTM46N / WGS84	N/A	N/A	100%	N/A
Lineage:		Up-to-date data extracts from the OpenStreetMap project. OSM data is crowd sourced data with contribution from volunteers and freely available.									

Source:	Http://www.download.geofabrik.de

2.1 EO Data Quality

Sensors ⁽⁸⁾		QuickBird, Pleiades		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Band Specifications			Projection / Spheroid ^(4,6)	Scene Location	Completeness of Additional Data	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Cloud Cover (internal check)	Radiometry	Topography	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
File Name [e.g yymmdd; tbd...]							No. of Bands	Band Registration	Spectral/Spatial Resolution											
QuickBird																				
1.	06JAN22045500-M2AS-056358854080_01_P002			Y - Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V V	Y-V V	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V V	N-V Q	Yes s	none
2.	06DEC07050109-M2AS-056358854080_01_P003			Y - Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V V	Y-V V	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V V	N-V Q	Yes es	none
3.	06DEC20045602-M2AS-056358854080_01_P001			Y - Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V V	Y-V V	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V V	N-V Q	Yes s	none
Pleiades																				
4.	PHR1B_MS_201701120450161_SEN_2226845101-002			Y - Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V V	Y-V V	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V V	N-V Q	Yes s	none
5.	PHR1B_P_201701120450161_SEN_2226845101-001			Y - Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V V	Y-V V	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V Q	Y-V V	N-V Q	Yes s	none

2.2 In-situ Data Quality

N/A

2.3 Ancillary Data Quality

Dataset 1 ⁽¹⁴⁾	DEM															
File Name		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Extent	Projection / Spheroid ^(4,6)	Spatial Resolution	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Location	Completeness	Geom. Misalignment	Plausibility	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
SRTM1arcsec		Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - Q	Y - Q	Y - V	Y - V	N - V	Y - V	N - V	Yes	none

Dataset 2 ⁽¹⁴⁾	OSM															
File Name		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Extent	Projection / Spheroid ^(4,6)	Spatial Resolution	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Location	Completeness	Geom. Misalignment	Plausibility	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
OSM roads/railways		Y - Q	Y - Q	N/A	Y - V Q	Y - V Q	N/A	Y - Q	N/A	Y - V Q	Y - V	N - V	Y - V	N - V	Yes	none

3.1 Geometric Correction

Sensors ⁽⁸⁾		QuickBird, Pleiades		Processing Date	AOI City / Region / Country	Projection / Spheroid ⁽¹⁶⁾	No. & RMS (m) of GCPs ⁽¹⁷⁾	No. & RMS (m) of TP _s ⁽¹⁸⁾	No. & RMS (m) of CP _s ⁽¹⁹⁾	Digital Elevation Model (DEM)	Model / Algorithm ⁽¹³⁾	Resampling Method ⁽²⁰⁾	Validation Reports	Acceptance Status	Output File Name [e.g yymmdd; tbd...]
No	File Name [e.g yymmdd; tbd...]														
QuickBird															
1.	06JAN22045500-M2AS-056358854080_01_P002	11.04. 2017	Dhaka/ Banglade sh	UTM46 N / WGS84	28; X RMSE:0 ,45;Y RMSE: 0,46	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006JAN22045500-M2AS- 056358854080_01_P002		
2.	06DEC07050109-M2AS-056358854080_01_P003	11.04. 2017	Dhaka/ Banglade sh	UTM46 N / WGS84	28; X RMSE:0 ,45;Y RMSE: 0,46	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006DEC07050109-M2AS- 056358854080_01_P003		
3.	06DEC20045602-M2AS-056358854080_01_P001	11.04. 2017	Dhaka/ Banglade sh	UTM46 N / WGS84	28; X RMSE:0 ,45;Y RMSE: 0,46	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006DEC20045602-M2AS- 056358854080_01_P001		
4.	06JAN22045500-P2AS-056358854080_01_P002	11.04. 2017	Dhaka/ Banglade sh	UTM46 N / WGS84	25; X RMSE:0 ,83;Y RMSE: 0,47	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006JAN22045500-P2AS- 056358854080_01_P002		
5.	06DEC07050109-P2AS-056358854080_01_P003	11.04. 2017	Dhaka/ Banglade sh	UTM46 N / WGS84	25; X RMSE:0 ,83;Y RMSE: 0,47	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006DEC07050109-P2AS- 056358854080_01_P003		

6.	o6DEC20045602-P2AS-056358854080_01_P001	11.04.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	25; X RMSE:0 ,83;Y RMSE: 0,47	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	o06DEC20045602-P2AS- 056358854080_01_P001
Pleiades													
7.	PHR1B_MS_201701120450161_SEN_2226845101-002	11.4.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	12; X RMSE:0 ,49;Y RMSE: 0,43	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	oPHR1B_MS_2017011204501 61_SEN_2226845101-002
8.	PHR1B_P_201701120450161_SEN_2226845101-001	11.4.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	9; X RMSE:0 ,32;Y RMSE: 0,08	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	oPHR1B_P_201701120450161 _SEN_2226845101-001

3.1.1 Data Fusion

Dataset 1 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_QB_20060122_MS		EO4SD_Dhaka_QB_20060122_PAN		EO4SD_Dhaka_QB_20060122_PS2Y			
Sensor	QuickBird		QuickBird		QuickBird			
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)			
Spatial Resolution / (MS/Pan)	2,4m	MS	0,6m	Pan	0,6m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

Dataset 2 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_QB_20061207_MS		EO4SD_Dhaka_QB_20061207_PAN		EO4SD_Dhaka_QB_20061207_PS2Y			
Sensor	QuickBird		QuickBird		QuickBird			
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)			
Spatial Resolution / (MS/Pan)	2,4m	MS	0,6m	Pan	0,6m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

Dataset 3 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_QB_20061220_MS		EO4SD_Dhaka_QB_20061220_PAN		EO4SD_Dhaka_QB_20061220_PS2Y			
Sensor	QuickBird		QuickBird		QuickBird			
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)			
Spatial Resolution / (MS/Pan)	2,4m	MS	0,6m	Pan	0,6m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

Dataset 4 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_PLE_20170112_MS		EO4SD_Dhaka_PLE_20170112_PAN		EO4SD_Dhaka_PLE_20170112_PS2Y			
Sensor	Pleiades		Pleiades		Pleiades			
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)			
Spatial Resolution / (MS/Pan)	2m	MS	0,5m	Pan	0,5m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

3.2 Data Processing

Sensors ⁽⁸⁾		QuickBird. Pleiades		Processing Date	Atmospheric Correction		Radiometric Processing ⁽¹⁵⁾		Topographic Normalisation		OTHER Calculation:		Acceptance Status	Backup	Comments
No.	File Name [e.g yymmdd; tbd...]	processed	Software / Method		processed	Software / Method	processed	Software / Method	processed	Software / Method					
QuickBird															
1.	o6JAN22o455oo-M2AS-056358854o8o_01_Poo2	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none		
2.	o6DECo7o5o1o9-M2AS-056358854o8o_01_Poo3	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none		
3.	o6DEC2oo456o2-M2AS-056358854o8o_01_Poo1	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none		
Pleiades															

4.	PHR1B_MS_201701120450161_SEN_2226845101-002	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none

4.1 Classification

Sensors ⁽⁸⁾		QuickBird, Pleiades	Processin g Date	Cloud Masking		Thematic Classification		Manual Enhancement		Mosaicking		Acceptance Status	Backup	Comment
No.	File Name [e.g yymmdd; tbd...]	processed		Software / Method	processed	Software / Method	processed	Software / Method	processed	Software / Method				
QuickBird														
1.	o6JAN22045500-M2AS-056358854080_01_P002 o6DEC07050109-M2AS-056358854080_01_P003 o6DEC20045602-M2AS-056358854080_01_P001	13.04.2017	N	N/A	Y	eCognition/semi-automatic approach	Y	Webim/GISA T proprietary solution	N	N/A	Y	Y		
Pleiades														
2.	PHR1B_MS_201701120450161_SEN_2226845101-002	13.04.2017	N	N/A	Y	eCognition/semi-automatic approach	Y	Webim/GISA T proprietary solution	N	N/A	Y	Y		

5.1 Thematic Accuracy

No.	Product Name	Processing Date	Area Coverage	No. of Classes	Reference Data	Sample Size	Sampling Unit ⁽²¹⁾	Sampling Design ⁽²²⁾	Sample Exclusion Criteria ⁽²³⁾	Thematic Accuracy ⁽²⁴⁾	Acceptance Status	Comment
1.	UrbanGreenAreas_2006	2.6.2017	Dhaka – core city	14	QuickBird	500 points	POINT	Stratified random sampling	None	84%	Yes	
2.	UrbanGreenAreas_2017	2.6.2017	Dhaka – core city	14	Pleiades	500 points	POINT	Stratified random sampling	None	84%	Yes	

6.1 Completeness

INPUT DATA									
No.	Item	AOI Coverage [km²]	Area Coverage [km²]	Completeness of Coverage ⁽²⁵⁾	No. of Scenes	Scenes used in Production	Completeness of Verification ⁽²⁷⁾	Metadata	Comments
2.	VHR EO Data	324 km²	324 km²	100%	4	4	100%	Yes	none
4.	Ancillary Data		N/A	N/A	N/A	N/A	N/A	Yes	none
PRODUCTS									
No.	Product ⁽²⁸⁾	AOI Coverage [km²]	Product Coverage [km²]	Completeness of Coverage ⁽²⁵⁾	Completeness of Classification ⁽²⁶⁾	Unclassifiable Area [%]	Completeness of Verification ⁽²⁷⁾	Metadata	Comments
1.	UrbanGreenAreas_2006	324 km²	324 km²	100%	100%	N/A	100%	Yes	none
2.	UrbanGreenAreas_2017		324 km²	100%	100%	N/A	100%	Yes	none

6.2 Compliancy					
Product 1 ⁽²⁸⁾		Urban Green Areas and Change (2006, 2017)			
Abstract					
Urban green areas refer to land within and on the edges of a city that is partly or completely covered with grass, trees, shrubs, or other vegetation. This includes public parks, private gardens, cemeteries, forested areas as well as trees, river alignments, hedges, etc. The product delivered within EO4SD-Urban project thus provides accurate information (1m resolution) on the spatial location and extent of the green areas located within the Urban Extent derived from baseline LULC information product.					
Service / Product Specifications					
Area Coverage					
Requirements		Achieved Specifications		Compliance	Comments
Country 1:	Bangladesh	Country 1:	Bangladesh	100 %	
Country 2:	N/A	Country 2:	N/A		
Country 3:	N/A	Country 3:	N/A		
Country 4:	N/A	Country 4:	N/A		
A) Wall-to-wall: Sub-national, Dhaka		A) Wall-to-wall: Sub-national, Dhaka		100 %	
Area of approx. 710 km ² , defined by the user.					
B) Sampling based: N/A		B) Sampling based: N/A			
Time Period - Update Frequency					
Requirements		Achieved Specifications		Compliance	Comments
Baseline Year(s): 2000 2015		Baseline Year(s): 2006 2017		Yes	EO data differs from requirements because of availability of VHR EO data.
B) Update Frequency N/A		B) Update Frequency N/A			
Geographic Reference System					
Requirements		Achieved Specifications		Compliance	Comments
EPSG: 32646. WGS84 / UTM 46N		EPSG: 32646. WGS84 / UTM 46N		Yes	-
Mapping Classes and Definitions (Definitions see REDD+ MRV Design Document)					
Requirements		Achieved Specifications			Compliance
N/A		All classes were agreed by the user.			Yes
Cloud and Shadow Detection and Removal					
Requirements		Achieved Specifications		Compliance	Comments
Cloud free.		Cloud free scenes used.		Yes	-

Spatial Resolution			
Requirements	Achieved Specifications	Compliance	Comments
VHR for core	VHR for core	Yes	-
Minimum Mapping Unit (MMU)			
Requirements	Achieved Specifications	Compliance	Comments
0.25 ha for Core City	0.1 - 0.25 ha for Core City	Yes	-
Data Type			
Requirements	Achieved Specifications	Compliance	Comments
GIS-ready format	ESRI Shapefile / Geodatabase	Yes	-
Bit Depth			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	N/A	N/A	-
Data Format			
Requirements	Achieved Specifications	Compliance	Comments
*.shp ESRI Shapefile (open cross-platform format)	ESRI Shapefile	Yes	-
Class Coding			
Requirements	Achieved Specifications	Compliance	
N/A	N/A		
Metadata			
Requirements	Achieved Specifications	Compliance	Comments
INSPIRE compliant	ISO/Inspire compliant	Yes	-
Service / Product Quality			
Thematic Accuracy			
Requirements	Achieved Specifications	Compliance	Comments
Overall Accuracy: 80%	84%	Yes	-
Positional Accuracy			
Requirements	Achieved Specifications	Compliance	Comments
RMSE < 2 m	RMSE < 2 m	Yes	Calculated from adjustment to OSM roads. Technical difficulties to co-registrate and find appropriate GCP/TPs due to

			extremely high amount of changes in the area
Delivery Procedure			
Service Provision			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	FTP	Yes	
Delivery Date			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	Sep 15 th 2017	Yes	Interim products delivered earlier on demand

Glossary

Quality Checks		
Prefix	Suffix	Explanation
Y (Yes)	- V	Visually checked
N (No)	- Q	Quantitatively/qualitatively checked
N/A	- V Q	Visually and quantitatively/qualitatively checked
	- N/A	Not applicable

Readability ⁽¹⁾	Check readability of all required input data. Can the data be stored again?
Header / Metadata ⁽²⁾	Check Image Header Information and/or Metadata for completeness / distinctive features.
Data Format ⁽³⁾	For digital data, please give file format (e.g. *.tiff, *.shp).
Data Type ⁽⁴⁾	Please specify the type of the data (e.g. raster, vector or analogue).
Bit Depth ⁽⁵⁾	Please give pixel depth and sign of raster data (e.g. 8 bit unsigned integer).
Dynamic Range ⁽⁶⁾	Check dynamic range of all image bands. Visual check of dynamic range should be accompanied by histograms and statistics.
Dropped Lines & Artefacts ⁽⁷⁾	Check Image for dropped lines and other artefacts. If such occurs, please give description of extent and influence in the Comments section.
Sensor 1, 2, ... ⁽⁸⁾	Please delete / add additional sensor sections as necessary.
Purpose ⁽⁹⁾	The purpose and use of the given auxiliary or reference data should be stated: ORHTOrectification, GEOmetric correction, REference, VERification source, POSitional ACCuracy assessment, THEMatic ACCuracy assessment.
Sampling Methodology ⁽¹⁰⁾	Methodology of in situ or reference/auxiliary data sampling scheme should be outlined. In case of sample plots, also state how the plot positions have been determined (e.g. from GPS measurements, topographic maps, terrestrial triangulation, EO data, etc.) and how the sampling grid was established.
Positional Accuracy ⁽¹¹⁾	Positional accuracy of collected in situ data should be given. For reference/auxiliary data it MUST be given. If unknown, the data's use must be explained. For DEM or other data with 3D information please specify both vertical and horizontal Positional Accuracy. For analogue data (e.g. maps) try to give approximate accuracy related to mapping scale.
Completeness ⁽¹²⁾	Data should be checked for spatial/temporal/content gaps.
Model / Algorithm ⁽¹³⁾	Give the name of the software and its version. Specify the software module/algorithm used for: a) geometric correction, e.g., Polynomial and its degree, Rational Functions, Thin Plate Spline, etc. b) classification, e.g., ISODATA, Maximum Likelihood, Neural Networks, etc.
Dataset 1, 2, ... ⁽¹⁴⁾	Please delete / add additional dataset sections as necessary.
Radiometric Processing ⁽¹⁵⁾	State whether (and which) radiometric processing was applied (e.g., contrast enhancement, histogram matching, bundle block adjustment, radiometric normalisation, filtering, etc.).
Projection; Spheroid / Ellipsoid ⁽¹⁶⁾	Always specify completely, i.e. at minimum the Projection (+Zone, if applicable), Spheroid / Ellipsoid, Map Datum. Give additional information if necessary to unambiguously define the reference system.

Ground Control Points (GCPs) ⁽¹⁷⁾	Give the number, distribution and RMS of used Ground Control Points (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of GCPs should be attached as a snapshot, or described in the Comments section.
Tie Points (TPs) ⁽¹⁸⁾	In case of mosaicking, give the number of used Tie Points and their total RMS (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of TP's should be attached as a snapshot, or described in the Comments section.
Check Points (CPs) ⁽¹⁹⁾	The real measure of positional accuracy and the only measure which should be examined as to its Acceptable Range. Give the independent Check Points' total RMS (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of CP's should be attached as a snapshot, or described in the Comments section.
Resampling Method ⁽²⁰⁾	Specify, if / which resampling algorithm has been used (e.g. NN=Nearest Neighbour, BIL=Bilinear Interpolation, CC=Cubic Convolution).
Sampling unit ⁽²¹⁾	Specify sampling unit of Accuracy Assessment as POINT, FRAME, POLYGON and how it is treated (e.g. pixel center, polygon centre, etc.).
Sampling Design ⁽²²⁾	SYST=Systematic, RAND=Random, STRAT=Stratified, SBCLASS=Stratified by class, SBAREA=Stratified by area
Sample exclusion criteria ⁽²³⁾	Describe which <u>criteria</u> you apply for sampling point selection resp. exclusion of certain points. For example, if the point is too close to a class boundary (less than 1 pixel), it is excluded. If the selected sample point is not representative of the class, it is excluded. For these reasons, it is recommended to oversample by 10% to compensate for sample point exclusion.
Thematic Accuracy ⁽²⁴⁾	Provide a detailed description of the Accuracy Assessment results in the form of error matrices showing commission and omission errors, user's, producer's and overall accuracies and other measures of Thematic Accuracy, as the confidence level (usually fixed at 95%) and the respective confidence interval, at least for the overall accuracy. If classification is done in a phased approach, e.g. if Forest Area and subsequently Forest Type are mapped, independent reports have to be produced.
Completeness of Coverage ⁽²⁵⁾	State whether the coverage is limited to a subset, or portion of the final product.
Completeness of Classification ⁽²⁶⁾	State whether classification was constrained to a subset, or portion of the final product.
Completeness of Verification ⁽²⁷⁾	State whether verification applied to lineage, positional, or thematic accuracy is constrained to a subset, or portion of the final product.
Product ⁽²⁸⁾	Please delete / add additional product sections as necessary.

Earth Observation for Sustainable Development - Urban

Quality Assurance and Quality Control Sheets

These QA/QC Templates were prepared by GAF AG compliant with ISO 9001:2008 Quality Management System standards and can only be used by Partners in the current EO4SD-Urban Project.

Project Title:	EO4SD-Urban		
Project Leader:	GAF AG		
Service Provider:	GISAT, s.r.o.	Editor:	Václav Stonáček
Client:	WB	Date:	15.9.2017

<i>Overview of QC-Sheets and Processing Steps</i>	<i>Sheet used</i>	<i>Sheet filled in</i>
---	-------------------	------------------------

Requirements

0.1 Requirements

Yes	Yes
-----	-----

Specifications of Input Data

1.1 List of EO Data

Yes	Yes
-----	-----

1.2 List of In-situ Data

Yes	No
-----	----

1.3 List of Ancillary Data

Yes	Yes
-----	-----

Data Quality Checks

2.1 EO Data Quality

Yes	Yes
-----	-----

2.2 In-situ Data Quality

Yes	No
-----	----

2.3 Ancillary Data Quality

Yes	Yes
-----	-----

Pre-Processing of EO Data

3.1 Geometric Correction

Yes	Yes
-----	-----

3.1.1 Data Fusion

Yes	Yes
-----	-----

3.2 Data Processing

Yes	Yes
-----	-----

Thematic Processing

4.1 Classification

Yes	Yes
-----	-----

Accuracy Assessment

5.1 Thematic Accuracy

Yes	Yes
-----	-----

5.2 Error Matrices

Yes	Yes
-----	-----

Delivery Checks / Delivery

6.1 Completeness

Yes	Yes
-----	-----

6.2 Compliancy

Yes	Yes
-----	-----

Glossary (index numbers in the QA/QC tables refer to the glossary at the end of this document)

Further QC-relevant Documents:

Comments / Characteristics:

o.1 Requirements		
Product 1 ⁽²⁸⁾	Informal Settlements and Change (2006, 2017)	
Abstract		
Informal Settlements Extent and Change product describe spatial location, extent, typology and development of building agglomerations that can be classified as informal settlements/slum areas.		
Service / Product Specifications		
Area Coverage		
Country:	Bangladesh	
City:	Dhaka	
Area km²	Core Urban:	324,34 km²
	Peri-Urban:	N/A
Time Period - Update Frequency		
A) Baseline Year(s): 2006, 2017		B) Update Frequency 2 time horizons, no future update
Comments:.		
Geographic Reference System		
EPSG: 32646. UTM 46N / WGS84		
Mapping Classes and Definitions		
Pocket slum	Fully surrounded by urban fabric	
Linear along rivers	Distance to river < 10m & linear shape	
Linear along railways	Distance to river < 10m & linear shape	
Low density slum	Sealing =< 0.79	
High density slum	Sealing > 0.8	
Under Trees	Greenness > 0.699	
Cloud and Cloud Shadow Detection and Removal		
EO data without clouds or shadows.		
Spatial Resolution		
n.a. (Product provided as Shapefile)		
Minimum Mapping Unit (MMU)		
Minimum Mapping Unit is 0.1 - 0.25 ha for Core City		
Data Type & Format		
Vector files (ESRI File Geodatabase, Shapefile), Maps (JPEG, TIFF, PDF, KMZ)		
Bit Depth		

N/A		
Class Coding		
<i>Class Code</i>	<i>Class Name</i>	<i>RGB Code</i>
N/A	N/A	N/A
Metadata		
Provided as INSPIRE conformant *.xml data set, covering at least the mandatory elements.		
Service / Product Quality		
Thematic Accuracy		
Overall Accuracy: 89%		
Positional Accuracy		
RMSE < 30 m.		
Delivery Procedure		
Service Provision		
Online via FTP		
Delivery Date		
End of June 2017		

1.1 List of EO Data

Sensors ⁽⁸⁾		QuickBird, Pleiades											
File Name		Incoming Date	Acquisition Date	Proc. Level	Path / Row	AOI - City / Region / Country	Spatial Res.	No. of Bands	Cloud Cover (Data Provider)	Projection / Spheroid ⁽¹⁶⁾	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Header / Metadata ⁽²⁾
QuickBird													
1.	o6JAN22045500-M2AS-056358854080_01_P002	03.04.2017	22.01.2006	LV2A	N/A	Dhaka	MUL:2.4 m PAN: 0.6 m	4	0%	UTM46N / WGS84	*.tif	16 Bit u	*.IMD / *.XML
2.	o6DEC07050109-M2AS-056358854080_01_P003	03.04.2017	07.12.2006	LV2A	N/A	Dhaka	MUL:2.4 m PAN: 0.6 m	4	0,4%	UTM46N / WGS84	*.tif	16 Bit u	*.IMD / *.XML
3.	o6DEC20045602-M2AS-056358854080_01_P001	03.04.2017	20.12.2006	LV2A	N/A	Dhaka	MUL:2.4 m PAN: 0.6 m	4	0%	UTM46N / WGS84	*.tif	16 Bit u	*.IMD / *.XML
Pleiades													
4.	PHR1B_MS_201701120450161_SEN_2226845101-002	23.03.2017	12.01.2017	Primary	N/A	Dhaka	MUL:2 m PAN: 0.5 m	4	0%	UTM46N / WGS84	*.tif	16 Bit u	*.XML

1.2 List of In-situ Data

N/A

1.3 List of Ancillary Data

Dataset 1 (1.4)	DEM	Incoming Date	Acquisition Date	AOI - City / Region / Country	Data Type (4)	Data Format (3)	Projection / Spheroid (1.6)	Positional Accuracy (1.1)	No. of Classes	Area Coverage	Header / Metadata (2)
SRTM1arcsec		23.03.2017	11.02.2000	Dhaka / Bangladesh	Raster	*.tif	WGS84	Horizontal: CE90 <20m Vertical: CE90 <16m	N/A	100%	*.mta
Lineage:		The Shuttle Radar Topography Mission (SRTM) was flown aboard the space shuttle Endeavour on February 11 - 22, 2000. The National Aeronautics and Space Administration (NASA) and the National Geospatial-Intelligence Agency (NGA) participated in an international project to acquire radar data which were used to create detailed topographic maps. SRTM data are intended for scientific use with a Geographic Information System (GIS) or other special application software. Endeavour orbited Earth 16 times each day during the 11-day mission completing 176 orbits. SRTM successfully collected radar data over 80% of the Earth's land surface between 60° north and 56° south latitude with data points posted every 1 arc-second (approximately 30 meters).									
Source:		USGS Earth Explorer https://earthexplorer.usgs.gov/									

Dataset 2 (1.4)	OSM	Incoming Date	Acquisition Date	AOI - City / Region / Country	Data Type (4)	Data Format (3)	Projection / Spheroid (1.6)	Positional Accuracy (1.1)	No. of Classes	Area Coverage	Header / Metadata (2)
OSM roads/railways		27.02.2017	N/A	Dhaka / Bangladesh	Vector	*.shp	UTM46N / WGS84	N/A	N/A	100%	N/A
Lineage:		Up-to-date data extracts from the OpenStreetMap project. OSM data is crowd sourced data with contribution from volunteers and freely available.									

Source:	Http://www.download.geofabrik.de

2.1 EO Data Quality

Sensors ⁽⁸⁾		QuickBird, Pleiades		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Band Specifications			Projection / Spheroid ^(4,6)	Scene Location	Completeness of Additional Data	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Cloud Cover (internal check)	Radiometry	Topography	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
File Name [e.g yymmdd; tbd...]		No. of Bands	Band Registration				Spectral/Spatial Resolution													
QuickBird																				
1.	06JAN22045500-M2AS-056358854080_01_P002	Y - Q	Y-V Q	Y-V Q	Y - V Q	Y-V Q	Y - Q	Y - V	Y - V	Y-V Q	Y - Q	Y - Q	Y-V Q	Y-V Q	Y - V	N - V Q	Yes s	none		
2.	06DEC07050109-M2AS-056358854080_01_P003	Y - Q	Y-V Q	Y-V Q	Y - V Q	Y-V Q	Y - Q	Y - V	Y - V	Y-V Q	Y - Q	Y - Q	Y-V Q	Y-V Q	Y - V	N - V Q	Yes es	none		
3.	06DEC20045602-M2AS-056358854080_01_P001	Y - Q	Y-V Q	Y-V Q	Y - V Q	Y-V Q	Y - Q	Y - V	Y - V	Y-V Q	Y - Q	Y - Q	Y-V Q	Y-V Q	Y - V	N - V Q	Yes s	none		
Pleiades																				
4.	PHR1B_MS_201701120450161_SEN_2226845101-002	Y - Q	Y-V Q	Y-V Q	Y - V Q	Y-V Q	Y - Q	Y - V	Y - V	Y-V Q	Y - Q	Y - Q	Y-V Q	Y-V Q	Y - V	N - V Q	Yes s	none		
5.	PHR1B_P_201701120450161_SEN_2226845101-001	Y - Q	Y-V Q	Y-V Q	Y - V Q	Y-V Q	Y - Q	Y - V	Y - V	Y-V Q	Y - Q	Y - Q	Y-V Q	Y-V Q	Y - V	N - V Q	Yes s	none		

2.2 In-situ Data Quality

N/A

2.3 Ancillary Data Quality

Dataset 1 ⁽¹⁴⁾	DEM															
File Name		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Extent	Projection / Spheroid ^(4,6)	Spatial Resolution	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Location	Completeness	Geom. Misalignment	Plausibility	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
SRTM1arcsec		Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - Q	Y - Q	Y - V	Y - V	N - V	Y - V	N - V	Yes	none

Dataset 2 ⁽¹⁴⁾	OSM															
File Name		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Extent	Projection / Spheroid ^(4,6)	Spatial Resolution	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Location	Completeness	Geom. Misalignment	Plausibility	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
OSM roads/railways		Y - Q	Y - Q	N/A	Y - V Q	Y - V Q	N/A	Y - Q	N/A	Y - V Q	Y - V	N - V	Y - V	N - V	Yes	none

3.1 Geometric Correction

Sensors ⁽⁸⁾		QuickBird, Pleiades		Processing Date	AOI City / Region / Country	Projection / Spheroid ⁽¹⁶⁾	No. & RMS (m) of GCPs ⁽¹⁷⁾	No. & RMS (m) of TP _s ⁽¹⁸⁾	No. & RMS (m) of CP _s ⁽¹⁹⁾	Digital Elevation Model (DEM)	Model / Algorithm ⁽¹³⁾	Resampling Method ⁽²⁰⁾	Validation Reports	Acceptance Status	Output File Name [e.g yymmdd; tbd...]
No	File Name [e.g yymmdd; tbd...]														
QuickBird															
1.	06JAN22045500-M2AS-056358854080_01_P002	11.04. 2017	Dhaka/ Banglade sh	UTM46 N / WGS84	28; X RMSE:0 ,45;Y RMSE: 0,46	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006JAN22045500-M2AS- 056358854080_01_P002		
2.	06DEC07050109-M2AS-056358854080_01_P003	11.04. 2017	Dhaka/ Banglade sh	UTM46 N / WGS84	28; X RMSE:0 ,45;Y RMSE: 0,46	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006DEC07050109-M2AS- 056358854080_01_P003		
3.	06DEC20045602-M2AS-056358854080_01_P001	11.04. 2017	Dhaka/ Banglade sh	UTM46 N / WGS84	28; X RMSE:0 ,45;Y RMSE: 0,46	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006DEC20045602-M2AS- 056358854080_01_P001		
4.	06JAN22045500-P2AS-056358854080_01_P002	11.04. 2017	Dhaka/ Banglade sh	UTM46 N / WGS84	25; X RMSE:0 ,83;Y RMSE: 0,47	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006JAN22045500-P2AS- 056358854080_01_P002		
5.	06DEC07050109-P2AS-056358854080_01_P003	11.04. 2017	Dhaka/ Banglade sh	UTM46 N / WGS84	25; X RMSE:0 ,83;Y RMSE: 0,47	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	006DEC07050109-P2AS- 056358854080_01_P003		

6.	o6DEC20045602-P2AS-056358854080_01_P001	11.04.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	25; X RMSE:0 ,83;Y RMSE: 0,47	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	o06DEC20045602-P2AS- 056358854080_01_P001
Pleiades													
7.	PHR1B_MS_201701120450161_SEN_2226845101-002	11.4.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	12; X RMSE:0 ,49;Y RMSE: 0,43	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	oPHR1B_MS_2017011204501 61_SEN_2226845101-002
8.	PHR1B_P_201701120450161_SEN_2226845101-001	11.4.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	9; X RMSE:0 ,32;Y RMSE: 0,08	N/A	N/A	SRTM30	polyno mic; 1st order	CU B	Ye s	Ye s	oPHR1B_P_201701120450161 _SEN_2226845101-001

3.1.1 Data Fusion

Dataset 1 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_QB_20060122_MS		EO4SD_Dhaka_QB_20060122_PAN		EO4SD_Dhaka_QB_20060122_PS2Y			
Sensor	QuickBird		QuickBird		QuickBird			
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)			
Spatial Resolution / (MS/Pan)	2,4m	MS	0,6m	Pan	0,6m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

Dataset 2 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_QB_20061207_MS		EO4SD_Dhaka_QB_20061207_PAN		EO4SD_Dhaka_QB_20061207_PS2Y			
Sensor	QuickBird		QuickBird		QuickBird			
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)			
Spatial Resolution / (MS/Pan)	2,4m	MS	0,6m	Pan	0,6m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

Dataset 3 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_QB_20061220_MS		EO4SD_Dhaka_QB_20061220_PAN		EO4SD_Dhaka_QB_20061220_PS2Y			
Sensor	QuickBird		QuickBird		QuickBird			
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)			
Spatial Resolution / (MS/Pan)	2,4m	MS	0,6m	Pan	0,6m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

Dataset 4 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments
Filename	EO4SD_Dhaka_PLE_20170112_MS		EO4SD_Dhaka_PLE_20170112_PAN		EO4SD_Dhaka_PLE_20170112_PS2Y			
Sensor	Pleiades		Pleiades		Pleiades			
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)			
Spatial Resolution / (MS/Pan)	2m	MS	0,5m	Pan	0,5m	PS		
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir			
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes	

3.2 Data Processing

Sensors ⁽⁸⁾		QuickBird. Pleiades		Processing Date	Atmospheric Correction		Radiometric Processing ⁽¹⁵⁾		Topographic Normalisation		OTHER Calculation:		Acceptance Status	Backup	Comment
No.	File Name [e.g yymmdd; tbd...]		Software / Method		processed	Software / Method	processed	Software / Method	processed	Software / Method	processed	Software / Method			
QuickBird															
1.	06JAN22045500-M2AS-056358854080_01_P002	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none
2.	06DEC07050109-M2AS-056358854080_01_P003	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none
3.	06DEC20045602-M2AS-056358854080_01_P001	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none

Pleiades														
4	PHR1B_MS_201701120450161_SEN_2226845101-002	11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	N/A	Yes	Yes	none
4.1 Classification														

Sensors ⁽⁸⁾		QuickBird, Pleiades		Processi ng Date	Cloud Masking		Thematic Classification		Manual Enhancement		Mosaicking		Acceptance Status	Backup	Comment
No.	File Name [e.g yymmdd; tbd...]	processed	Software / Method		processed	Software / Method	processed	Softwar e / Method	processed	Softwar e / Method					
QuickBird															
1.	06JAN22045500-M2AS-056358854080_01_P002 06DEC07050109-M2AS-056358854080_01_P003 06DEC20045602-M2AS-056358854080_01_P001	13.04.2017	N	N/A	Y	eCognition/se mi-automatic approach	Y	Webim/ GISAT propriet ary solution	N	N/A	Y	Y			
Pleiades															
2.	PHR1B_MS_201701120450161_SEN_2226845101-002	13.04.2017	N	N/A	Y	eCognition/se mi-automatic approach	Y	Webim/ GISAT propriet ary solution	N	N/A	Y	Y			
5.1 Thematic Accuracy															

No.	Product Name	Processing Date	Area Coverage	No. of Classes	Reference Data	Sample Size	Sampling Unit ⁽²¹⁾	Sampling Design ⁽²²⁾	Sample Exclusion Criteria ⁽²³⁾	Thematic Accuracy ⁽²⁴⁾	Acceptance Status	Comment
1.	Slums2006	2.6.2017	Dhaka – core city	6	QuickBird	200 points	POINT	Stratified random sampling	None	>80%	Yes	
2.	Slums2017	2.6.2017	Dhaka – core city	6	Pleiades	200 points	POINT	Stratified random sampling	None	>80%	Yes	

6.1 Completeness

INPUT DATA									
No .	Item	AOI Coverage [km²]	Area Coverage [km²]	Completeness of Coverage ⁽²⁵⁾	No. of Scenes	Scenes used in Production	Completeness of Verification ⁽²⁷⁾	Metadata	Comments
2.	VHR EO Data	324 km²	324 km²	100%	4	4	100%	Yes	none
4.	Ancillary Data		N/A	N/A	N/A	N/A	N/A	Yes	none
PRODUCTS									
No .	Product ⁽²⁸⁾	AOI Coverage [km²]	Product Coverage [km²]	Completeness of Coverage ⁽²⁵⁾	Completeness of Classification ⁽²⁶⁾	Unclassifiable Area [%]	Completeness of Verification ⁽²⁷⁾	Metadata	Comments
1.	Slums2006	324 km²	324 km²	100%	100%	N/A	100%	Yes	none
2.	Slums2016		324 km²	100%	100%	N/A	100%	Yes	none

6.2 Compliancy					
Product 1 ⁽²⁸⁾		Informal Settlements and Change (2006, 2017)			
Abstract					
<i>Informal Settlements Extent and Change product describe spatial location, extent, typology and development of building agglomerations that can be classified as informal settlements/slum areas.</i>					
Service / Product Specifications					
Area Coverage					
Requirements		Achieved Specifications		Compliance	Comments
Country 1:	Bangladesh	Country 1:	Bangladesh	100 %	
Country 2:	N/A	Country 2:	N/A		
Country 3:	N/A	Country 3:	N/A		
Country 4:	N/A	Country 4:	N/A		
A) Wall-to-wall: Sub-national, Dhaka		A) Wall-to-wall: Sub-national, Dhaka		100 %	
Area of approx. 710 km ² , defined by the user.					
B) Sampling based: N/A		B) Sampling based: N/A			
Time Period - Update Frequency					
Requirements		Achieved Specifications		Compliance	Comments
Baseline Year(s): 2000 2015		Baseline Year(s): 2006 2017		Yes	EO data differs from requirements because of availability of VHR EO data.
B) Update Frequency N/A		B) Update Frequency N/A			
Geographic Reference System					
Requirements		Achieved Specifications		Compliance	Comments
EPSG: 32646. WGS84 / UTM 46N		EPSG: 32646. WGS84 / UTM 46N		Yes	-
Mapping Classes and Definitions (Definitions see REDD+ MRV Design Document)					
Requirements		Achieved Specifications			Compliance
N/A		All classes were agreed by the user.			Yes
Cloud and Shadow Detection and Removal					
Requirements		Achieved Specifications		Compliance	Comments
Cloud free.		Cloud free scenes used.		Yes	-

Spatial Resolution			
Requirements	Achieved Specifications	Compliance	Comments
VHR for core	VHR for core	Yes	-
Minimum Mapping Unit (MMU)			
Requirements	Achieved Specifications	Compliance	Comments
0.25 ha for Core City	Minimum Mapping Unit is 0.1 - 0.25 ha for Core City	Yes	-
Data Type			
Requirements	Achieved Specifications	Compliance	Comments
GIS-ready format	ESRI Shapefile / Geodatabase	Yes	-
Bit Depth			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	N/A	N/A	-
Data Format			
Requirements	Achieved Specifications	Compliance	Comments
*.shp ESRI Shapefile (open cross-platform format)	ESRI Shapefile	Yes	-
Class Coding			
Requirements	Achieved Specifications	Compliance	
N/A	N/A		
Metadata			
Requirements	Achieved Specifications	Compliance	Comments
INSPIRE compliant	ISO/Inspire compliant	Yes	-
Service / Product Quality			
Thematic Accuracy			
Requirements	Achieved Specifications	Compliance	Comments
Overall Accuracy: 85%	89%	Yes	-
Positional Accuracy			
Requirements	Achieved Specifications	Compliance	Comments
RMSE < 2 m	RMSE < 2 m	Yes	

			Calculated from adjustment to OSM roads. Technical difficulties to co-registrate and find appropriate GCP/TPs due to extremely high amount of changes in the area
Delivery Procedure			
Service Provision			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	FTP	Yes	
Delivery Date			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	Sep 15 th 2017	Yes	Interim products delivered earlier on demand

Glossary

Quality Checks		
Prefix	Suffix	Explanation
Y (Yes)	- V	Visually checked
N (No)	- Q	Quantitatively/qualitatively checked
N/A	- V Q	Visually and quantitatively/qualitatively checked
	- N/A	Not applicable

Readability ⁽¹⁾	Check readability of all required input data. Can the data be stored again?
Header / Metadata ⁽²⁾	Check Image Header Information and/or Metadata for completeness / distinctive features.
Data Format ⁽³⁾	For digital data, please give file format (e.g. *.tiff, *.shp).
Data Type ⁽⁴⁾	Please specify the type of the data (e.g. raster, vector or analogue).
Bit Depth ⁽⁵⁾	Please give pixel depth and sign of raster data (e.g. 8 bit unsigned integer).
Dynamic Range ⁽⁶⁾	Check dynamic range of all image bands. Visual check of dynamic range should be accompanied by histograms and statistics.
Dropped Lines & Artefacts ⁽⁷⁾	Check Image for dropped lines and other artefacts. If such occurs, please give description of extent and influence in the Comments section.
Sensor 1, 2, ... ⁽⁸⁾	Please delete / add additional sensor sections as necessary.
Purpose ⁽⁹⁾	The purpose and use of the given auxiliary or reference data should be stated: ORHTOrectification, GEOmetric correction, REference, VERification source, POSitional ACCuracy assessment, THEmatic ACCuracy assessment.
Sampling Methodology ⁽¹⁰⁾	Methodology of in situ or reference/auxiliary data sampling scheme should be outlined. In case of sample plots, also state how the plot positions have been determined (e.g. from GPS measurements, topographic maps, terrestrial triangulation, EO data, etc.) and how the sampling grid was established.
Positional Accuracy ⁽¹¹⁾	Positional accuracy of collected in situ data should be given. For reference/auxiliary data it MUST be given. If unknown, the data's use must be explained. For DEM or other data with 3D information please specify both vertical and horizontal Positional Accuracy. For analogue data (e.g. maps) try to give approximate accuracy related to mapping scale.
Completeness ⁽¹²⁾	Data should be checked for spatial/temporal/content gaps.
Model / Algorithm ⁽¹³⁾	Give the name of the software and its version. Specify the software module/algorithm used for: a) geometric correction, e.g., Polynomial and its degree, Rational Functions, Thin Plate Spline, etc. b) classification, e.g., ISODATA, Maximum Likelihood, Neural Networks, etc.
Dataset 1, 2, ... ⁽¹⁴⁾	Please delete / add additional dataset sections as necessary.
Radiometric Processing ⁽¹⁵⁾	State whether (and which) radiometric processing was applied (e.g., contrast enhancement, histogram matching, bundle block adjustment, radiometric normalisation, filtering, etc.).
Projection; Spheroid / Ellipsoid ⁽¹⁶⁾	Always specify completely, i.e. at minimum the Projection (+Zone, if applicable), Spheroid / Ellipsoid, Map Datum. Give additional information if necessary to unambiguously define the reference system.

Ground Control Points (GCPs) ⁽¹⁷⁾	Give the number, distribution and RMS of used Ground Control Points (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of GCPs should be attached as a snapshot, or described in the Comments section.
Tie Points (TPs) ⁽¹⁸⁾	In case of mosaicking, give the number of used Tie Points and their total RMS (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of TP's should be attached as a snapshot, or described in the Comments section.
Check Points (CPs) ⁽¹⁹⁾	The real measure of positional accuracy and the only measure which should be examined as to its Acceptable Range. Give the independent Check Points' total RMS (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of CP's should be attached as a snapshot, or described in the Comments section.
Resampling Method ⁽²⁰⁾	Specify, if / which resampling algorithm has been used (e.g. NN=Nearest Neighbour, BIL=Bilinear Interpolation, CC=Cubic Convolution).
Sampling unit ⁽²¹⁾	Specify sampling unit of Accuracy Assessment as POINT, FRAME, POLYGON and how it is treated (e.g. pixel center, polygon centre, etc.).
Sampling Design ⁽²²⁾	SYST=Systematic, RAND=Random, STRAT=Stratified, SBCLASS=Stratified by class, SBAREA=Stratified by area
Sample exclusion criteria ⁽²³⁾	Describe which <u>criteria</u> you apply for sampling point selection resp. exclusion of certain points. For example, if the point is too close to a class boundary (less than 1 pixel), it is excluded. If the selected sample point is not representative of the class, it is excluded. For these reasons, it is recommended to oversample by 10% to compensate for sample point exclusion.
Thematic Accuracy ⁽²⁴⁾	Provide a detailed description of the Accuracy Assessment results in the form of error matrices showing commission and omission errors, user's, producer's and overall accuracies and other measures of Thematic Accuracy, as the confidence level (usually fixed at 95%) and the respective confidence interval, at least for the overall accuracy. If classification is done in a phased approach, e.g. if Forest Area and subsequently Forest Type are mapped, independent reports have to be produced.
Completeness of Coverage ⁽²⁵⁾	State whether the coverage is limited to a subset, or portion of the final product.
Completeness of Classification ⁽²⁶⁾	State whether classification was constrained to a subset, or portion of the final product.
Completeness of Verification ⁽²⁷⁾	State whether verification applied to lineage, positional, or thematic accuracy is constrained to a subset, or portion of the final product.
Product ⁽²⁸⁾	Please delete / add additional product sections as necessary.

Earth Observation for Sustainable Development - Urban

Quality Assurance and Quality Control Sheets

These QA/QC Templates were prepared by GAF AG compliant with ISO 9001:2008 Quality Management System standards and can only be used by Partners in the current EO4SD-Urban Project.

Project Title:	EO4SD-Urban		
Project Leader:	GAF AG		
Service Provider:	GISAT, s.r.o.	Editor:	Václav Stonáček
Client:	WB	Date:	15.9.2017

<i>Overview of QC-Sheets and Processing Steps</i>	<i>Sheet used</i>	<i>Sheet filled in</i>
---	-------------------	------------------------

Requirements

0.1 Requirements

Yes	Yes
-----	-----

Specifications of Input Data

1.1 List of EO Data

Yes	Yes
-----	-----

1.2 List of In-situ Data

Yes	No
-----	----

1.3 List of Ancillary Data

Yes	Yes
-----	-----

Data Quality Checks

2.1 EO Data Quality

Yes	Yes
-----	-----

2.2 In-situ Data Quality

Yes	No
-----	----

2.3 Ancillary Data Quality

Yes	Yes
-----	-----

Pre-Processing of EO Data

3.1 Geometric Correction

Yes	Yes
-----	-----

3.1.1 Data Fusion

Yes	Yes
-----	-----

3.2 Data Processing

Yes	Yes
-----	-----

Thematic Processing

4.1 Classification

Yes	Yes
-----	-----

Accuracy Assessment

5.1 Thematic Accuracy

Yes	Yes
-----	-----

5.2 Error Matrices

Yes	Yes
-----	-----

Delivery Checks / Delivery

6.1 Completeness

Yes	Yes
-----	-----

6.2 Compliancy

Yes	Yes
-----	-----

Glossary (index numbers in the QA/QC tables refer to the glossary at the end of this document)

Further QC-relevant Documents:

Comments / Characteristics:

0.1 Requirements		
Product 1 ⁽²⁸⁾	Transportation network 2017	
Abstract		
Transportation network in Core City of Dhaka, Bangladesh. Extracted from OpenStreetMap layers and improved by visual re-interpretation. Harmonization of road classes a update of missing ones.		
Service / Product Specifications		
Area Coverage		
Country:	Bangladesh	
City:	Dhaka	
Area km ²	Core Urban:	324,34 km ²
	Peri-Urban:	N/A
Time Period - Update Frequency		
A) Baseline Year(s): 2017		B) Update Frequency no future update
Comments:.		
Geographic Reference System		
EPSG: 32646. UTM 46N / WGS84		
Mapping Classes and Definitions		
Highway / Motorway		
Primary road		
Secondary road		
Tertiary / Local road		
Cloud and Cloud Shadow Detection and Removal		
EO data without clouds or shadows.		
Spatial Resolution		
n.a. (Product provided as Shapefile)		
Minimum Mapping Unit (MMU)		
N/A		
Data Type & Format		
Vector files (ESRI File Geodatabase, Shapefile), Maps (JPEG, TIFF, PDF, KMZ)		
Bit Depth		
N/A		
Class Coding		

<i>Class Code</i>	<i>Class Name</i>	<i>RGB Code</i>
N/A	N/A	N/A
Metadata		
Provided as INSPIRE conformant *.xml data set, covering at least the mandatory elements.		
Service / Product Quality		
Thematic Accuracy		
Overall Accuracy: N/A		
Positional Accuracy		
RMSE < 30 m.		
Delivery Procedure		
Service Provision		
Online via FTP		
Delivery Date		
End of June 2017		

1.1 List of EO Data

Sensor ⁽⁸⁾	Pleiades	Incoming Date	Acquisition Date	Proc. Level	Path / Row	AOI - City / Region / Country	Spatial Res.	No. of Bands	Cloud Cover (Data Provider)	Projection / Spheroid ⁽¹⁶⁾	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Header / Metadata ⁽²⁾
Pleiades													
1.	PHR1B_MS_201701120450161_SEN_2226845101-002	23.03.2017	12.01.2017	Primary	N/A	Dhaka	MUL: 2 m PAN: 0.5 m	4	0%	UTM46N / WGS84	*.tif	16 Bit u	*.XML

1.2 List of In-situ Data

N/A

1.3 List of Ancillary Data

Dataset 1 (1.4)	DEM	Incoming Date	Acquisition Date	AOI - City / Region / Country	Data Type (4)	Data Format (3)	Projection / Spheroid (1.6)	Positional Accuracy (1.1)	No. of Classes	Area Coverage	Header / Metadata (2)
SRTM1arcsec		23.03.2017	11.02.2000	Dhaka / Bangladesh	Raster	*.tif	WGS84	Horizontal: CE90 <20m Vertical: CE90 <16m	N/A	100%	*.mta
Lineage:		The Shuttle Radar Topography Mission (SRTM) was flown aboard the space shuttle Endeavour on February 11 - 22, 2000. The National Aeronautics and Space Administration (NASA) and the National Geospatial-Intelligence Agency (NGA) participated in an international project to acquire radar data which were used to create detailed topographic maps. SRTM data are intended for scientific use with a Geographic Information System (GIS) or other special application software. Endeavour orbited Earth 16 times each day during the 11-day mission completing 176 orbits. SRTM successfully collected radar data over 80% of the Earth's land surface between 60° north and 56° south latitude with data points posted every 1 arc-second (approximately 30 meters).									
Source:		USGS Earth Explorer https://earthexplorer.usgs.gov/									

Dataset 2 (1.4)	OSM	Incoming Date	Acquisition Date	AOI - City / Region / Country	Data Type (4)	Data Format (3)	Projection / Spheroid (1.6)	Positional Accuracy (1.1)	No. of Classes	Area Coverage	Header / Metadata (2)
OSM roads/railways		27.02.2017	N/A	Dhaka / Bangladesh	Vector	*.shp	UTM46N / WGS84	N/A	N/A	100%	N/A
Lineage:		Up-to-date data extracts from the OpenStreetMap project. OSM data is crowd sourced data with contribution from volunteers and freely available.									

Source:	Http://www.download.geofabrik.de

2.1 EO Data Quality

Sensor ⁽⁸⁾		Pleiades		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Band Specifications			Projection / Spheroid ^(4,6)	Scene Location	Completeness of Additional Data	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Cloud Cover (internal check)	Radiometry	Topography	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
File Name [e.g yymmdd; tbd...]							No. of Bands	Band Registration	Spectral/Spatial Resolution											
Pleiades																				
1.	PHR1B_MS_201701120450161_SEN_2226845101-002			Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - V	Y - V Q	Y - Q	Y - Q	Y - V Q	Y - V Q	Y - V	N - V Q	Yes s	none	
2.	PHR1B_P_201701120450161_SEN_2226845101-001			Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - V	Y - V	Y - V Q	Y - Q	Y - Q	Y - V Q	Y - V Q	Y - V	N - V Q	Yes s	none	

2.2 In-situ Data Quality

N/A

2.3 Ancillary Data Quality

Dataset 1 ⁽¹⁴⁾	DEM															
File Name		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Extent	Projection / Spheroid ^(4,6)	Spatial Resolution	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Location	Completeness	Geom. Misalignment	Plausibility	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
SRTM1arcsec		Y - Q	Y - V Q	Y - V Q	Y - V Q	Y - V Q	Y - Q	Y - Q	Y - Q	Y - V	Y - V	N - V	Y - V	N - V	Yes	none

Dataset 2 ⁽¹⁴⁾	OSM															
File Name		Backup	Readability ⁽¹⁾	Check Header / Metadata ⁽²⁾	Extent	Projection / Spheroid ^(4,6)	Spatial Resolution	Data Format ⁽³⁾	Bit Depth ⁽⁵⁾	Location	Completeness	Geom. Misalignment	Plausibility	Dropped Lines / Artefacts ⁽⁷⁾	Acceptance Status	Comments
OSM roads/railways		Y - Q	Y - Q	N/A	Y - V Q	Y - V Q	N/A	Y - Q	N/A	Y - V Q	Y - V	N - V	Y - V	N - V	Yes	none

3.1 Geometric Correction

Sensor ⁽⁸⁾		Pleiades		Processing Date	AOI City / Region / Country	Projection / Spheroid ⁽¹⁶⁾	No. & RMS (m) of GCPs ⁽¹⁷⁾	No. & RMS (m) of TPs ⁽¹⁸⁾	No. & RMS (m) of CPs ⁽¹⁹⁾	Digital Elevation Model (DEM)	Model / Algorithm ⁽¹³⁾	Resampling Method ⁽²⁰⁾	Validation Reports	Acceptance Status	Output File Name [e.g yymmdd; tbd...]
No	File Name [e.g yymmdd; tbd...]														
Pleiades															
1.	PHR1B_MS_201701120450161_SEN_2226845101-002	11.4.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	12; X RMSE:0 ,49; Y RMSE: 0,43	N/A	N/A	SRTM30	polynomial; 1st order	CUB	Yes	Yes	oPHR1B_MS_201701120450161_SEN_2226845101-002		
2.	PHR1B_P_201701120450161_SEN_2226845101-001	11.4.2017	Dhaka/ Bangladesh	UTM46 N / WGS84	9; X RMSE:0 ,32; Y RMSE: 0,08	N/A	N/A	SRTM30	polynomial; 1st order	CUB	Yes	Yes	oPHR1B_P_201701120450161_SEN_2226845101-001		

3.1.1 Data Fusion									
Dataset 1 ⁽¹⁴⁾	Input A		Input B		Output		Acceptance Status	Comments	
Filename	EO4SD_Dhaka_PLE_20170112_MS		EO4SD_Dhaka_PLE_20170112_PAN		EO4SD_Dhaka_PLE_20170112_PS2Y				
Sensor	Pleiades		Pleiades		Pleiades				
Method	PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)		PANSHARP ₂ (PCI)				
Spatial Resolution / (MS/Pan)	2m	MS	0,5m	Pan	0,5m	PS			
Band Combination	R,G,B,Nir		PAN		R,G,B,Nir				
Data Format ⁽³⁾ & Bit Depth ⁽⁵⁾	*.img	16 Bit u	*.img	16 Bit u	*.tif	16 Bit u	Yes		

3.2 Data Processing																
Sensor ⁽⁸⁾		Pleiades		Processing Date	Atmospheric Correction		Radiometric Processing ⁽¹⁵⁾		Topographic Normalisation		OTHER Calculation:		Acceptance Status	Backup	Comments	
No.	File Name [e.g yymmdd; tbd...]				processed	Software / Method	processed	Software / Method	processed	Software / Method	processed	Software / Method				
Pleiades																
1.	PHR1B_MS_201701120450161_SEN_2226845101-002			11.04.2017	N	N/A	N	N/A	N	N/A	N	N/A	Yes	Yes	none	

4.1 Classification

Sensor ⁽⁸⁾		Pleiades		Processing Date	Cloud Masking		Thematic Classification		Manual Enhancement		Mosaicking		Acceptance Status	Backup	Comment
No.	File Name [e.g yymmdd; tbd...]		processed		Software / Method	processed	Software / Method	processed	Software / Method	processed	Software / Method				
Pleiades															
1.	PHR1B_MS_201701120450161_SEN_2226845101-002		13.04.2017	N	N/A	Y	eCognition/semi-automatic approach	Y	Webim/GISAT proprietary solution	N	N/A	Y	Y		

5.1 Thematic Accuracy

No.	Product Name	Processing Date	Area Coverage	No. of Classes	Reference Data	Sample Size	Sampling Unit ⁽²¹⁾	Sampling Design ⁽²²⁾	Sample Exclusion Criteria ⁽²³⁾	Thematic Accuracy ⁽²⁴⁾	Acceptance Status	Comment
1.	Transportation network 2006	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

6.1 Completeness

INPUT DATA									
No .	Item	AOI Coverage [km²]	Area Coverage [km²]	Completeness of Coverage ⁽²⁵⁾	No. of Scenes	Scenes used in Production	Completeness of Verification ⁽²⁷⁾	Metadata	Comments
2.	VHR EO Data	324 km²	324 km²	100%	4	4	100%	Yes	none
4.	Ancillary Data		N/A	N/A	N/A	N/A	N/A	Yes	none
PRODUCTS									
No .	Product ⁽²⁸⁾	AOI Coverage [km²]	Product Coverage [km²]	Completeness of Coverage ⁽²⁵⁾	Completeness of Classification ⁽²⁶⁾	Unclassifiable Area [%]	Completeness of Verification ⁽²⁷⁾	Metadata	Comments
1.	Transportation network 2017	324 km²	324 km²	100%	100%	N/A	100%	Yes	none

6.2 Compliance					
Product 1 ⁽²⁸⁾		Transportation network 2017			
Abstract					
Transportation network in Core City of Dhaka, Bangladesh. Extracted from OpenStreetMap layers and improved by visual re-interpretation. Harmonization of road classes a update of missing ones.					
Service / Product Specifications					
Area Coverage					
Requirements		Achieved Specifications		Compliance	Comments
Country 1:	Bangladesh	Country 1:	Bangladesh	100 %	
Country 2:	N/A	Country 2:	N/A		
Country 3:	N/A	Country 3:	N/A		
Country 4:	N/A	Country 4:	N/A		
A) Wall-to-wall: Sub-national, Dhaka		A) Wall-to-wall: Sub-national, Dhaka		100 %	
Area of approx. 710 km ² , defined by the user.					
B) Sampling based: N/A		B) Sampling based: N/A			
Time Period - Update Frequency					
Requirements		Achieved Specifications		Compliance	Comments
Baseline Year(s): 2017		Baseline Year(s): 2017		Yes	
B) Update Frequency N/A		B) Update Frequency N/A			
Geographic Reference System					
Requirements		Achieved Specifications		Compliance	Comments
EPSG: 32646. WGS84 / UTM 46N		EPSG: 32646. WGS84 / UTM 46N		Yes	-
Mapping Classes and Definitions (Definitions see REDD+ MRV Design Document)					
Requirements		Achieved Specifications		Compliance	
N/A		All classes were agreed by the user.		Yes	
Cloud and Shadow Detection and Removal					
Requirements		Achieved Specifications		Compliance	Comments
Cloud free.		Cloud free scenes used.		Yes	-

Spatial Resolution			
Requirements	Achieved Specifications	Compliance	Comments
VHR for core	VHR for core	Yes	-
Minimum Mapping Unit (MMU)			
Requirements	Achieved Specifications	Compliance	Comments
N/A	N/A	Yes	-
Data Type			
Requirements	Achieved Specifications	Compliance	Comments
GIS-ready format	ESRI Shapefile / Geodatabase	Yes	-
Bit Depth			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	N/A	N/A	-
Data Format			
Requirements	Achieved Specifications	Compliance	Comments
*.shp ESRI Shapefile	ESRI Shapefile	Yes	-
(open cross-platform format)			
Class Coding			
Requirements	Achieved Specifications		Compliance
N/A	N/A		
Metadata			
Requirements	Achieved Specifications	Compliance	Comments
INSPIRE compliant	ISO/Inspire compliant	Yes	-
Service / Product Quality			
Thematic Accuracy			
Requirements	Achieved Specifications	Compliance	Comments
Overall Accuracy: 85%	N/A	Yes	-
Positional Accuracy			
Requirements	Achieved Specifications	Compliance	Comments
RMSE < 2 m	RMSE < 2 m	Yes	Calculated from adjustment to OSM roads. Technical difficulties to co-register and find appropriate GCP/TPs due to extremely high amount of changes in the area

Delivery Procedure			
Service Provision			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	FTP	Yes	
Delivery Date			
Requirements	Achieved Specifications	Compliance	Comments
Not specified	Sep 15 th 2017	Yes	Interim products delivered earlier on demand

Glossary

Quality Checks		
Prefix	Suffix	Explanation
Y (Yes)	- V	Visually checked
N (No)	- Q	Quantitatively/qualitatively checked
N/A	- V Q	Visually and quantitatively/qualitatively checked
	- N/A	Not applicable

Readability ⁽¹⁾	Check readability of all required input data. Can the data be stored again?
Header / Metadata ⁽²⁾	Check Image Header Information and/or Metadata for completeness / distinctive features.
Data Format ⁽³⁾	For digital data, please give file format (e.g. *.tiff, *.shp).
Data Type ⁽⁴⁾	Please specify the type of the data (e.g. raster, vector or analogue).
Bit Depth ⁽⁵⁾	Please give pixel depth and sign of raster data (e.g. 8 bit unsigned integer).
Dynamic Range ⁽⁶⁾	Check dynamic range of all image bands. Visual check of dynamic range should be accompanied by histograms and statistics.
Dropped Lines & Artefacts ⁽⁷⁾	Check Image for dropped lines and other artefacts. If such occurs, please give description of extent and influence in the Comments section.
Sensor 1, 2, ... ⁽⁸⁾	Please delete / add additional sensor sections as necessary.
Purpose ⁽⁹⁾	The purpose and use of the given auxiliary or reference data should be stated: ORHTOrectification, GEOmetric correction, REference, VERification source, POSitional ACCuracy assessment, THEmatic ACCuracy assessment.
Sampling Methodology ⁽¹⁰⁾	Methodology of in situ or reference/auxiliary data sampling scheme should be outlined. In case of sample plots, also state how the plot positions have been determined (e.g. from GPS measurements, topographic maps, terrestrial triangulation, EO data, etc.) and how the sampling grid was established.
Positional Accuracy ⁽¹¹⁾	Positional accuracy of collected in situ data should be given. For reference/auxiliary data it MUST be given. If unknown, the data's use must be explained. For DEM or other data with 3D information please specify both vertical and horizontal Positional Accuracy. For analogue data (e.g. maps) try to give approximate accuracy related to mapping scale.
Completeness ⁽¹²⁾	Data should be checked for spatial/temporal/content gaps.
Model / Algorithm ⁽¹³⁾	Give the name of the software and its version. Specify the software module/algorithm used for: a) geometric correction, e.g., Polynomial and its degree, Rational Functions, Thin Plate Spline, etc. b) classification, e.g., ISODATA, Maximum Likelihood, Neural Networks, etc.
Dataset 1, 2, ... ⁽¹⁴⁾	Please delete / add additional dataset sections as necessary.
Radiometric Processing ⁽¹⁵⁾	State whether (and which) radiometric processing was applied (e.g., contrast enhancement, histogram matching, bundle block adjustment, radiometric normalisation, filtering, etc.).
Projection; Spheroid / Ellipsoid ⁽¹⁶⁾	Always specify completely, i.e. at minimum the Projection (+Zone, if applicable), Spheroid / Ellipsoid, Map Datum. Give additional information if necessary to unambiguously define the reference system.

Ground Control Points (GCPs) ⁽¹⁷⁾	Give the number, distribution and RMS of used Ground Control Points (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of GCPs should be attached as a snapshot, or described in the Comments section.
Tie Points (TPs) ⁽¹⁸⁾	In case of mosaicking, give the number of used Tie Points and their total RMS (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of TP's should be attached as a snapshot, or described in the Comments section.
Check Points (CPs) ⁽¹⁹⁾	The real measure of positional accuracy and the only measure which should be examined as to its Acceptable Range. Give the independent Check Points' total RMS (as average per scene) as obtained from the Geometric Correction, in meters [m]. Optionally, also RMSx and RMSy may be given. Distribution of CP's should be attached as a snapshot, or described in the Comments section.
Resampling Method ⁽²⁰⁾	Specify, if / which resampling algorithm has been used (e.g. NN=Nearest Neighbour, BIL=Bilinear Interpolation, CC=Cubic Convolution).
Sampling unit ⁽²¹⁾	Specify sampling unit of Accuracy Assessment as POINT, FRAME, POLYGON and how it is treated (e.g. pixel center, polygon centre, etc.).
Sampling Design ⁽²²⁾	SYST=Systematic, RAND=Random, STRAT=Stratified, SBCLASS=Stratified by class, SBAREA=Stratified by area
Sample exclusion criteria ⁽²³⁾	Describe which <u>criteria</u> you apply for sampling point selection resp. exclusion of certain points. For example, if the point is too close to a class boundary (less than 1 pixel), it is excluded. If the selected sample point is not representative of the class, it is excluded. For these reasons, it is recommended to oversample by 10% to compensate for sample point exclusion.
Thematic Accuracy ⁽²⁴⁾	Provide a detailed description of the Accuracy Assessment results in the form of error matrices showing commission and omission errors, user's, producer's and overall accuracies and other measures of Thematic Accuracy, as the confidence level (usually fixed at 95%) and the respective confidence interval, at least for the overall accuracy. If classification is done in a phased approach, e.g. if Forest Area and subsequently Forest Type are mapped, independent reports have to be produced.
Completeness of Coverage ⁽²⁵⁾	State whether the coverage is limited to a subset, or portion of the final product.
Completeness of Classification ⁽²⁶⁾	State whether classification was constrained to a subset, or portion of the final product.
Completeness of Verification ⁽²⁷⁾	State whether verification applied to lineage, positional, or thematic accuracy is constrained to a subset, or portion of the final product.
Product ⁽²⁸⁾	Please delete / add additional product sections as necessary.